Title: SPICE Models for TS18 Standard Logic Process

Purpose and Scope:

Purpose: This document contains SPICE electrical specification parameters for CMOS transistors and analog components as supported by DR2 0018 SL_SCL process.

Scope:

Spice models for 0.18-micron Standard Logic CMOS process (for DR2 0018 SL_SCL technology)

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List of Legal Devices Covered in this document:

Table 2-1

S. No	Device Name	Type	Device Description	SPICE Name	P-cell	LVS
1	MOSFETs	1.8V,Svt	1.8V SVt NMOS Transistor	n18	nmos_18	N
2	MOSFETs	1.8V,Svt	1.8V SVt PMOS Transistor	p18	pmos_18	P
3	MOSFETs	1.8V,Svt	1.8V SVt NMOS Transistor in IPW	n18	nmos_18_iso	N_ISO
4	MOSFETs	1.8V,Hvt-Low Leakage	1.8V HVt NMOS transistor for Low Leakage	n18hvtb3	n18hvt	N18HVT
5	MOSFETs	1.8V,Hvt-Low Leakage	1.8V HVt PMOS transistor for Low leakage	p18hvtb3	p18hvt	P18HVT
6	MOSFETs	1.8V,Hvt-Low Leakage	1.8V HVt NMOS transistor for Low leakage in IPW	n18hvtb3	n18hvt_iso	N18HVT_ISO
7	MOSFETs	1.8V,Native	1.8V NMOS Native Transistor	natlv	nmos_native	NA
8	MOSFETs	3.3V,Svt	3.3V SVt NMOS Transistor	nhv	nmos_33	NH
9	MOSFETs	3.3V,Svt	3.3V SVt PMOS Transistor	phv	pmos_33	PH
10	MOSFETs	3.3V,Svt	3.3V SVt NMOS Transistor in IPW	nhv	nmos_33_iso	NH_ISO
11	MOSFETs	3.3V,Native	3.3V NMOS Native Transistor	nathv	nmos_native_33	NB
12	MOSFETs	5VSvt	5V SVt NMOS transistor	n5v	nmos_5V	N5
13	MOSFETs	5VSvt	5V SVt PMOS transistor	p5v	pmos_5V	P5
14	MOSFETs	5VSvt	5V SVt NMOS transistor in IPW	n5v	nmos_5V_iso	N5_ISO
15	MOSFETs	5V,Native	5V NMOS Native Transistor	nat5v	nmos_native_5	NB5
16	BJT	1.8V, BJT	1.8V Vertical PNP 10um x 10um BJT	vd10	vpnp18_10	VD10

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17	BJT	1.8V, BJT	1.8V Vertical PNP 5um x 5um BJT	vd5	vpnp18_5	VD5
18	BJT	1.8V, BJT	1.8V Vertical PNP 2um x 2um BJT	vd2	vpnp18_2	VD
19	BJT	1.8/3.3/5V, BJT	1.8/3.3/5V Vertical NPN 5um x 5um BJT	vdn5	vnpn18_5	VDN5
20	BJT	3.3/5V, BJT	3.3/5 Vertical PNP 10um x 10um BJT	vd10	vpnp33_10	VH10
21	BJT	3.3/5V, BJT	3.3/5 Vertical PNP 5um x 5um BJT	vd5	vpnp33_5	VH5
22	BJT	3.3/5V, BJT	3.3/5 Vertical PNP 2um x 2um BJT	vd2	vpnp33_2	VH
23	Capacitors		1.8V Scalable Accumulation capacitor N+ Gate/ NW, 2 terminals	nwcap2t	cnwnmos_thn2t	CL
24	Capacitors		1.8V Scalable Accumulation capacitor N+ Gate/NW)3 terminals	nwcap3t	cnwnmos_thn3t	CL3
25	Capacitors		3.3V Scalable Accumulation capacitor, N+ Gate/NW,2 terminals)	nwcaph2t	cnwnmos_thk2t	CH
26	Capacitors	3.3V, Acc.Capacitor	3.3V Scalable Accumulation capacitor, N+ Gate/NW,3 terminals	nwcaph3t	cnwnmos_thk3t	CH3
27	Capacitors	5V, Acc.Capacitor	5V Accumulation capacitor, N+ Gate/NW.	ch53t	ch53t	CH53T
28	Capacitors	MIM cap	MIM Capacitor, 1fF/um2 capacitor	cmim_sq	cmim_sq	CB
29	Capacitors	MFC in USG	M15 USG 2T MFC	usgcm52t	usgcm52t	usgcm52t
30	Capacitors	MFC in USG	M13 USG 2T MFC	usgcm32t	usgcm32t	usgcm32t
31	Capacitors	MFC in USG	M15 USG 3T MFC	usgcm53t	usgcm53t	usgcm53t
32	Capacitors	MFC in USG	M13 USG 3T MFC	usgcm33t	usgcm33t	usgcm33t

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33	Resistors	poly resistor	HIPO-1k High Ohmic P-type Poly w/o salicide 2 terminal resistor	rphpoly2t	rphpoly2t	RC
34	Resistors	poly resistor	HIPO-1k High Ohmic P-type Poly w/o salicide 3 terminal resistor	rphpoly3t	rphpoly3t	RC3
35	Resistors	poly resistor	2k High Ohmic P-type Poly w/o salicide 2 terminal resistor	mr22t	mr22t	MR22T
36	Resistors	poly resistor	2k High Ohmic P-type Poly w/o salicide 3 terminal resistor	mr23t	mr23t	MR23T
37	Resistors	poly resistor	NMOPO–N-type Poly w/o salicide 2 terminals resistor	rnmpoly2t	rnmpoly2t	RE
38	Resistors	poly resistor	NMOPO–N-type Poly w/o salicide 3 terminals resistor	rnmpoly3t	rnmpoly3t	RE3
39	Resistors	poly resistor	PMOPO–P-type Poly w/o salicide 2 terminals resistor	rpmpoly2t	rpmpoly2t	RD
40	Resistors	poly resistor	PMOPO–P-type Poly w/o salicide 3 terminals resistor	rpmpoly3t	rpmpoly3t	RD3
41	Resistors	poly resistor	N-poly salicide 2 terminals resistor	rnlpoly2t	rnlpoly2t	RH
42	Resistors	poly resistor	N-poly salicide 3 terminals resistor	rnlpoly3t	rnlpoly3t	RH3
43	Resistors	poly resistor	P-poly salicide 2 terminals resistor	rplpoly2t	rplpoly2t	RG
44	Resistors	poly resistor	P-poly salicide 3 terminals resistor	rplpoly3t	rplpoly3t	RG3
45	Resistors	diffusion resistor	N+ AA Diffusion w/o salicide 2 terminals resistor	rnplus2t	rnplus2t	RB
46	Resistors	diffusion resistor	N+ AA Diffusion w/o salicide 3 terminals resistor,	rnplus3t	rnplus3t	RB3
47	Resistors	diffusion resistor	P+ AA Diffusion w/o salicide 2 terminals resistor	rpplus2t	rpplus2t	RA

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48	Resistors	diffusion resistor	P+ AA Diffusion w/o salicide 3 terminals resistor	rpplus3t	rpplus3t	RA3
49	Resistors	diffusion resistor	N+ AA Diffusion salicide 2 terminals resistor	nasr2t	rnplus_sal2t	RK
50	Resistors	diffusion resistor	N+ AA Diffusion salicide 3 terminals resistor	nasr3t	rnplus_sal3t	RK3
51	Resistors	diffusion resistor	P+ AA Diffusion salicide 2 terminals resistor	pasr2t	rpplus_sal2t	RM
52	Resistors	diffusion resistor	P+ AA Diffusion salicide 3 terminals resistor	pasr3t	rpplus_sal3t	RM3
53	Resistors	well resistor	N-Well under AA 2 terminals resistor	rnwellaa2t	rnwell_AA2t	RF
54	Resistors	well resistor	N-Well under AA 3 terminals resistor	rnwellaa3t	rnwell_AA3t	RF3
55	Resistors	well resistor	N-Well under STI 2 terminals resistor	rnwellsti2t	rnwell_STI2t	RW
56	Resistors	well resistor	N-Well under STI 3 terminals resistor	rnwellsti3t	rnwell_STI3t	RW3
57	Resistors	Metal resistor	M1 resistor	rm1	rm1	R1
58	Resistors	Metal resistor	M2 resistor	rm2	rm2	R2
59	Resistors	Metal resistor	M3 resistor	rm3	rm3	R3
60	Resistors	Metal resistor	M4 resistor	rm4	rm4	R4
61	Resistors	Metal resistor	M5 resistor	rm5	rm5	R5
62	Resistors	Metal resistor	ML resistor	rml	rmL	RL
63	Diodes	1.8V diode	1.8V N+ /Pwell Diode,	dn18	ndio_sal	DN
64	Diodes	1.8V diode	1.8V P+ /Nwell Diode	dp18	pdio_sal	DP
65	Diodes	diode	3.3/5V N+ /Pwell Diode	dnh	ndio_33	N3
66	Diodes	diode	3.3/5V P+ /Nwell Diode	dph	pdio_33	P3
67	Diodes	1.8V diode	1.8V Nwell/Psub Diode	dnwell	nwldio	DW
68	Diodes	diode	3.3/5V Nwell/Psub Diode	dnwell33	nwldio_33	W3

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69	Diodes	diode	1.8/3.3/5V Deep-N-Well/Psub Diode	ddwnps18	ddwnps18	DDWNPS18
70	Diodes	diode	1.8/3.3/5V Isolated-P-Well / Deep N-Well diode	ddwnpw18	ddwnpw18	DDWNPW18
71	Diodes	diode	N+ in 5V Pwell diode	dn50	dn50	DN50
72	Diodes	diode	P+ in 5V Nwell diode	dp50	dp50	DP50

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TABLE 14-2 VALUES FOR MODEL PARAMETERS - MIM CAPACITOR MODELS 117

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2 Referenced Documents

Table 2-1: REFERENCED DOCUMENTS

Document Title	Document Name
SPICE Models for TS18 Std. Logic Process	DRS2_0018_B
Design Rules for TS18, Standard Logic Process	DR2 0018 SL_SCL

3 Definitions

Table 3-1: ABBREVATIONS AND TERMS DESCRIPTIONS

Abbreviations/Terms	Description
None	

4 Equipments

N/A

5 Supplies and Materials

N/A

6 Safety

All personnel shall read and comply with Semi-Conductor Laboratory. Safety Procedures, as specified in SOP 09008

7 Environmental protection

All personnel must review their operations with reference to the requirements as noted in Tower's Environmental Quality Master Specifications (QMS).

8 Extraction and Simulation Tools

The SPICE models were measured and extracted from the Engineering Vehicle PYTHON using ICCAP & BsimProPlus Extraction Tools. Synopsys HSPICE (ver. 2005.03) was used to verify and QA the BSIM3v3 model, and other device models. Accuracy and QA on other simulators are not guaranteed.

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9 MOS Model

Table 8-1: LIST OF LEGAL DEVICES - MOSFETS

Device	Spice Name	Voltage Range
N18 (1.8V NMOS General, Std Vt Transistor)	n18	-1.8 to 1.8V
P18 (1.8V PMOS General Std Vt Transistor)	p18	-1.8 to 1.8V
N18HVT (1.8V NMOS Low Leakage Transistor)	n18hvtb3	-1.8 to 1.8V
P18HVT (1.8V PMOS Low Leakage Transistor)	p18hvtb3	-1.8 to 1.8V
NATLV (1.8V NMOS Native Vt Transistor)	natlv	-1.8 to 1.8V
NHV (3.3V NMOS Transistor)	nhv	-3.3 to 3.3V
PHV (3.3V PMOS Transistor)	phv	-3.3 to 3.3V
NATHV (3.3V NMOS Native Vt Transistor)	nathv	-3.3 to 3.3V
N5V (5V NMOS)	n5v	-5V to 5V
P5V (5V PMOS)	p5v	-5V to 5V
NATH5 (5V NMOS Native Vt Transistor)	nath5v	-5V to 5V

*Isolated transistors use same model as non-isolated transistors. For more information, refer to Design rule document.

9.1 Compatibility

Temperature coefficients are also included in the model. The model valid in the temperature range: -40°C to 125°C. The values of the temperature coefficient are listed in the model card.

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9.2 Model Usage Guidelines

- The 1.8 V N/PMOS model with model name n18.x and p18.x are valid for length: 0.18um to 20um and width 0.22um to 100um.

- The 1.8 V Low Leakage N/PMOS model with model name n18hvtb3.x and p18hvtb3.x are valid for length: 0.18um to 20um and width 0.22um to 100um.

- The 1.8 V Super Low Leakage N/PMOS model with model name n18shvtb3.x and p18shvtb3.x are valid for length: 0.18um to 20um and width 0.22um to 100um.

- The 1.8 V Low Vt N/PMOS model with model name n18lvt.x and p18lvt.x are valid for length: 0.18um to 20um and width 0.22um to 100um.

- The 1.8V native NMOS model with model name natlv is valid for length: 0.5um to 20um and width: 0.22um to 100um.

- The 3.3V NMOS model with model name nhv.x is valid for length: 0.35um to 20um and width: 0.22um to 100um . The 3.3V PMOS model with model name phv.x is valid for length: 0.3um to 20um and width: 0.22um to 100um.

- The 3.3V native NMOS model with model name nathv is valid for length: 1.2um to 20um and width: 0.22um to 100um.

- The 5V NMOS BSIM model with model name n5v is valid for length: 0.6um to 20um and width: 0.6um to 100um. The 5V PMOS BSIM model with model name p5v is valid for length: 0.5um to 20um and width: 0.6um to 100um.

- The 5V NMOS PSP model with model name n5v is valid for length: 0.6um to 20um and width: 0.22um to 100um. The 5V PMOS PSP model with model name p5v is valid for length: 0.5um to 20um and width: 0.22um to 100um.

- The 5V native NMOS model with model name nath5v is valid for length: 1.5um to 20um and width: 0.6um to 100um.

NOTE: Using models for devices out of the above valid ranges may cause large simulation error. For best accuracy it is recommended to use L ≤ 10um.

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- The operating voltage for the transistors are:

1.8V	|Vgs|:0~1.8V, |Vds|:0~1.8V, |Vbs|:0~1.8V
3.3V	|Vgs|:0~3.3V, |Vds|:0~3.3V, |Vbs|:0~3.3V
5V	|Vgs|:0~5V, |Vds|:0~5, |Vbs|:0~5V

- The MOSFET models are valid for base-band frequencies, that are not higher than 0.1*Ft. The Ft can be estimated as Gm/(2*PI*Cgg). Since Cgg is MOSFET geometry dependent and Gm is voltage dependent Ft for large devices may vary from few MHz up to few hundreds of MHz (see figures below). The figures below represent the simulated values for Ft as function of L for specific operating point and Vg for specific geometry.

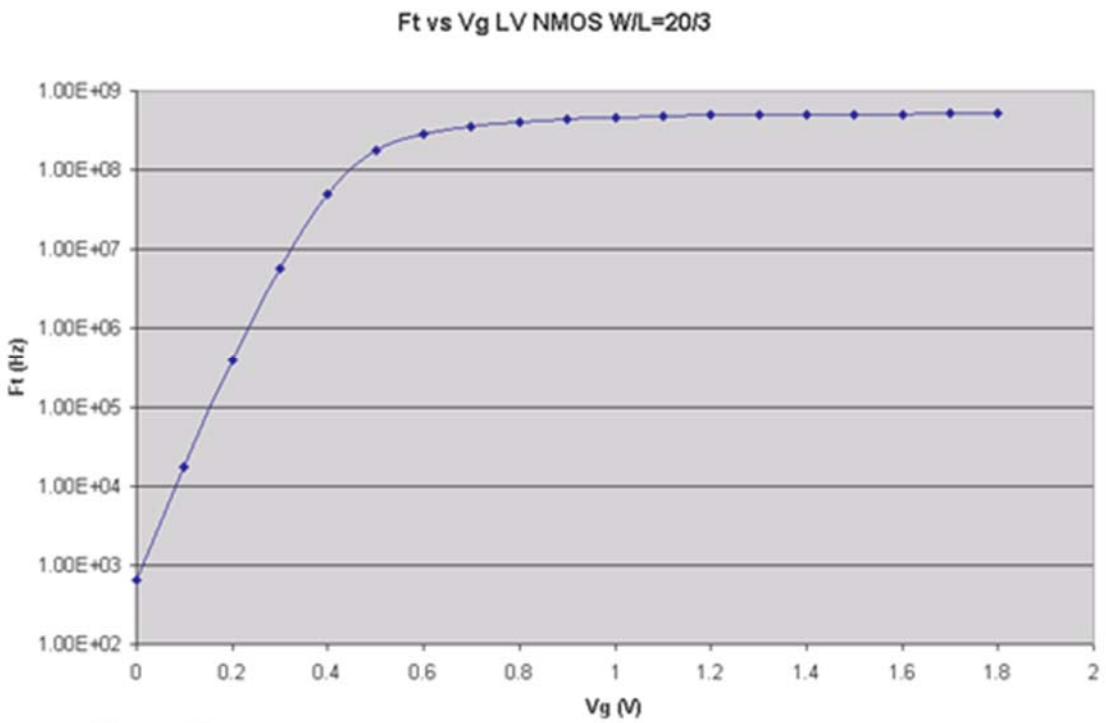

Figure 9-1 ***FT OF 1.8V NMOSFET (STD VTH) W/L=20/3 AS FUNCTION OF VG***

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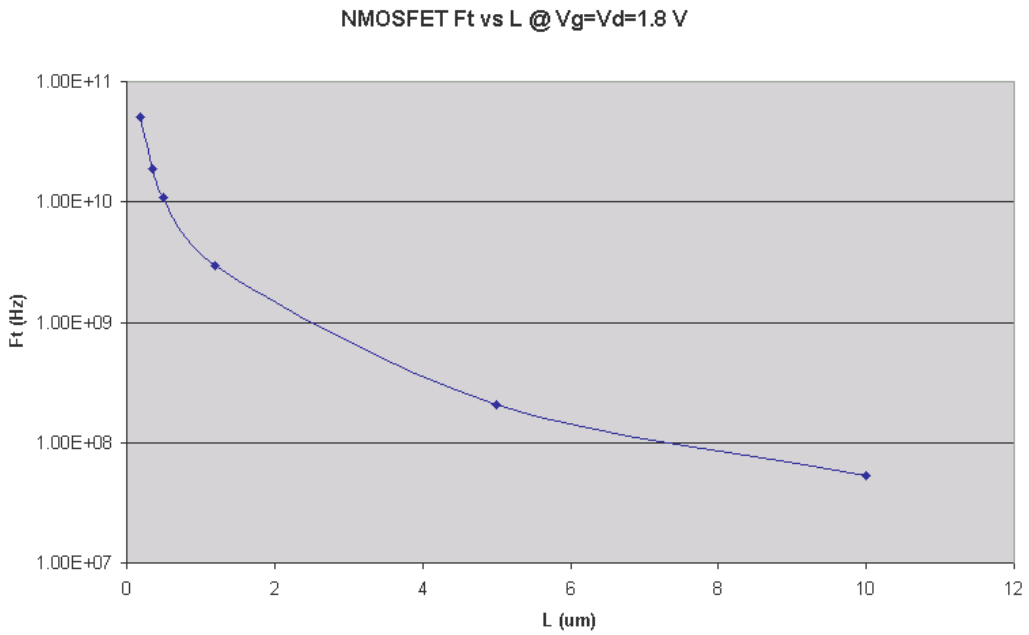

Figure 9-2 ***FT OF 1.8V NMOSFET (STD VTH) @ VD=VG=VDD=1.8 V AND W=20UM AS FUNCTION OF L***

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9.3 Model Description

Multi-device models are defined as follows:

There are nine models for 1.8V N and PMOS. Twenty-seven different n-channel and twenty different p- channel devices were characterized and model parameters were extracted for temperatures at –40, 25, and 125°C. The transistor sizes for extraction of NMOS and PMOS (W/L in um) are listed in tables below:

Table 9-1 1.8V NMOS (STD VTH)

Type	Geometries	No. of Geometries
NMOS	10.0/10.0 10.0/5.0 10.0/2.0 10.0/1.2 10.0/0.8 10.0/0.5 10.0/0.35 10.0/0.25 10.0/0.18 5.0/10.0 5.0/0.18 1.2/10.0 1.2/1.2 1.2/0.5 1.2/0.18 0.5/10.0 0.5/1.2 0.5/0.5 0.5/0.18 0.35/10.0 0.35/0.18 0.28/10.0 0.28/0.18 0.22/10.0 0.22/1.2 0.22/0.5 0.22/0.18	2 7

Table 9-2 1.8V PMOS (STD VTH)

Type	Geometries	No. of Geometries
PMOS	10.0/10.0 10.0/5.0 10.0/2.0 10.0/1.2 10.0/0.8 10.0/0.5 10.0/0.35 10.0/0.25 10.0/0.18 5.0/10.0 5.0/0.18 1.2/10.0 1.2/1.2 1.2/0.5 1.2/0.18 0.5/10.0 0.5/1.2 0.5/0.5 0.5/0.18 0.35/10.0 0.35/0.18 0.28/10.0 0.28/0.18 0.22/10.0 0.22/1.2 0.22/0.5 0.22/0.18	27

There are nine models for 1.8V Low Leakage N and PMOS. Twenty-seven different n-channel and twenty different p-channel devices were characterized and model parameters were extracted for temperatures at –40, 25, and 125°C. The transistor sizes for extraction of NMOS and PMOS (W/L in um) are listed in tables below:

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Table 9-3 1.8V LOW LEAKAGE NMOS

T	Geometries	No. of Geometries
NMOS	10.0/10.0 10.0/5.0 10.0/2.0 10.0/1.2 10.0/0.8 10.0/0.5 10.0/0.35 10.0/0.25 10.0/0.18 5.0/10.0 5.0/0.18 1.2/10.0 1.2/1.2 1.2/0.5 1.2/0.18 0.5/10.0 0.5/1.2 0.5/0.5 0.5/0.18 0.35/10.0 0.35/0.18 0.28/10.0 0.28/0.18 0.22/10.0 0.22/1.2 0.22/0.5	27
Type	Geometries	No. of Geometries
PMOS	10.0/10.0 10.0/5.0 10.0/2.0 10.0/1.2 10.0/0.8 10.0/0.5 10.0/0.35 10.0/0.25 10.0/0.18 5.0/10.0 5.0/0.18 1.2/10.0 1.2/1.2 1.2/0.5 1.2/0.18 0.5/10.0 0.5/1.2 0.5/0.5 0.5/0.18 0.35/10.0 0.35/0.18 0.28/10.0 0.28/0.18 0.22/10.0 0.22/1.2 0.22/0.5 0.22/0.18	27

NOTE: 1.8V Low Leakage MOS models are not characterized and modeled for Mismatch and 1/f noise analysis.

There are eight model cards for 1.8V Native NMOS transistors. Ten different n-channel devices were characterized and model parameters were extracted for temperatures at –40, 25, and 125C. The transistor sizes for extraction (W/L in um) are listed in table below:

Table 9-4 1.8V NATIVE NMOS

Type	Geometries	No. of Geometries
NMOS	100/10 100/1.2 100/0.5 10/10 10/1.2 10/0.5 1.2/10 1.2/1.2 1.2/0.5 0.5/10 0.5/1.2 0.5/0.5 0.22/10 0.22/1.2 0.22/0.5	15

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There are nine models for 3.3V N and PMOS. Twenty-six different n-channel and twenty-six different p-channel devices were characterized and model parameters were extracted for temperatures at 25, and 125C. The transistor sizes for extraction of NMOS and PMOS (W/L in um) are listed in tables below:

Table 9-5

Type	Geometries	No. of Geometries
NMOS	10.0/10.0 10.0/5.0 10.0/2.0 10.0/1.2 10.0/0.8 10.0/0.5 10.0/0.4 10.0/0.35 5.0/10.0 5.0/0.35 2.0/10 2.0/0.35 1.2/5.0 1.2/1.2 1.2/0.5 1.2/0.35 0.5/10 0.5/1.2 0.5/0.5 0.5/0.35 0.35/10.0 0.35/0.35 0.22/10.0 0.22/1.2 0.22/0.5 0.22/0.35	26

Table 9-6

Type	Geometries	No. of Geometries
PMOS	10.0/10.0 10.0/5.0 10.0/2.0 10.0/1.2 10.0/0.8 10.0/0.5 10.0/0.4 10.0/0.3 5.0/10.0 5.0/0.3 2.0/10 2.0/0.3 1.2/5.0 1.2/1.2 1.2/0.5 1.2/0.3 0.5/10 0.5/1.2 0.5/0.5 0.5/0.3 0.35/10.0 0.35/0.3 0.22/10.0 0.22/1.2 0.22/0.5 0.22/0.3	26

There are four model cards for 3.3V Native NMOS transistors. Ten different n-channel devices were characterized and model parameters were extracted for temperatures at 25, and 125C. The transistor sizes for extraction (W/L in um) are listed in table below:

Table 9-7

Type	Geometries	No. of Geometries
NMOS	100/10 100/1.2 10/10 10/1.2 1.2/5 1.2/1.2 0.5/10 0.5/1.2 0.22/10 0.22/1.2	10

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N5V PSP: There is a global model card for 5V NMOS transistors. 46 different n-channel devices were characterized and model parameters were extracted at temperatures –40C, 25C, 125C and 175C. The transistor sizes for extraction _W/L in um) are listed below:

Table 9-8 -5V LOW VT NMOS

Type	Geometries	No. of Geometries
NMOS	10/10 10/5 10/2 10/1.5 10/1.2 10/0.9 10/0.8 10/0.6 10/0.55 5/10 5/0.5 2/10 2/0.5 1.5/ 0.5 1.2/5 1.2/2 1.2/1.5 1.2/1.2 1.2/0.8 1.2/0.6 0.8/10 0.8 /0.5 0.7/10 0.7/0.5 0.65 /10 0.65/0.5 0.6/10 0.6/0.5 0.55/10 0.55/0.5 0.5/10 0.5/2 0.5/1.5 0.5/1.2 0.5/0.8 0.5/0.5 0.35/1.2 0.22/10 0.22/5 0.22/2 0.22/1.5 0.22/1.2 0.22/0.8 0.22/0.6 0.22/0.5 0.21/0.6	46

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P5V PSP: There is a global model card for 5V PMOS transistors. 56 different p-channel devices were characterized and model parameters were extracted at temperatures –40C, 25C, 125C and 175C. The transistor sizes for extraction _W/L in um) are listed below:

Table 9-9 5V LOW VT PMOS

Type	Geometries	No. of Geometries
PMOS	10/10 10/5 10/2 10/1.2 10/0.8 10/0.6 10/0.55 10/0.45 5/10 5/5 5/1.2 5/0.5 2/10 2/1.2 2/0.5 1.5/10 1.5/2 1.5/1.2 1.5/0.8 1.5/0.5 1.2/10 1.2/5 1.2/2 1.2/0.8 1.2/0.6 1.2/0.5 0.8/10 0.8 /0.5 0.7/10 0.7/0.5 0.65/10 0.65/0.5 0.6/10 0.6/0.5 0.55/10 0.55/0.5 0.5/10 0.5/2 0.5/1.2 0.5/0.8 0.5/0.5 0.35/10 0.35/1.2 0.28/10 0.24/10 0.24/2 0.24/1.2 0.24/0.8 0.24/0.5 0.22/10 0.22/2 0.22/1.2 0.22/0.8 0.22/0.6 0.22/0.5 0.21/0.5	56

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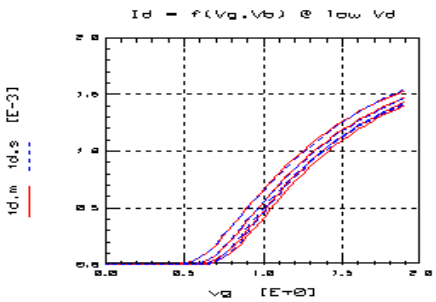
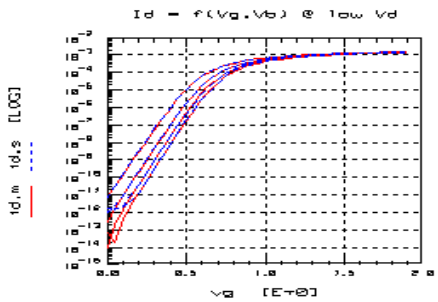
9.4 NMOS and PMOS Extracted Model vs. Measured Data

9.4.1 DC Analysis

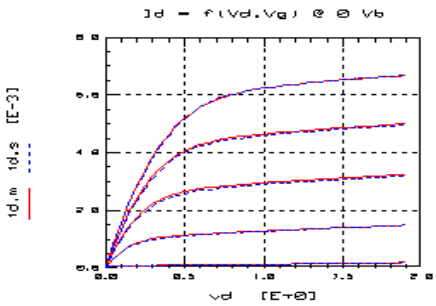
9.4.1.1 1.8V NMOS Standard Vt



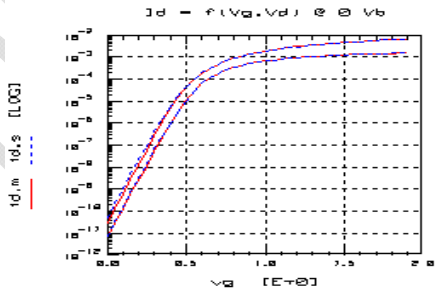
Id = f (Vg,Vb) @ low Vd



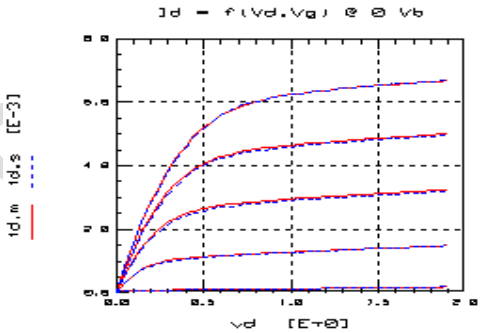
log(Id) = f (Vg,Vb) @ low Vd



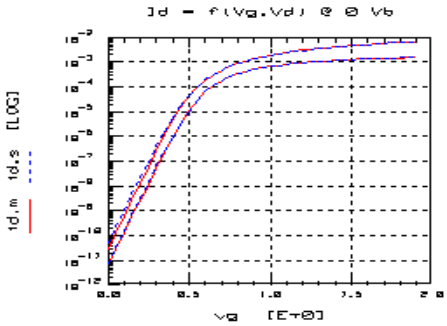
Id = f (Vd,Vg) @ 0 Vb



log(Id) = f(Vg,Vd) @ 0 Vb



Id = f (Vd,Vg) @ 0 Vb



log(Id) = f(Vg,Vd) @ 0 Vb

Figure 9-3 ***1.8V NMOS STANDARD VT T=25 C W/L=10/0.18***

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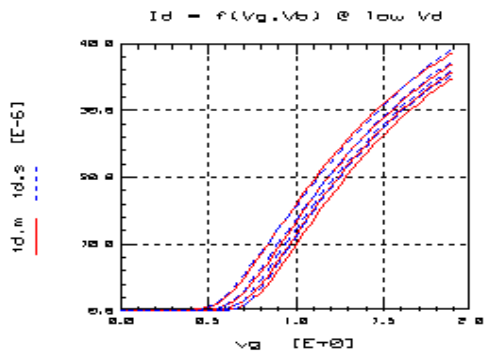
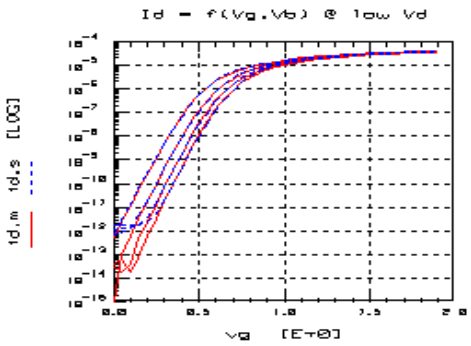
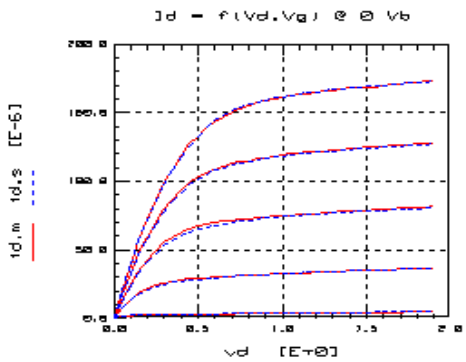


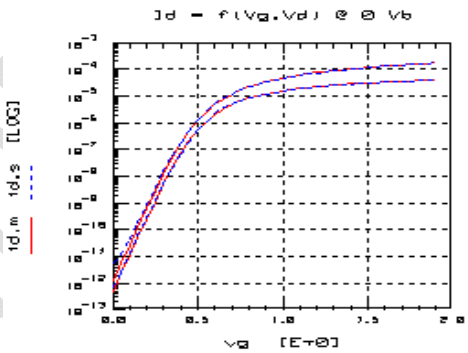
Id = f (Vg,Vb) @ low Vd



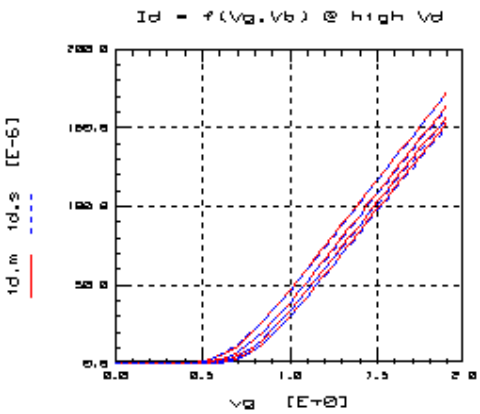
log(Id) = f (Vg,Vb) @ low Vd



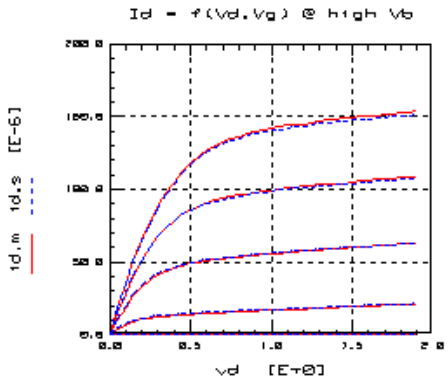
Id = f (Vd,Vg) @ 0 Vb



log(Id) = f(Vg,Vd) @ 0 Vb



Id = f (Vg,Vb) @ high Vd



Id = f (Vd,Vg) @ high Vb

Figure 9-4 1.8V NMOS STANDARD VT T=25 C W/L=0.22/0.18

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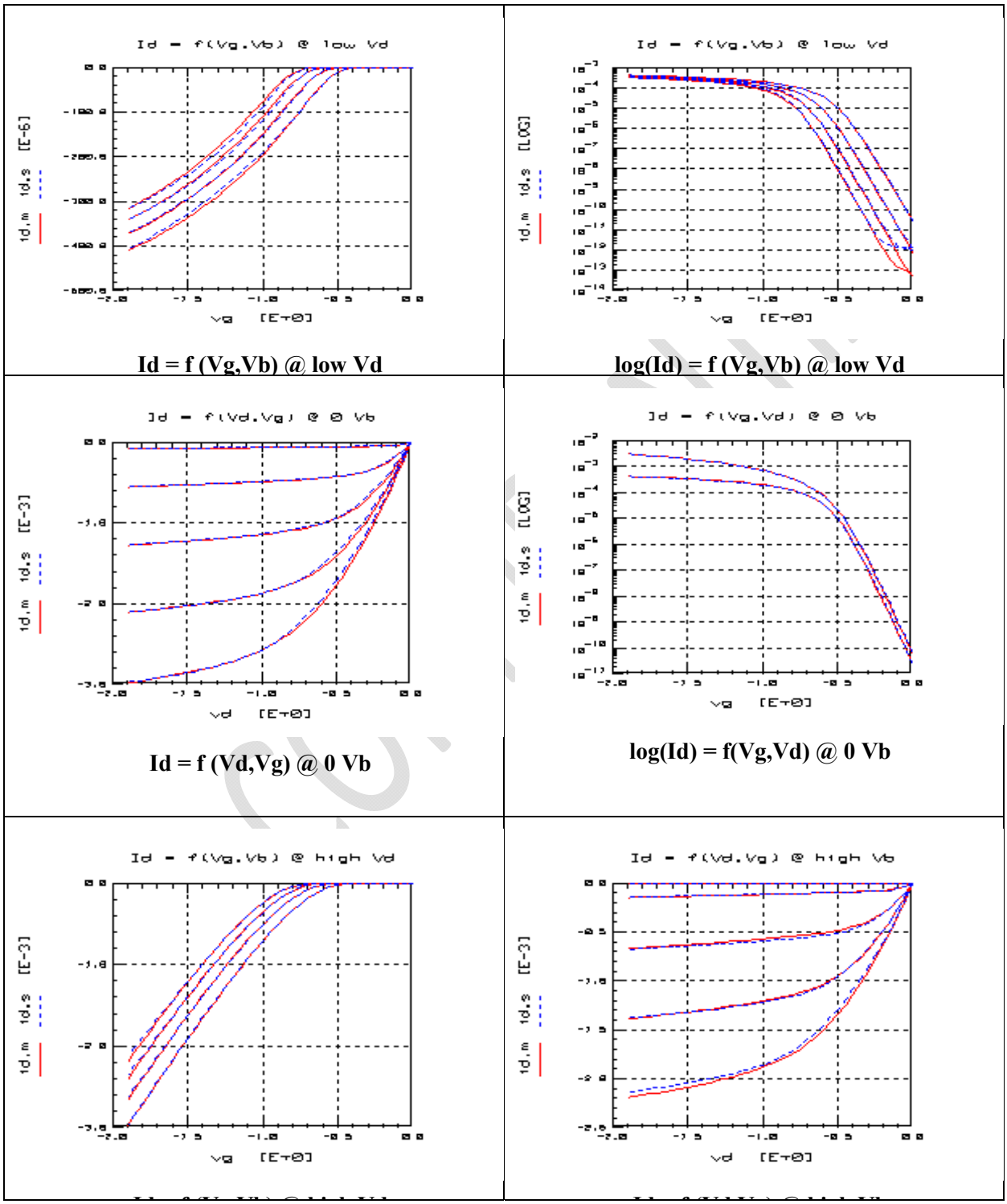
9.4.1.2 1.8V PMOS Standard Vt



Id = f (Vg,Vb) @ low Vd

log(Id) = f (Vg,Vb) @ low Vd

Id = f (Vd,Vg) @ 0 Vb

log(Id) = f(Vg,Vd) @ 0 Vb

Figure 9-5 1.8V PMOS STANDARD VT T=25 C W/L=10/0.18

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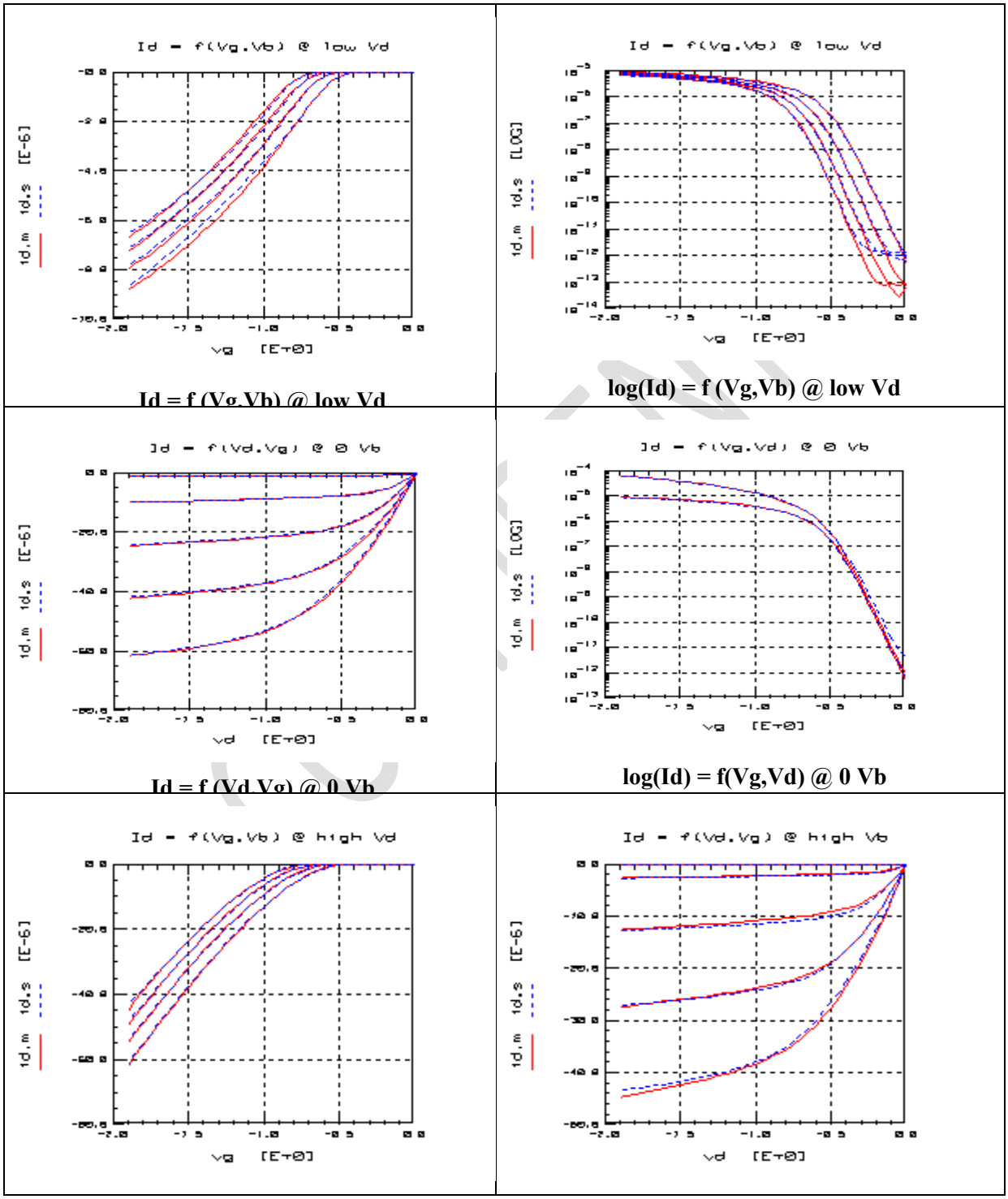

Figure 9-6 ***1.8V PMOS STANDARD VT T=25 C W/L=0.22/0.18***

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9.4.1.3 1.8V Low Leakage NMOS

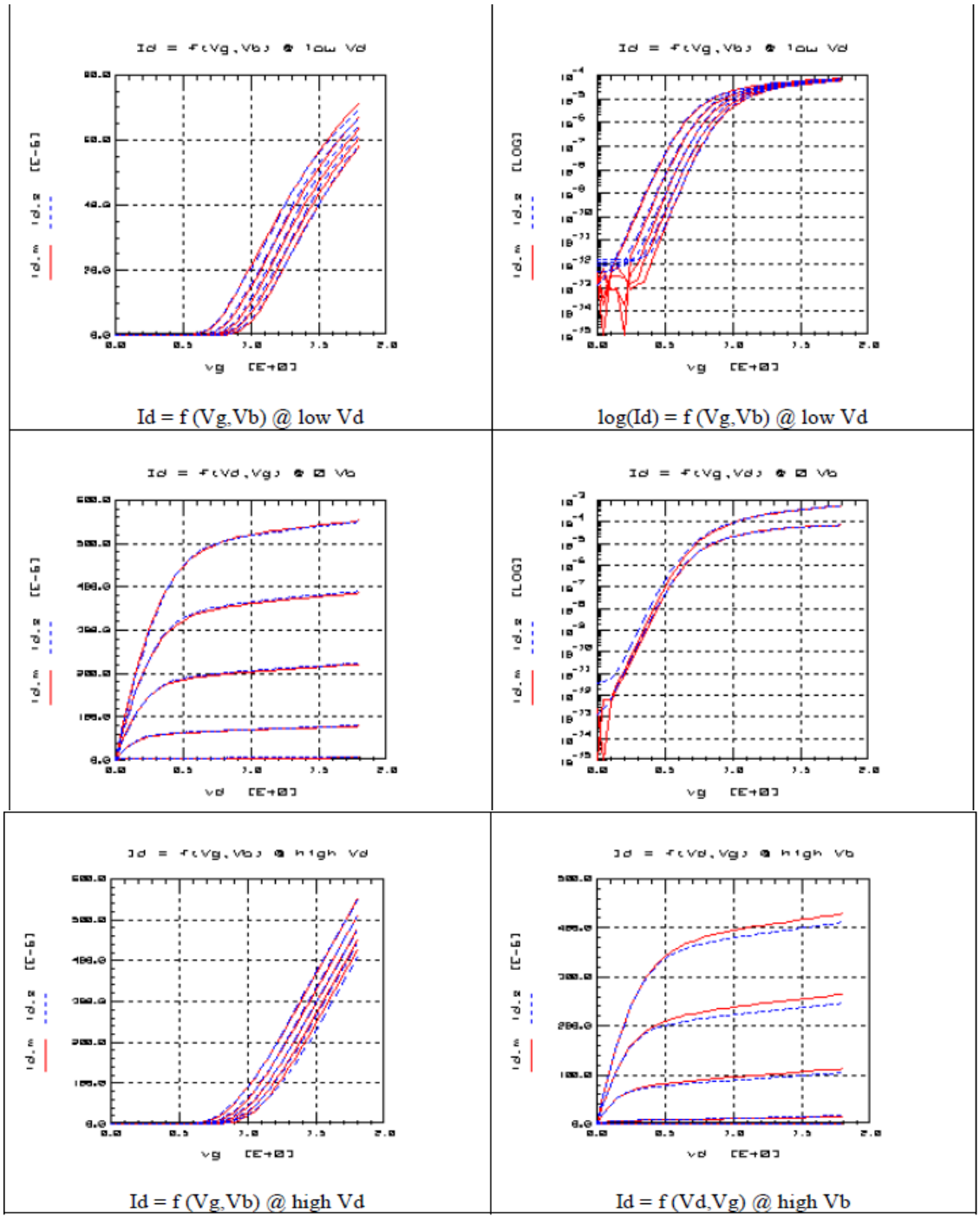

Figure 9-71.8V LOW LEAKAGE NMOS T=25 C W/L=10/0.18

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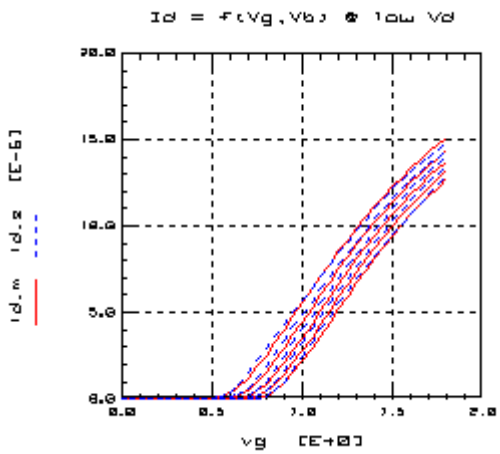
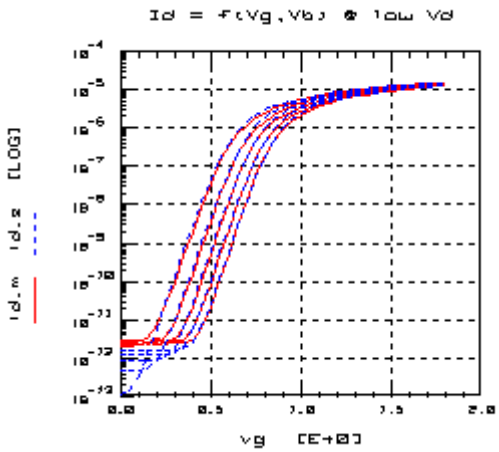
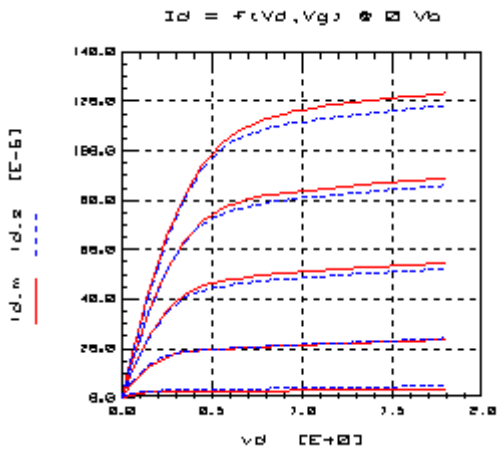


Id = f (Vg,Vb) @ low Vd



log(Id) = f (Vg,Vb) @ low Vd



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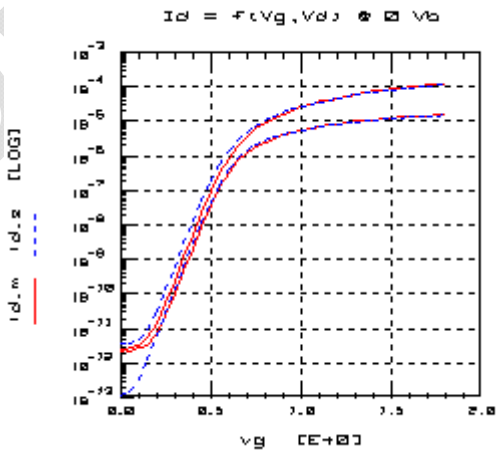

log(Id) = f(Vg,Vd) @ 0 Vb

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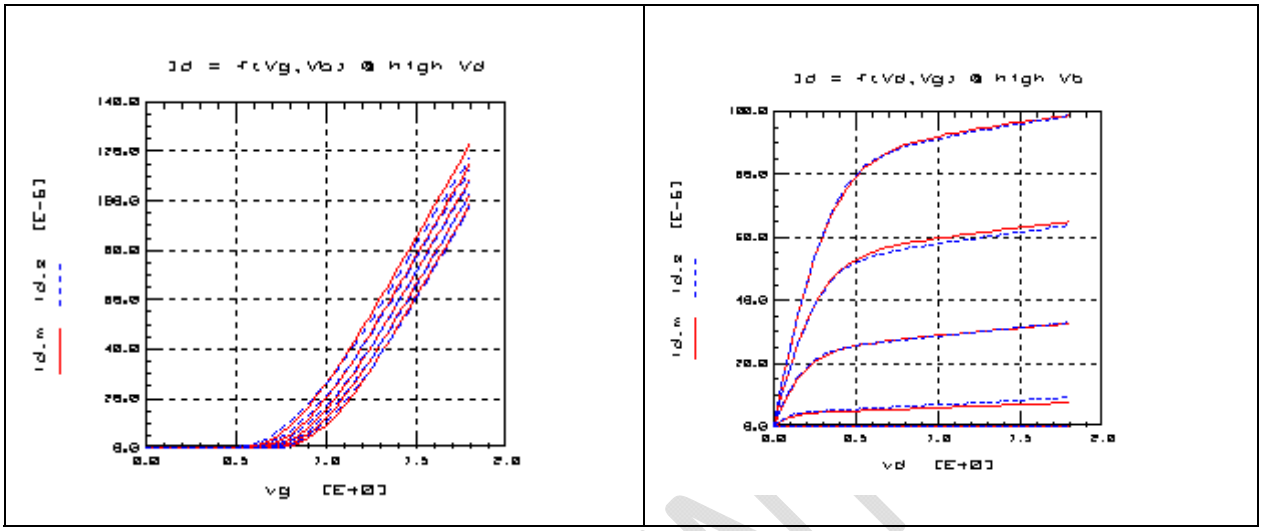

Figure 9-81.8V LOW LEAKAGE NMOS T=25 C W/L=0.22/0.18

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9.4.1.4 1.8V Low Leakage PMOS

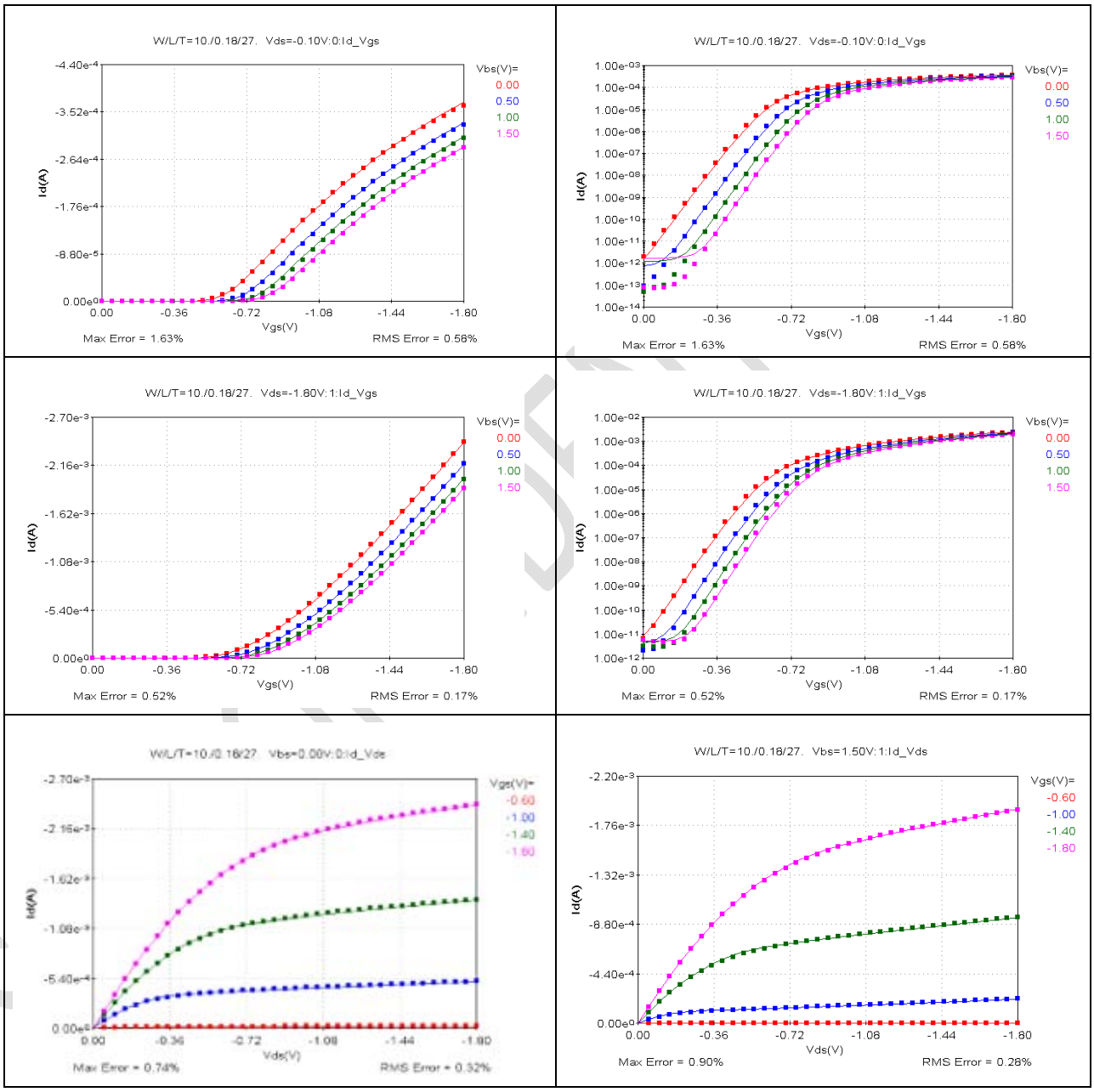

Figure 9-9 1.8V LOW LEAKAGE PMOS T=25 C W/L=10/0.18

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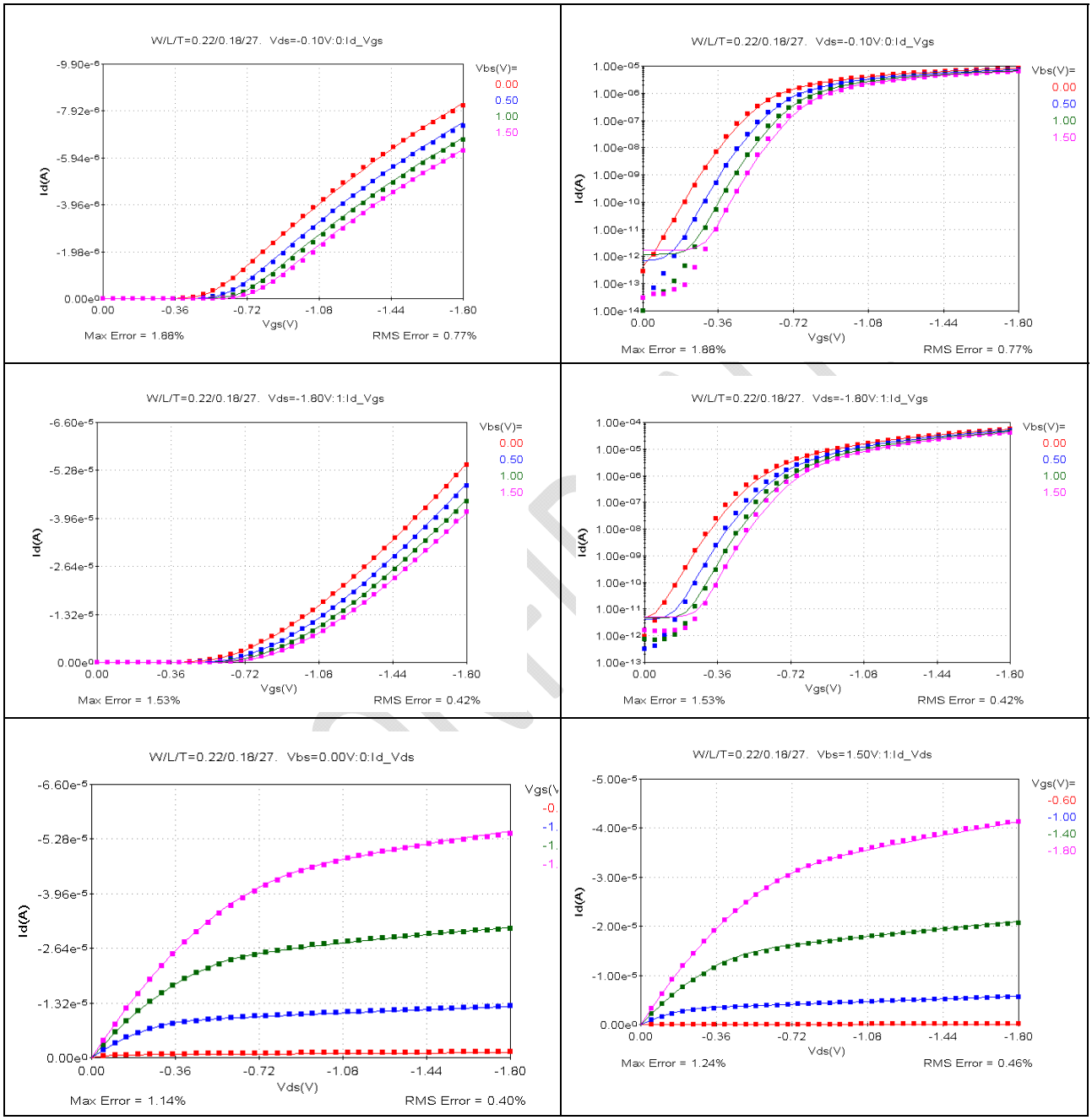

Figure 9-10 1.8V LOW LEAKAGE PMOS T=25 C W/L=0.22/0.18

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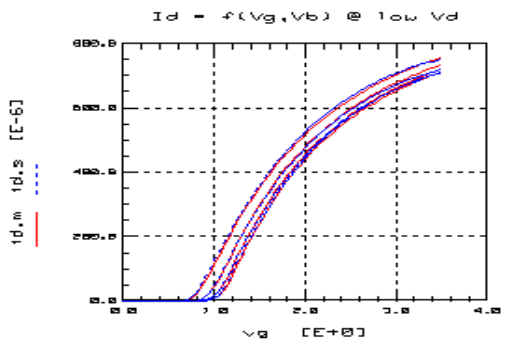
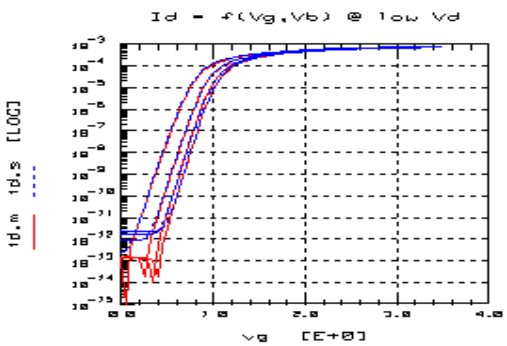
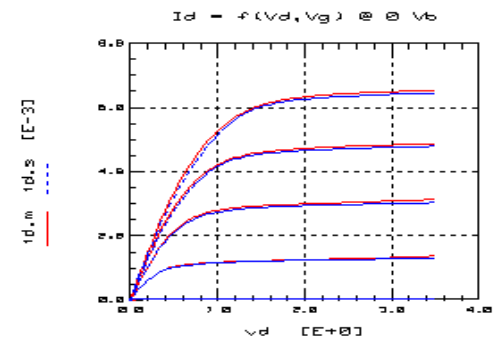
9.4.1.5 3.3V NMOS



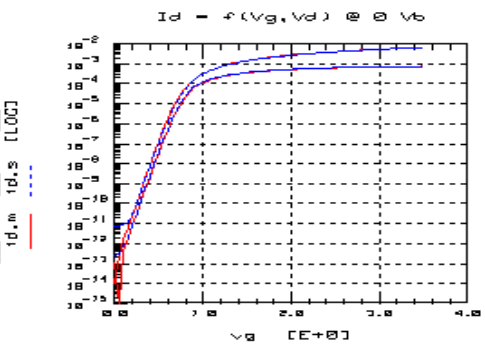
Id = f (Vg,Vb) @ low Vd



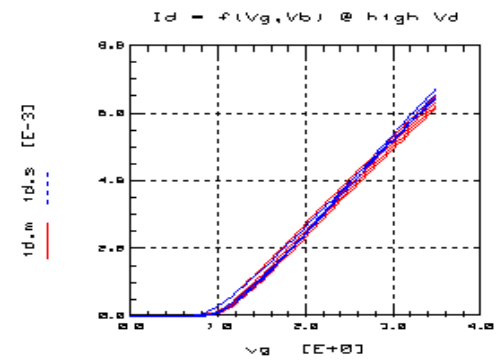
log(Id) = f (Vg,Vb) @ low Vd



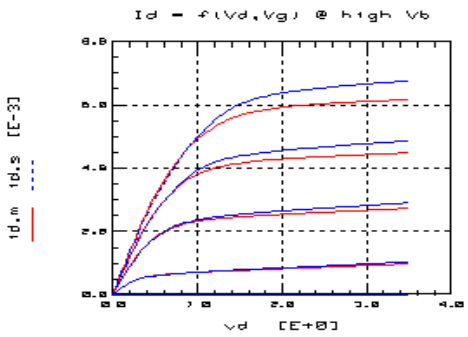
Id = f (Vd,Vg) @ 0 Vb



log(Id) = f(Vg,Vd) @ 0 Vb



Id = f (Vg,Vb) @ high Vd



Id = f (Vd,Vg) @ high Vb

Figure 9-11 3.3V NMOS T=25 C W/L=10.0/0.35

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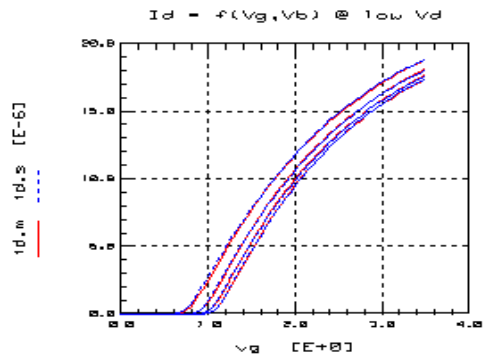

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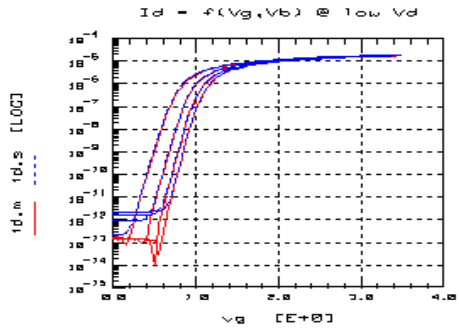

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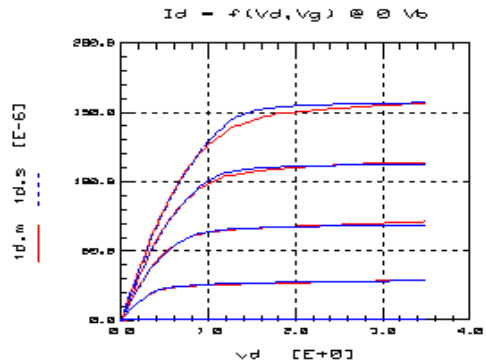
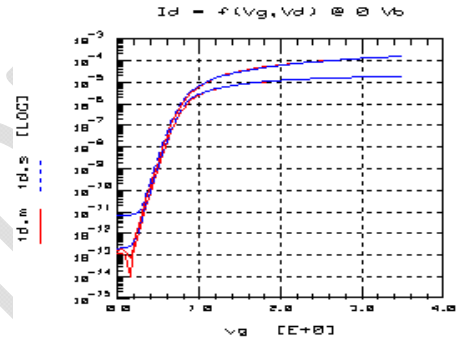
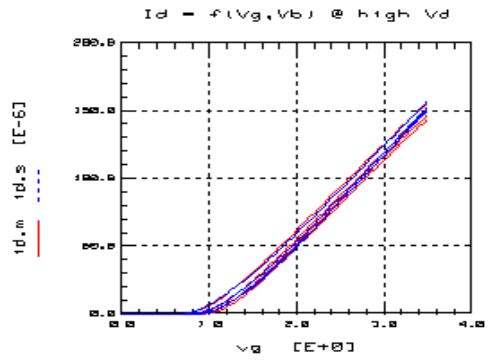


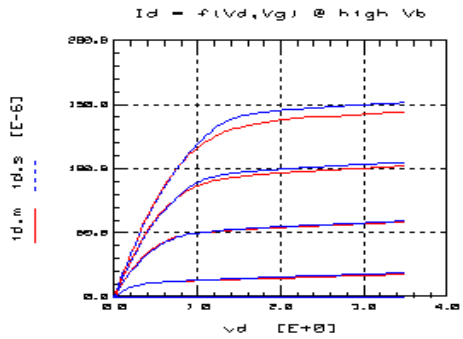
Id = f (Vd,Vg) @ 0 Vb



log(Id) = f(Vg,Vd) @ 0 Vb



Id = f (Vg,Vb) @ high Vd



Id = f (Vd,Vg) @ high Vb

Figure 9-12 3.3V NMOS T=25 C W/L=0.22/0.35

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9.4.1.6 3.3V PMOS

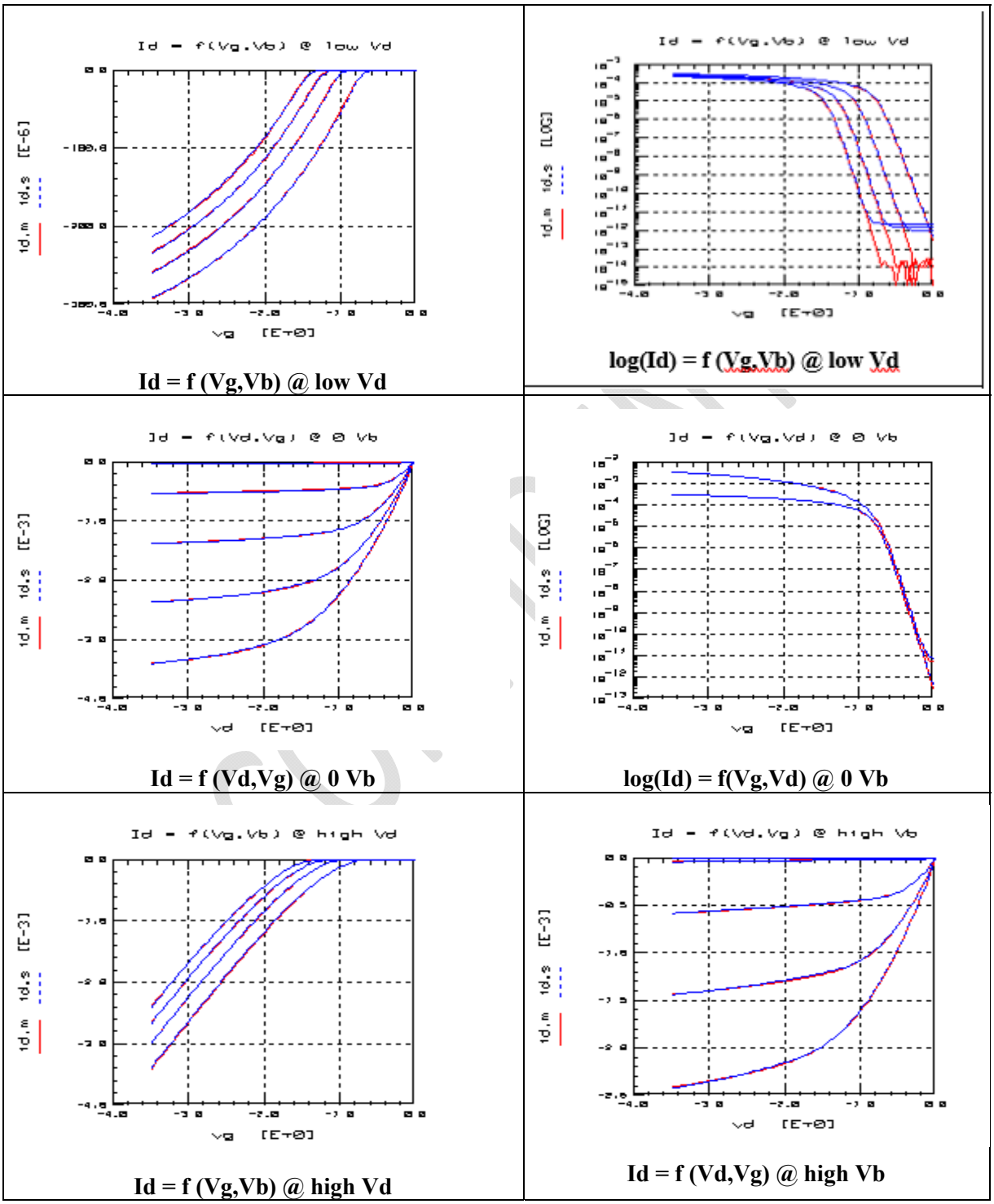

Figure 9-13 3.3V PMOS T=25 C W/L=10.0/0.3

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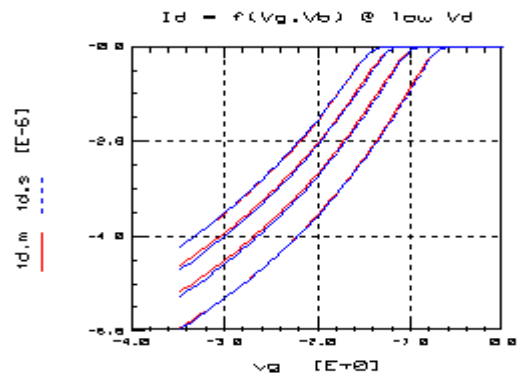
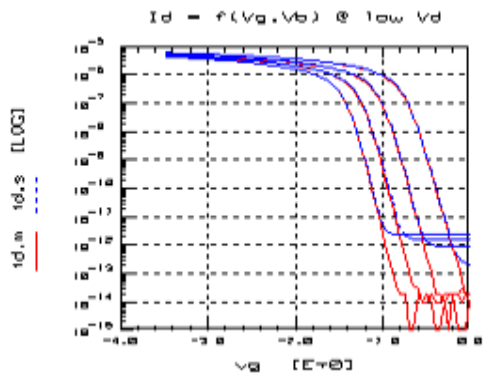
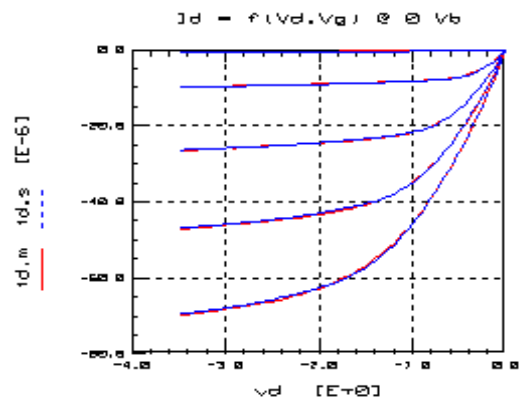


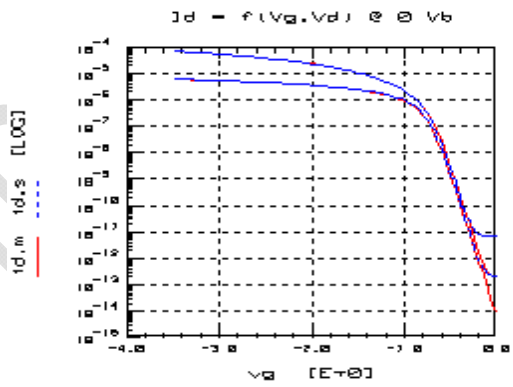
Id = f (Vg,Vb) @ low Vd



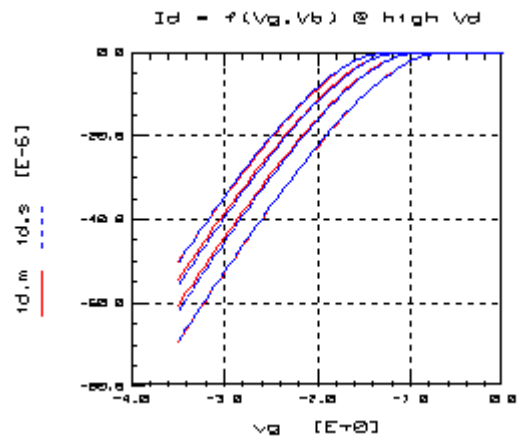
log(Id) = f (Vg,Vb) @ low Vd



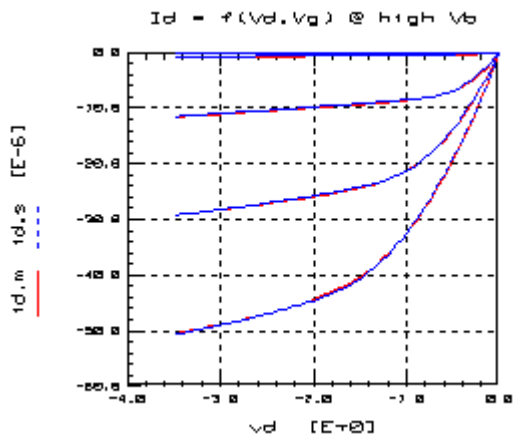
Id = f (Vd,Vg) @ 0 Vb



log(Id) = f(Vg,Vd) @ 0 Vb



Id = f (Vg,Vb) @ high Vd



Id = f (Vd,Vg) @ high Vb

Figure 9-14 3.3V PMOS T=25 C W/L=0.22/0.3

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9.4.1.7 5V NMOS – PSP

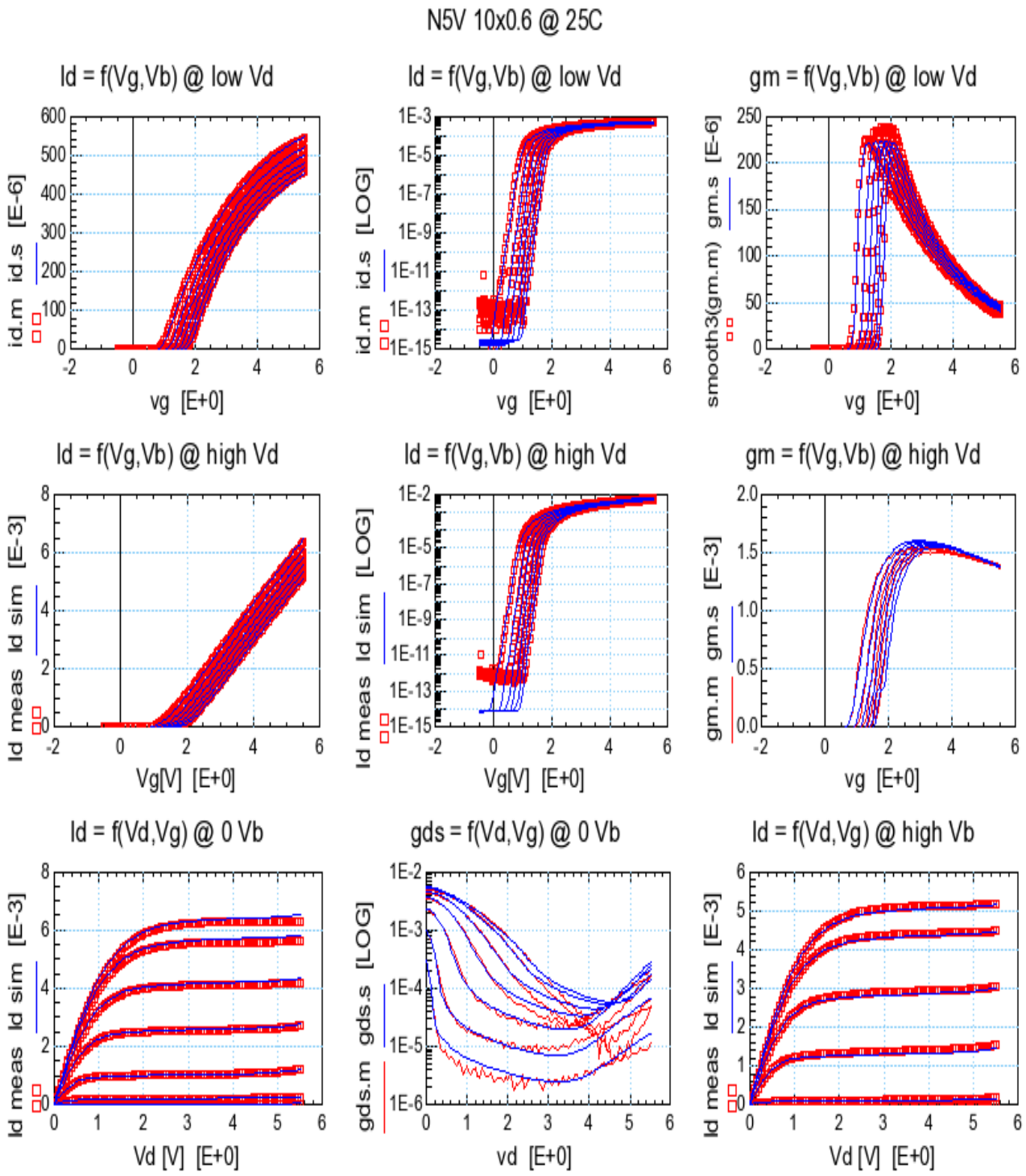

Figure 9-15 : ***5V PSP NMOS T=25 C W/L=10/0.6***

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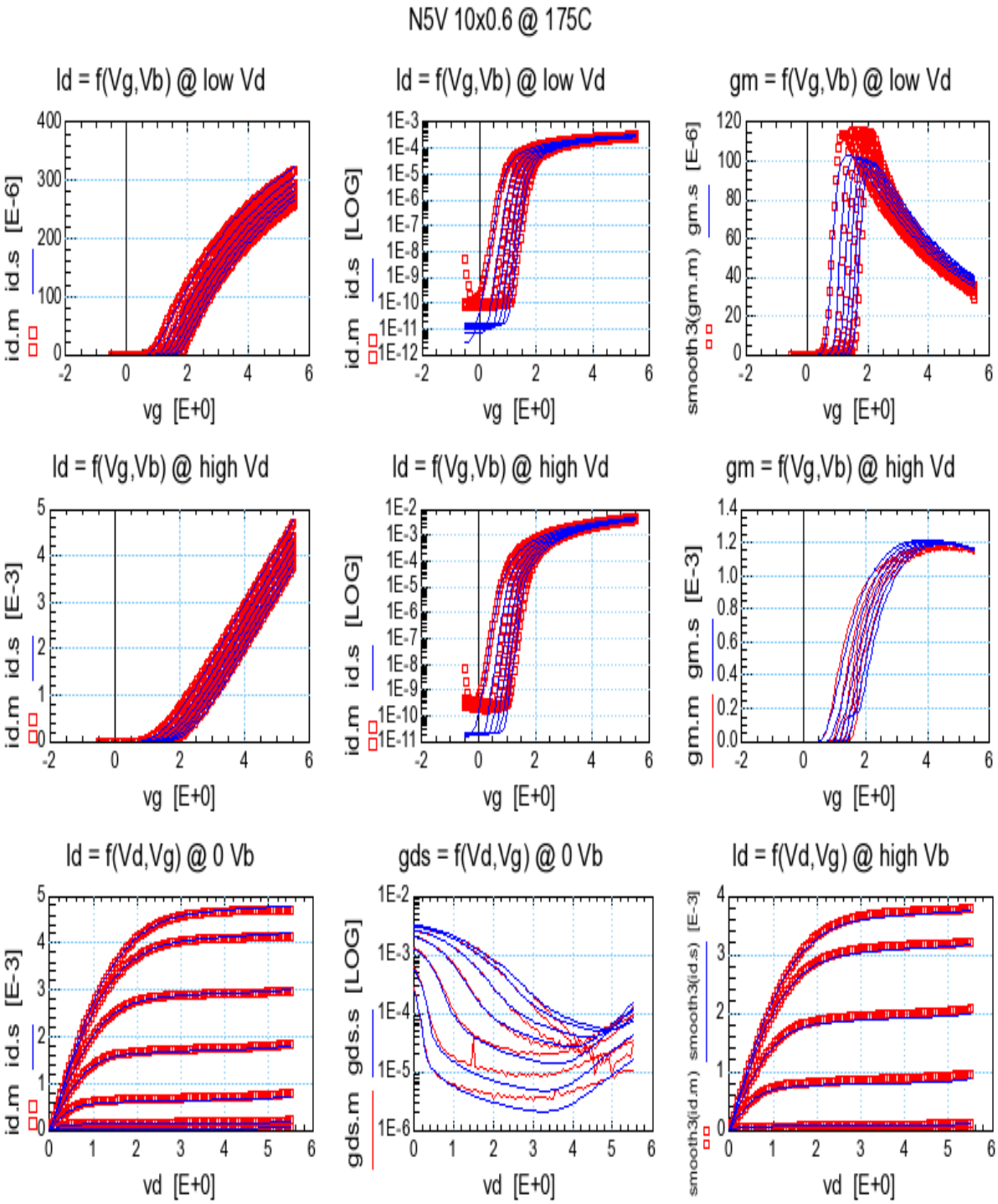

Figure 9-16 5V PSP NMOS T=175 C W/L=10/0.6

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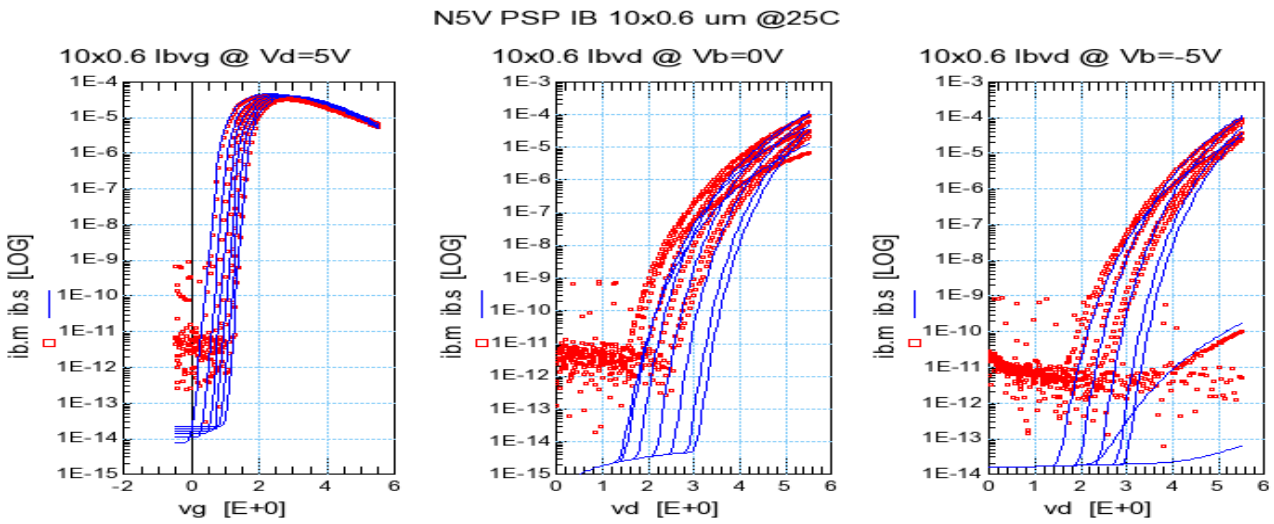

Figure 9-18 5V PSP NMOS SUBSTRATE CURRENT T=25 C W/L=10/0.6

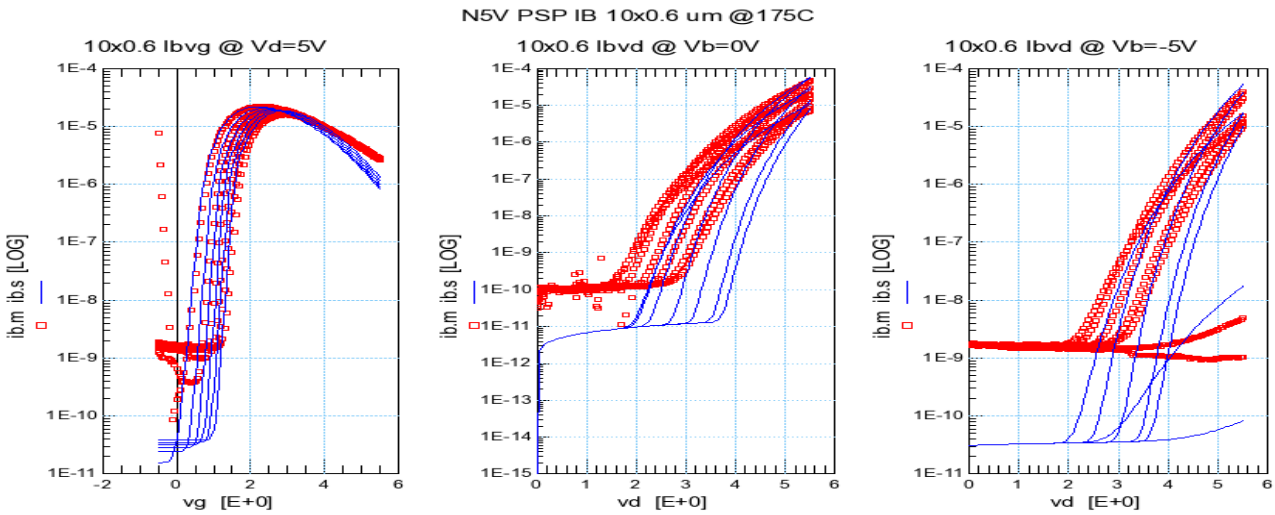

Figure 9-17 5V PSP NMOS SUBSTRATE CURRENT T=175 C W/L=10/0.6

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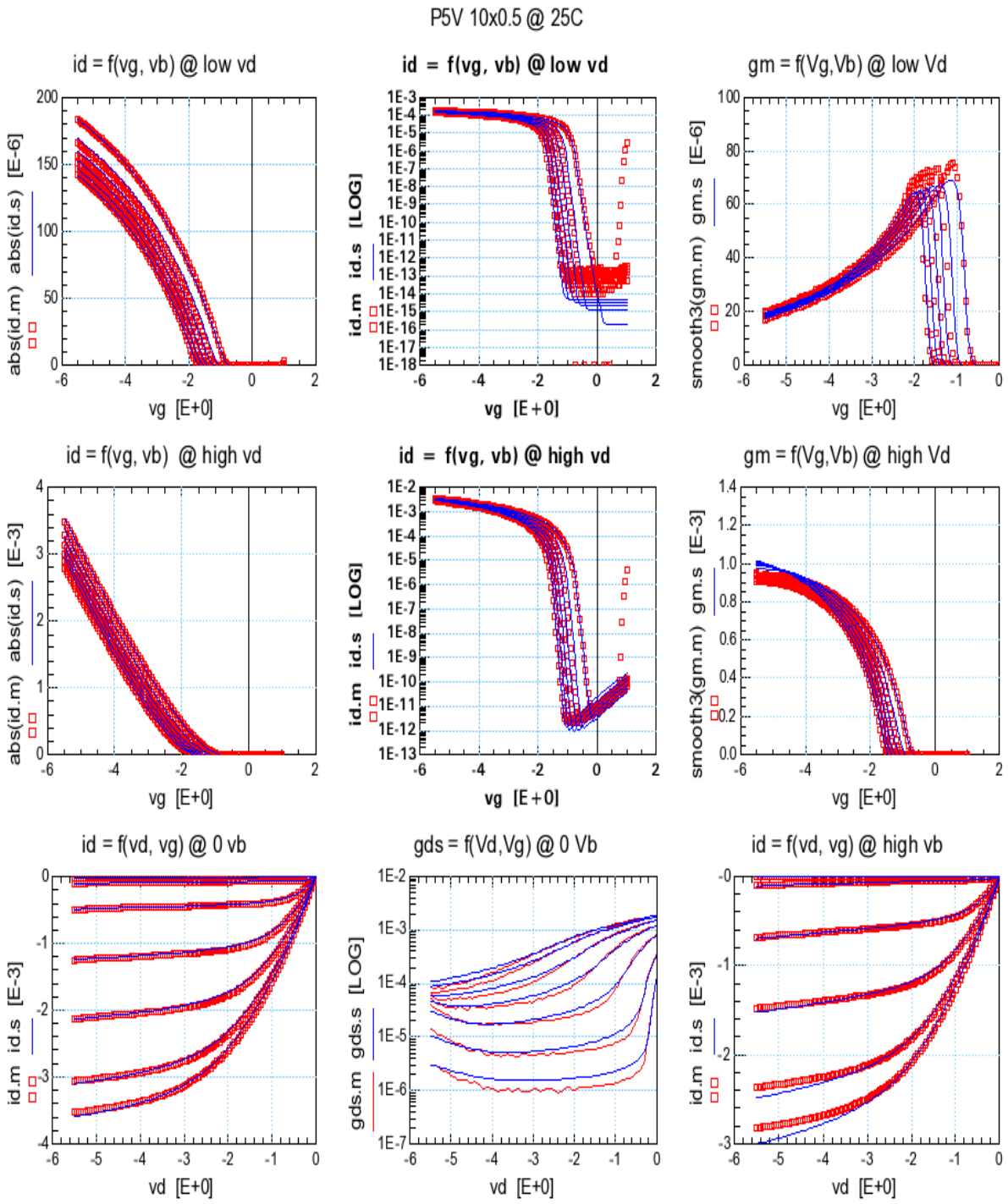

Figure 9-19 5V PSP PMOS T=25 C W/L=10/0.5

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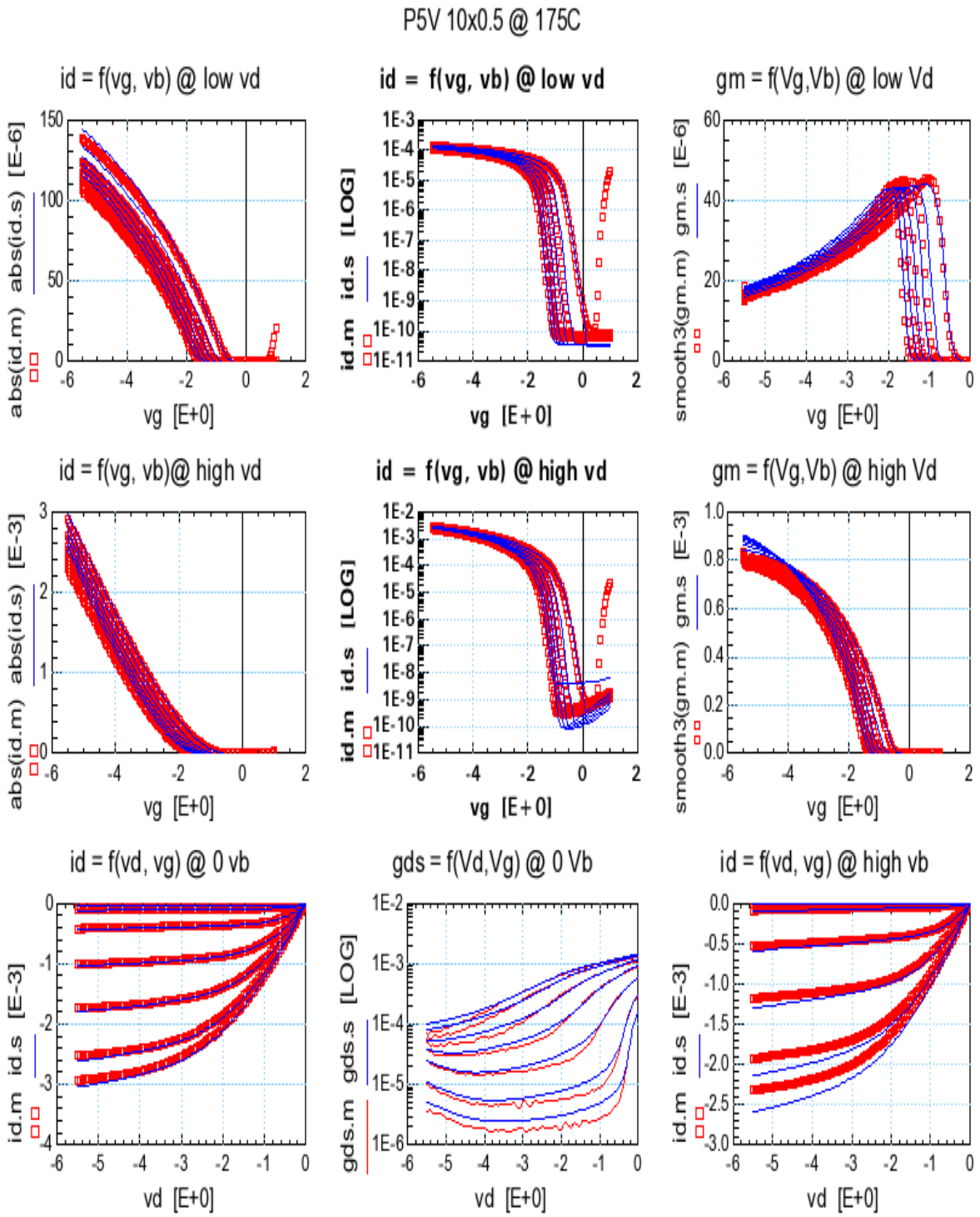

Figure 9-20 5V PSP PMOS T=175 C W/L=10/0.5

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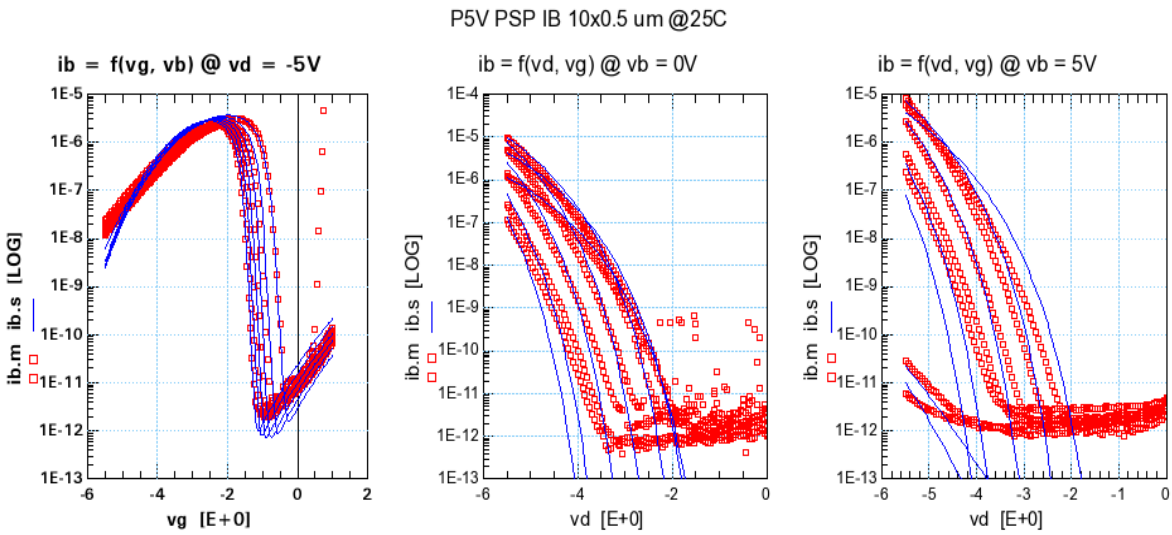

Figure 9-22 5V PSP PMOS SUBSTRATE CURRENT T=25 C W/L=10/0.5

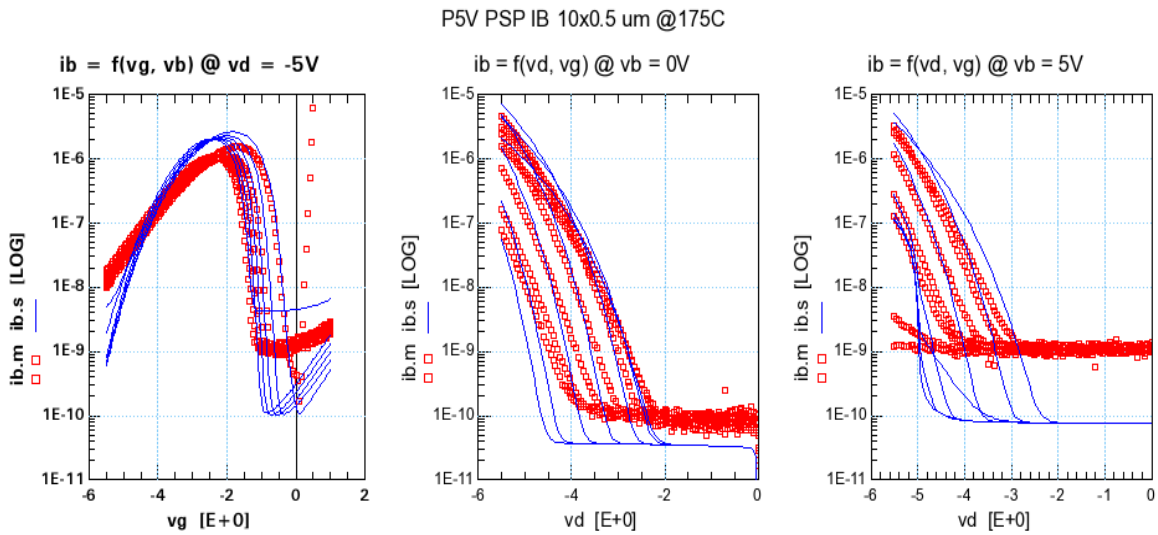

Figure 9-21 5V PSP PMOS SUBSTRATE CURRENT T=175 C W/L=10/0.5

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9.4.1.9 3.3V Native NMOS

Table 9-10 3.3V NATIVE NMOS

Wdrawn (um)	Ldrawn (um)	Idsat Meas. [uA/um]	Idsat Sim. [uA/um]
100	10	97	97
100	1.2	476	476
10	10	100	98
10	1.2	563	563
1.2	5	186.7	186.7
1.2	1.2	572	566
0.5	10	98	98
0.5	1.2	590	590
0.22	10	100	100
0.22	1.2	622	627

9.4.1.10 AC Analysis

The parasitic capacitor values of the transistors included in model were extracted and verified by the Inverter Ring Oscillators (RO's) measurements. The comparison between simulated and measured results of RO's for different temperatures is presented in tables below.

The simulation was performed with default hdif=0.15e-6. No source/drain junction dimensions were specified in the netlist

9.4.1.11 1.8V MOSFET's - T=25 C

Table 9-11

INV (Fanout=1)	Simulated with hdif (ps)	Simulated with S/D dimensions as measured (ps)	Measured (ps)
NMOS W/L=5/0.18 PMOS W/L=10/0.18	24.232	28	27.329
NMOS W/L=5/0.27 PMOS W/L=10/0.27	48.766	54.729	52.287

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9.4.1.12 1.8V MOSFET's - T=125 C

Table 9-12

INV (Fanout=1)	Simulated with hdif (ps)	Simulated with S/D dimensions as measured (ps)	Measured (ps)
NMOS W/L=5/0.18 PMOS W/L=10/0.18	28.088	32.7	32.059
NMOS W/L=5/0.27 PMOS W/L=10/0.27	57.256	64.1	62.744

9.4.1.13 1.8V Low Leakage MOSFET's - T=25 C

Table 9-13 1.8V LOW LEAKAGE MOSFET'S - T=25 C

INV (Fanout=1)	Simulated with hdif (ps)	Simulated with S/D dimensions as measured (ps)	Measured (ps)
NMOS W/L=5/0.18 PMOS W/L=10/0.18	24.8	28.8	29.4
NMOS W/L=5/0.27 PMOS W/L=10/0.27	54	60	63.2

9.4.1.14 1.8V Low Leakage MOSFET's - T=125 C

Table 9-14 1.8V LOW LEAKAGE MOSFET'S - T=125 C

INV (Fanout=1)	Simulated with hdif (ps)	Simulated with S/D dimensions as measured (ps)	Measured (ps)
NMOS W/L=5/0.18 PMOS W/L=10/0.18	28.2	32	33.7
NMOS W/L=5/0.27 PMOS W/L=10/0.27	62	69	74.4

9.4.1.15 1.8V Low Leakage MOSFET's - T=-40 C

Table 9-15 1.8V LOW LEAKAGE MOSFET'S - T=-40 C

INV (Fanout=1)	Simulated with hdif (ps)	Simulated with S/D dimensions as measured (ps)	Measured (ps)
NMOS W/L=5/0.18 PMOS W/L=10/0.18	23	26	26.2
NMOS W/L=5/0.27 PMOS W/L=10/0.27	47.5	52.7	54.3

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9.4.1.16 3.3V MOSFET's - T=25 C

Table 9-16 3.3V MOSFET'S - T=25 C

INV (Fanout=1)	Simulated with hdif (ps)	Simulated with S/D dimensions as measured (ps)	Measured (ps)
NMOS W/L=5/0.35 PMOS W/L=10/0.3	43.1	44.77	43.395
NMOS W/L=5/0.53 PMOS W/L=10/0.45	80.2	82.6	83.177

9.4.1.17 3.3V MOSFET's - T=125 C

Table 9-17 3.3V MOSFET'S - T=125 C

INV (Fanout=1)	Simulated with hdif (ps)	Simulated with S/D dimensions as measured	Measured (ps)
NMOS W/L=5/0.35 PMOS W/L=10/0.3	50	52	50.90
NMOS W/L=5/0.53 PMOS W/L=10/0.45	97.15	100	102

9.4.1.18 5V PSP MOSFET's T=25 C

Table 9-18 5V PSP MOSFET'S - T=25 C

VDD	INV (Fanout=1)	NOM (ps)	SLOW (ps)	FAST (ps)
VDD=5	NMOS W/L=5/0.6 PMOS W/L=10/0.5	62.59	77.43	53.83

9.4.1.19 5V PSP MOSFET's T=175 C

Table 9-19 5V PSP MOSFET'S - T=150 C

VDD	INV (Fanout=1)	NOM (ps)	SLOW (ps)	FAST (ps)
VDD=5	NMOS W/L=5/0.6 PMOS W/L=10/0.5	82.50	100.58	75.31

9.4.1.20 5V PSP MOSFET's T=-40 C

Table 9-20 5V PSP MOSFET'S - T=-40 C

VDD	INV (Fanout=1)	NOM (ps)	SLOW (ps)	FAST (ps)
VDD=5	NMOS W/L=5/0.6 PMOS W/L=10/0.5	56.37	73.86	45.97

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9.5 Comparison Between DR20018SL_SCL Main ET Parameters and Simulated Results

Table 9-21 1.8V(STD VTH), TYPICAL SIZE: L=0.18UM, W=10UM, MOS, THIN GATE OXIDE=40A

Parameter	Device Size	Unit	Conditions/Criteria	Typical	Low Limit	High Limit
VthN(S)	W/L=10/0.18	V	Constant current method: Vth=Vg @Id=0.1[uA]*(W/L)	0.42	0.32	0.52
VthP(S)	W/L=10/0.18	V	Constant current method: Vth=Vg @Id=0.1[uA]*(W/L)	-0.5	-0.4	-0.6
IdsatN	W/L=10/0.18	uA/um	Vds=1.8V, Vgs=1.8V	600	510	690
IdsatP	W/L=10/0.18	uA/um	Vds=1.8V, Vgs=1.8V	-260	-220	-300

Table 9-22 1.8V(LOW LEAKAGE), TYPICAL SIZE: L=0.18UM, W=10UM, MOS, THIN GATE OXIDE=40A

Parameter	Device Size	Unit	Conditions/Criteria	Typical	Low Limit	High Limit
VthN(S)	W/L=10/0.18	V	Constant current method: Vth=Vg @Id=0.1[uA]*(W/L)	0.54	0.47	0.61
VthP(S)	W/L=10/0.18	V	Constant current method: Vth=Vg @Id=0.1[uA]*(W/L)	-0.56	-0.43	-0.69
IdsatN	W/L=10/0.18	uA/um	Vds=1.8V, Vgs=1.8V	520	430	610
IdsatP	W/L=10/0.18	uA/um	Vds=1.8V, Vgs=1.8V	225	180	270

Table 9-23 3.3V(STD VTH), TYPICAL SIZE: L=0.35UM, NMOS; L=0.3UM, PMOS, W=10UM, MOSFET, THIN GATE OXIDE=70A

Parameter	Device Size	Unit	Conditions/Criteria	Typical	Low Limit	High Limit
VthN(S)	W/L=10/0.35	V	Linear Vth: Gm max extrapolation, -Vds/2 (Vds=0.1V)	0.72	0.62	0.82
VthP(S)	W/L=10/0.3	V	Linear Vth: Gm max extrapolation, -Vds/2 (Vds=0.1V)	-0.68	-0.58	-0.78
IdsatN	W/L=10/0.35	uA/um	Vds=3.3V, Vgs=3.3V	600	510	690
IdsatP	W/L=10/0.30	uA/um	Vds=3.3V, Vgs=3.3V	-300	-255	-345

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Table 9-24 1.8V NATIVE, TYPICAL SIZE: L=0.5UM, NMOS; W=10UM, MOSFET, THIN GATE OXIDE=40A

Parameter	Device Size	Unit	Conditions/Criteria	Typical	Low Limit	High Limit
VthN(S)	W/L=10/0.5	V	Linear Vth: Gm max extrapolation, -Vds/2 (Vds=0.1V)	-0.05	-0.30	0.10
IdsatN	W/L=10/0.5	uA/um	Vds=1.8V, Vgs=1.8V	620	530	710

3.3V NATIVE, TYPICAL SIZE: L=1.2UM, NMOS; W=10UM, MOSFET, THIN GATE OXIDE=70A

Parameter	Device Size	Unit	Conditions/Criteria	Typical	Low Limit	High Limit
VthN(S)	W/L=10/1.2	V	Linear Vth: Gm max extrapolation, -Vds/2 (Vds=0.1V)	-0.15	-0.25	-0.05
IdsatN	W/L=10/1.2	uA/um	Vds=3.3V, Vgs=3.3V	565	480	650

Table 9-25 5V (STD VTH) PSP, TYPICAL SIZE: L=0.6UM; W=10UM, NMOS, L=0.5UM, W=10UM, PMOS, GATE OXIDE=115A

Parameter	Device Size	Unit	Conditions/Criteria	Typical	Low Limit	High Limit
VthN(S)	W/L=10/0.6	V	Linear Vth: Gm max extrapolation, -Vds/2 (Vds=0.1V)	0.85	0.75	0.95
VthP(S)	W/L=10/0.5	V	Linear Vth: Gm max extrapolation, -Vds/2 (Vds=0.1V)	-0.79	-0.69	-0.89
IdsatN	W/L=10/0.6	uA/um	Vds=3.3V, Vgs=3.3V	565	465	665
IdsatP	W/L=10/0.5	uA/um	Vds=3.3V, Vgs=3.3V	-298	-255	-341

Table 9-26 5V NATIVE, TYPICAL SIZE: L=1.5UM; W=10UM, NMOS, GATE OXIDE=115A

Parameter	Device Size	Unit	Conditions/Criteria	Typical	Low Limit	High Limit
VthN(S)	W/L=10/1.5	V	Linear Vth: Gm max extrapolation, -Vds/2 (Vds=0.1V)	-0.09	-0.3	0.1
IdsatN	W/L=10/1.5	uA/um	Vds=3.3V, Vgs=3.3V	520	360	680

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9.6 1/F Noise Analysis

The 1/f Noise data was collected for 24 devices (5 nmos 1.8V, 5 pmos 1.8V, 3 nmos 3.3V, 3 pmos 3.3V, 4 nmos 5V, 4 pmos 5V) for room temperature at a frequency range of 1Hz to 100kHz.

The floor noise of the measurement system is presented in the graph below. It can be seen that system floor noise is between 1E-24 to 1E-25 A2/Hz.

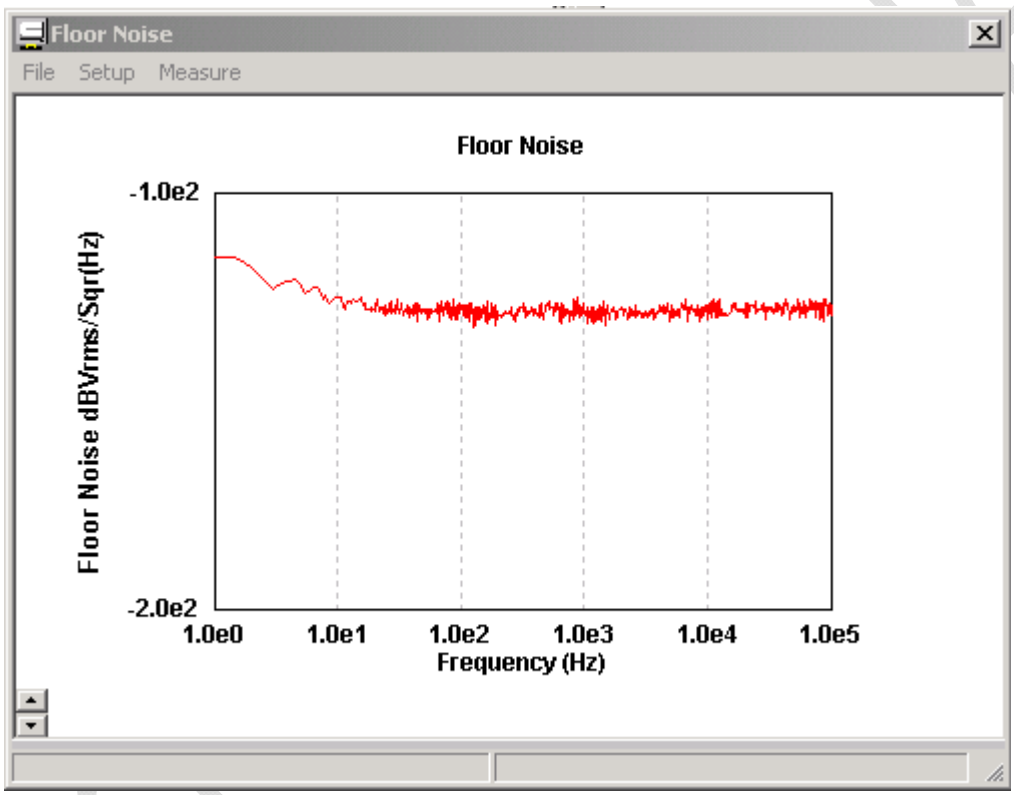

Figure 9-23 FLOOR NOISE

The 1/f noise parameters where extracted for noimod=1 and noimod=2 cases. The model default is noimod=2, since it provides the most precision. In the case there is interest to simulate using noimod=1 one has to uncomment relevant parameters in the model card and comment out those parameters related to noimod=2.

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9.6.1 1.8V MOST (Std Vth)

Table 9-27 1.8V MOST LIST OF DUT'S WAS MEASURED

Index	W/L	Device Type
1	10/1	NMOS
2	10/0.18	NMOS
3	1/10	NMOS
4	0.22/10	NMOS
5	0.4/0.4	NMOS
6	10/1	PMOS
7	10/0.18	PMOS
8	1/10	PMOS
9	0.22/10	PMOS
10	0.4/0.4	PMOS

Table 9-28 1.8V MOST (STD VTH) BIASES APPLIED DURING MEASUREMENTS

Index	Vgs (V)	Vds (V)
1	0.8	0.1
2	1.2	0.1
3	1.8	0.1
4	0.8	1
5	1.2	1
6	1.8	1
7	0.8	1.8
8	1.2	1.8
9	1.8	1.8

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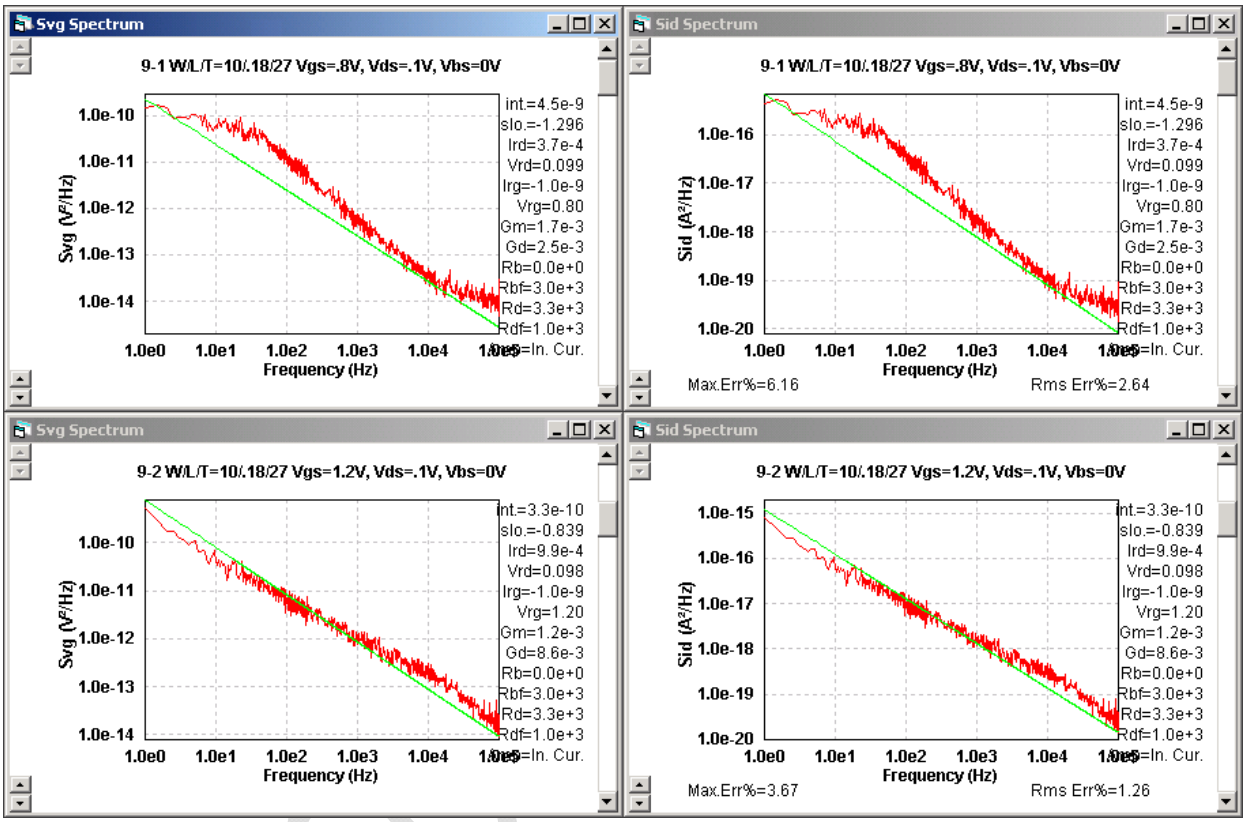

Figure 9-24 INDEX #2 W/L = 10/0.18 NMOS (STD VTH),- BIAS #1 AND #2

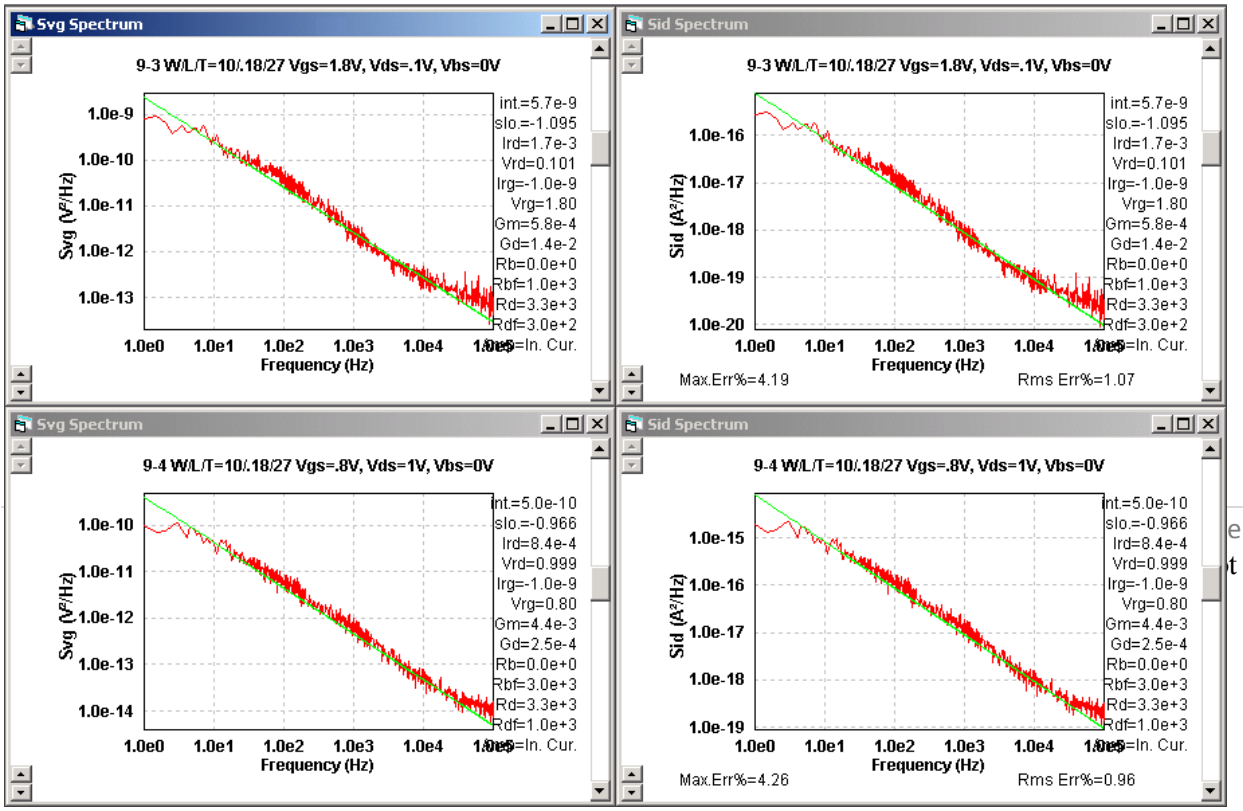

Figure 9-25INDEX #2 W/L = 10/0.18 NMOS (STD VTH)- BIAS #3 AND #4

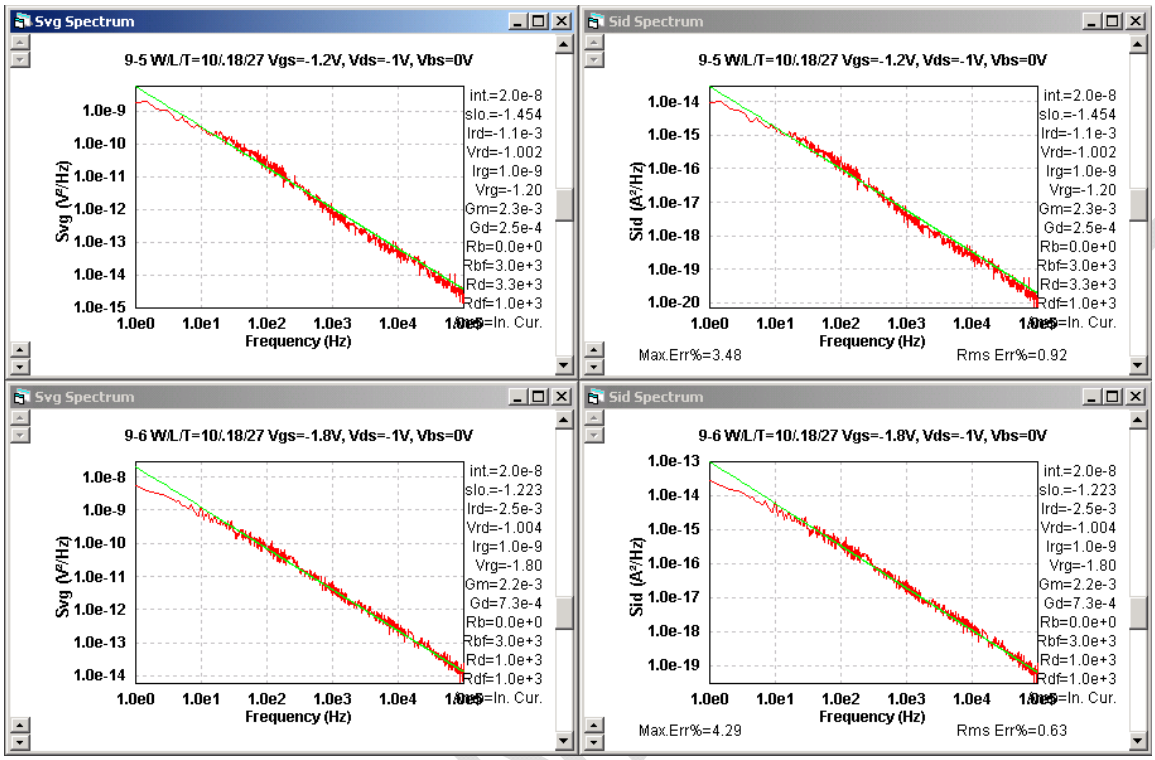

Figure 9-26 INDEX #7 W/L = 10/0.18 PMOS (STD VTH) - BIAS #5 AND #6

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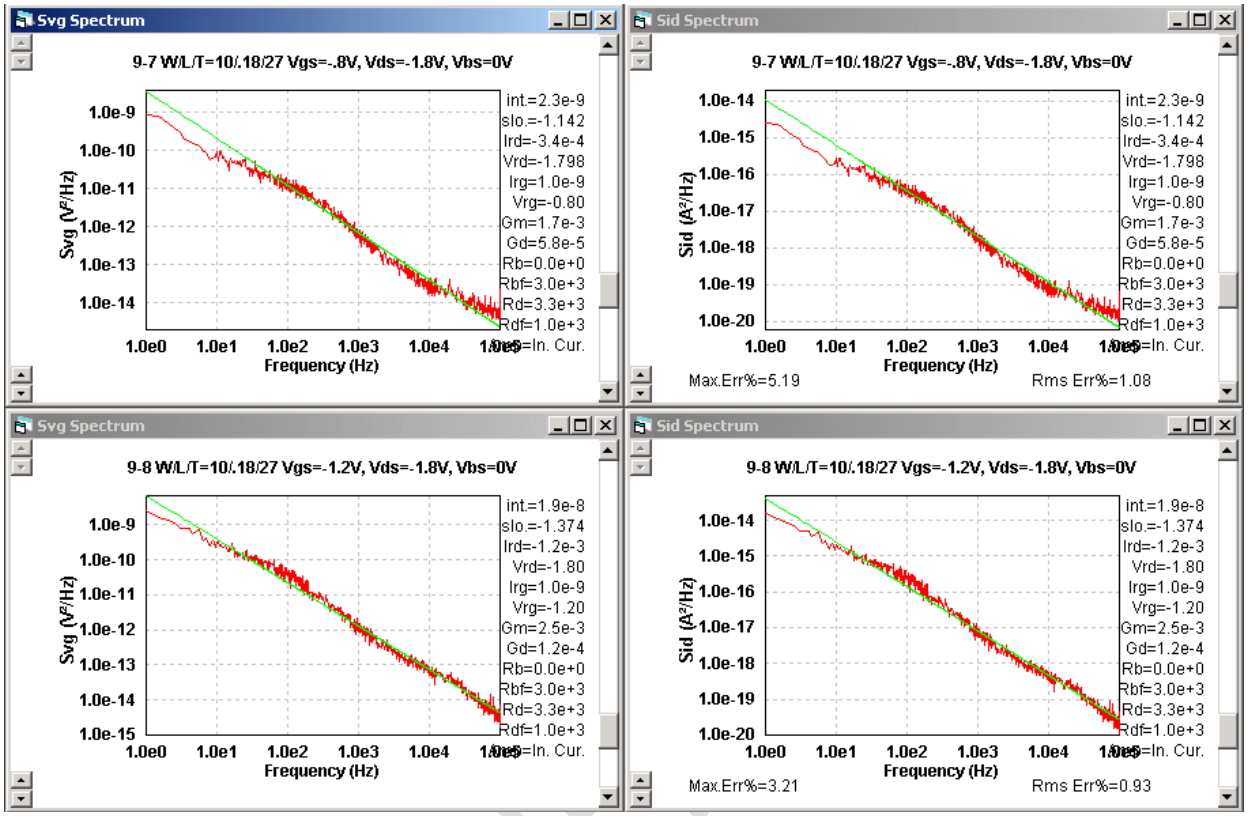

Figure 9-27 INDEX #7 W/L = 10/0.18 PMOS (STD VTH) - BIAS #7 AND #8

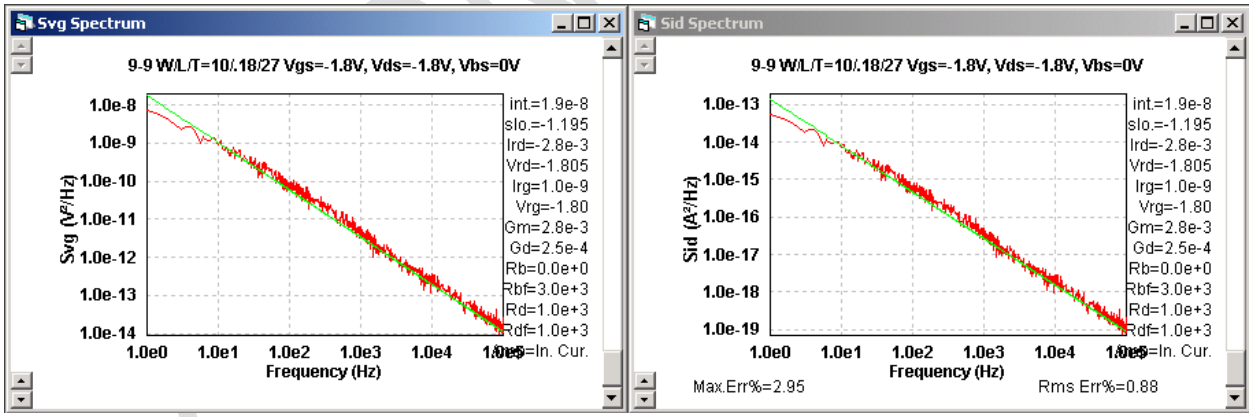

Figure 9-28 INDEX #7 W/L = 10/0.18 PMOS (STD VTH) - BIAS #9

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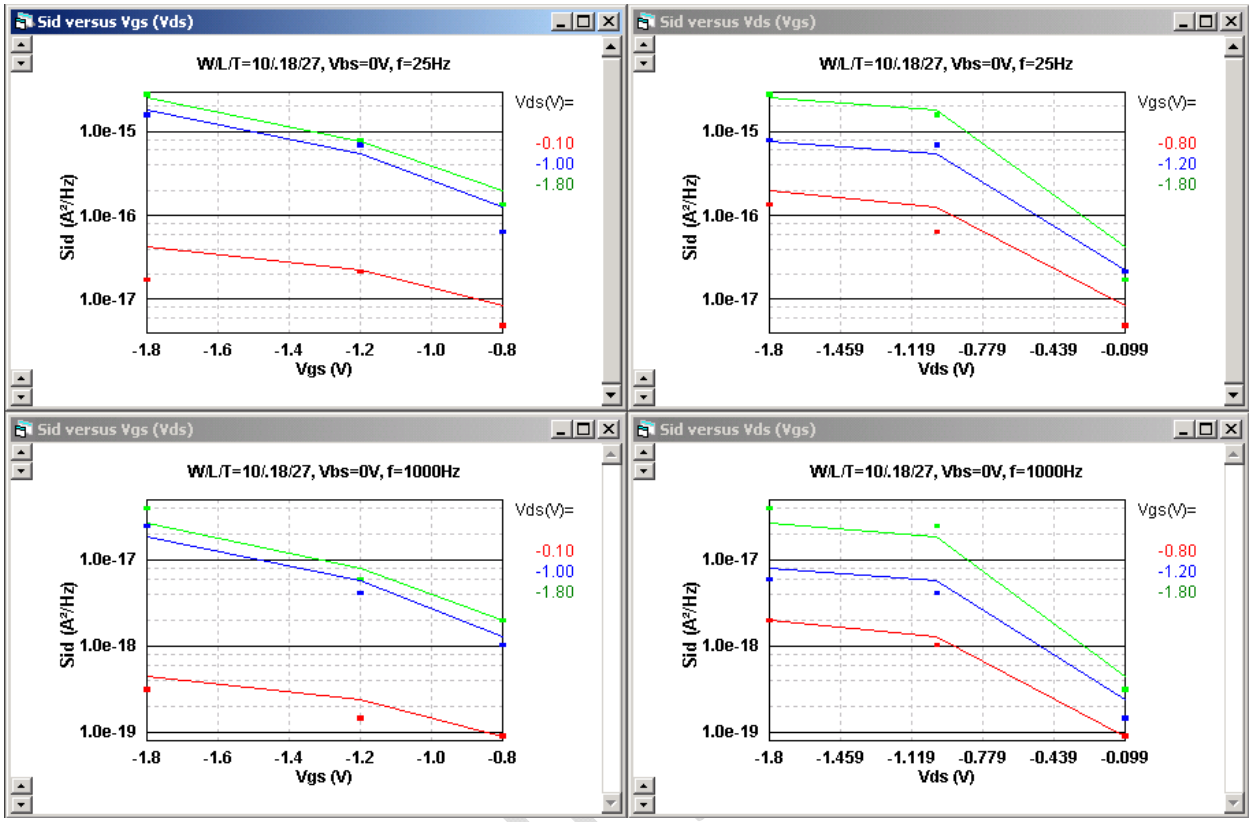

Figure 9-29 SID AT SAMPLING FREQUENCIES OF 25HZ AND 1000HZ

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9.6.2 3.3V MOST

Table 9-29 3.3V MOST LIST OF DUT'S WAS MEASURED

Index	W/L	Device Type
1	10/0.5	NMOS
2	0.42/10	NMOS
3	0.4/0.4	NMOS
4	10/0.5	PMOS
5	0.42/10	PMOS
6	0.4/0.4	PMOS

Table 9-30 3.3V BIASES APPLIED DURING MEASUREMENTS

Index	Vgs (V)	Vds (V)
1	0.8	0.1
2	2.5	0.1
3	3.3	0.1
4	0.8	1.5
5	2.5	1.5
6	3.3	1.5
7	0.8	3.3
8	2.5	3.3
9	3.3	3.3

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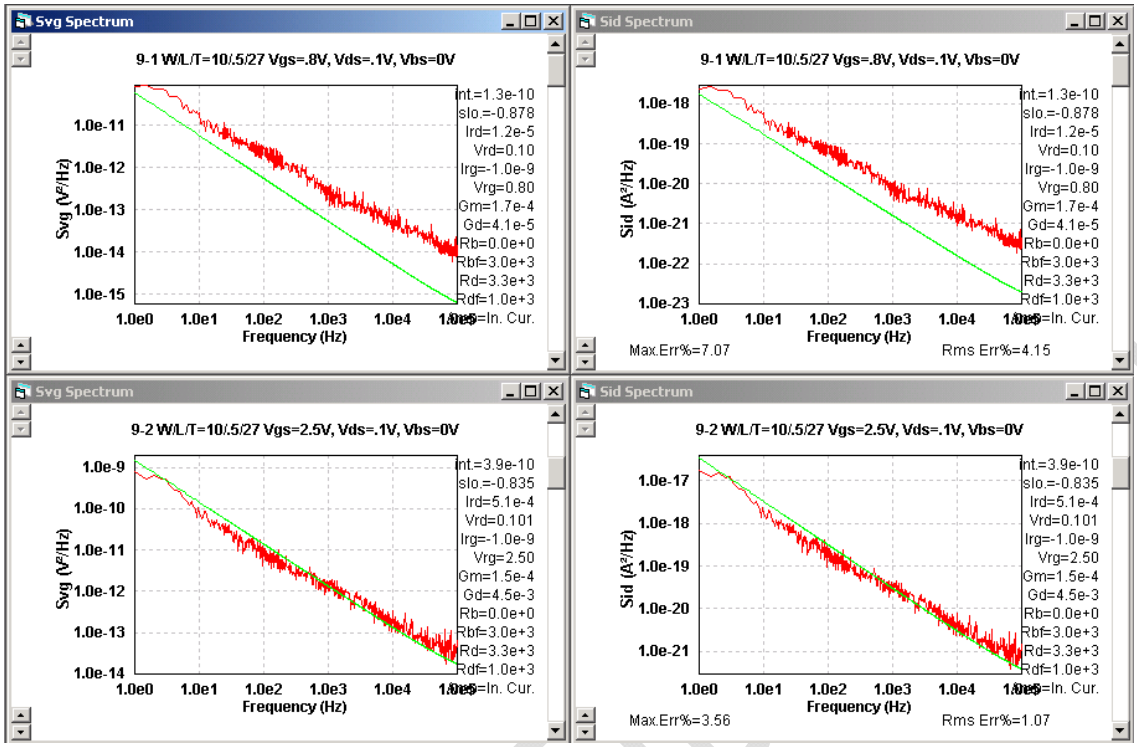

Figure 9-30 INDEX #1 W/L = 10/0.5 NMOS- BIAS #1 AND #2

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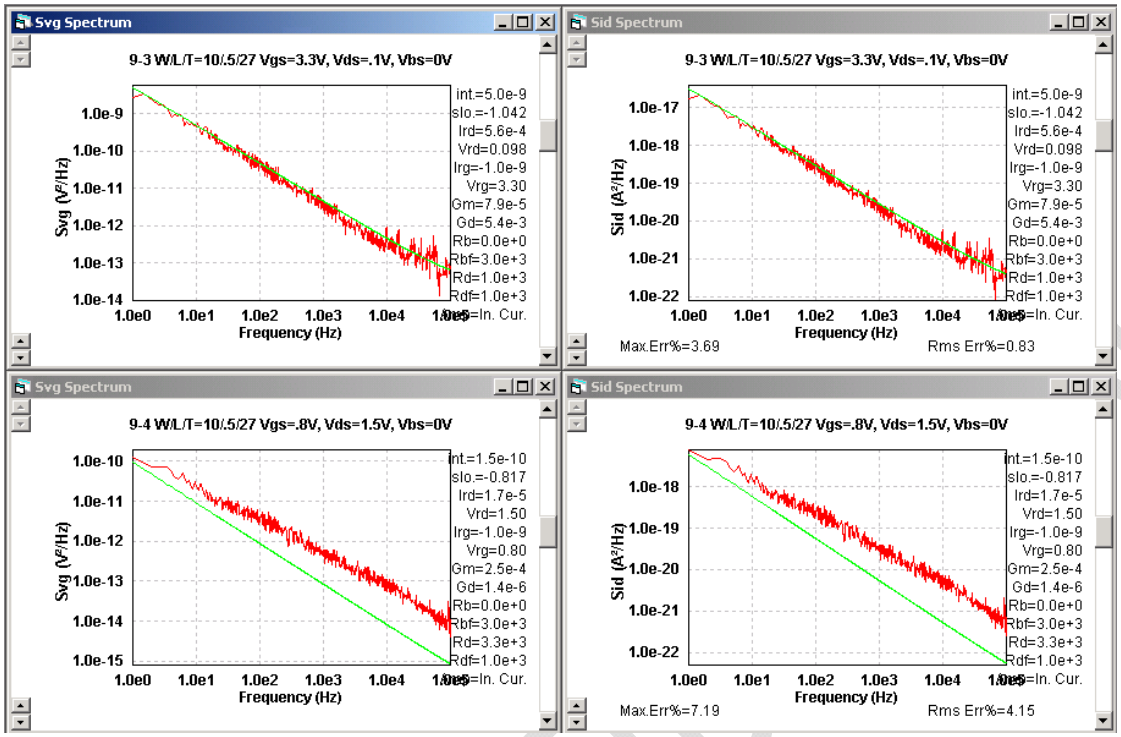

Figure 9-31 INDEX #1 W/L = 10/0.5 NMOS BIAS #3 AND #4

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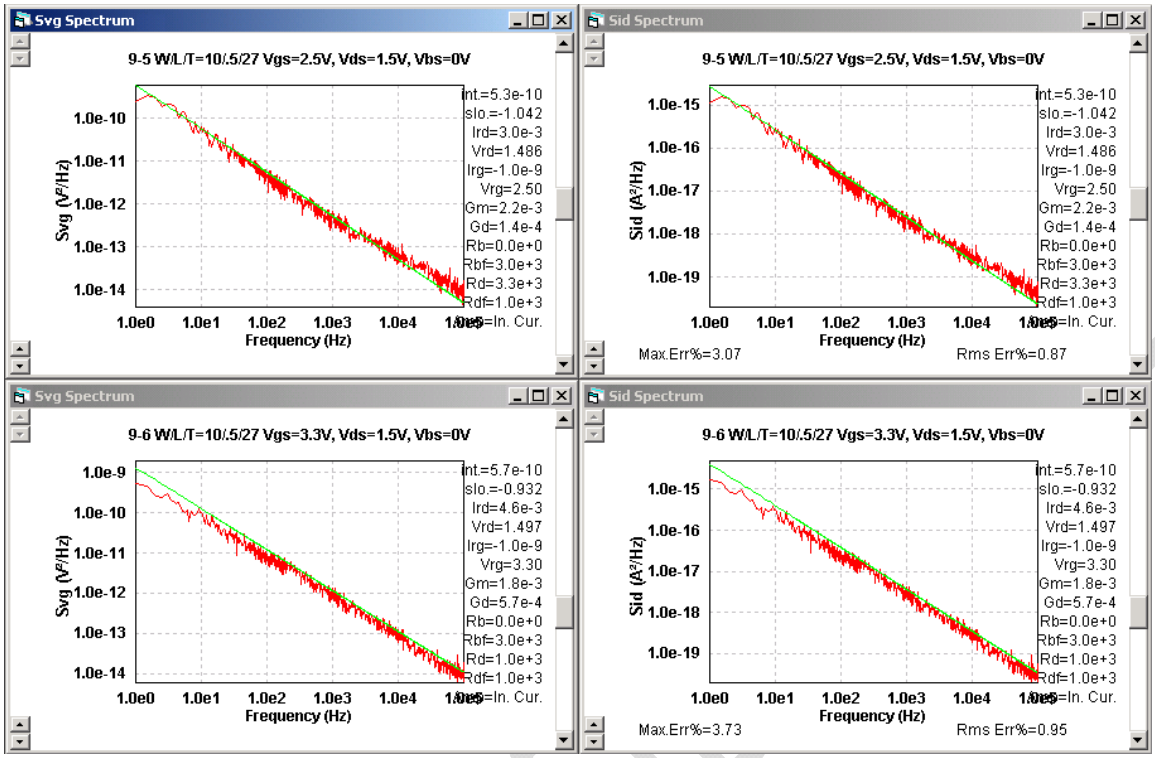

Figure 9-32 INDEX #1 W/L = 10/0.5 NMOS- BIAS #5 AND #6

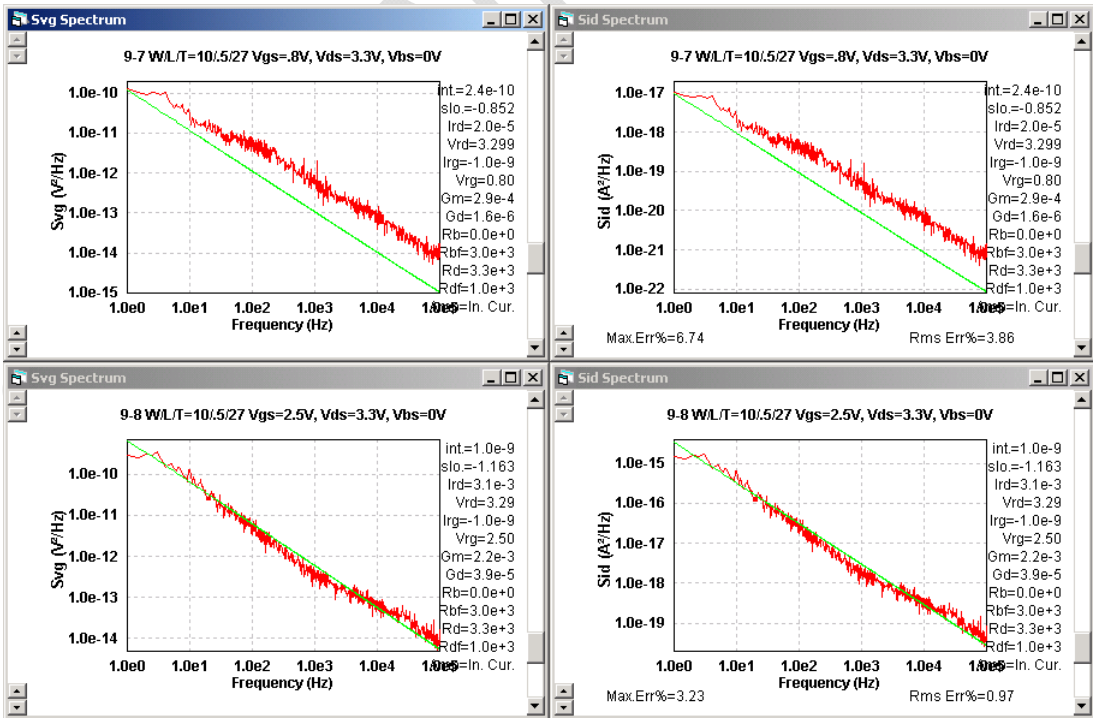

Figure 9-33 INDEX #1 W/L = 10/0.5 NMOS- BIAS #7 AND #8

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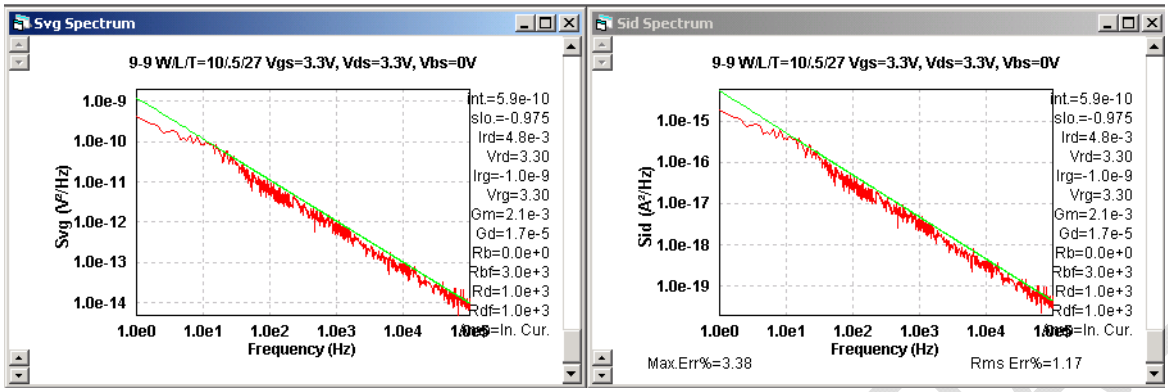

Figure 9-34 INDEX #1 W/L = 10/0.5 NMOS- BIAS #9

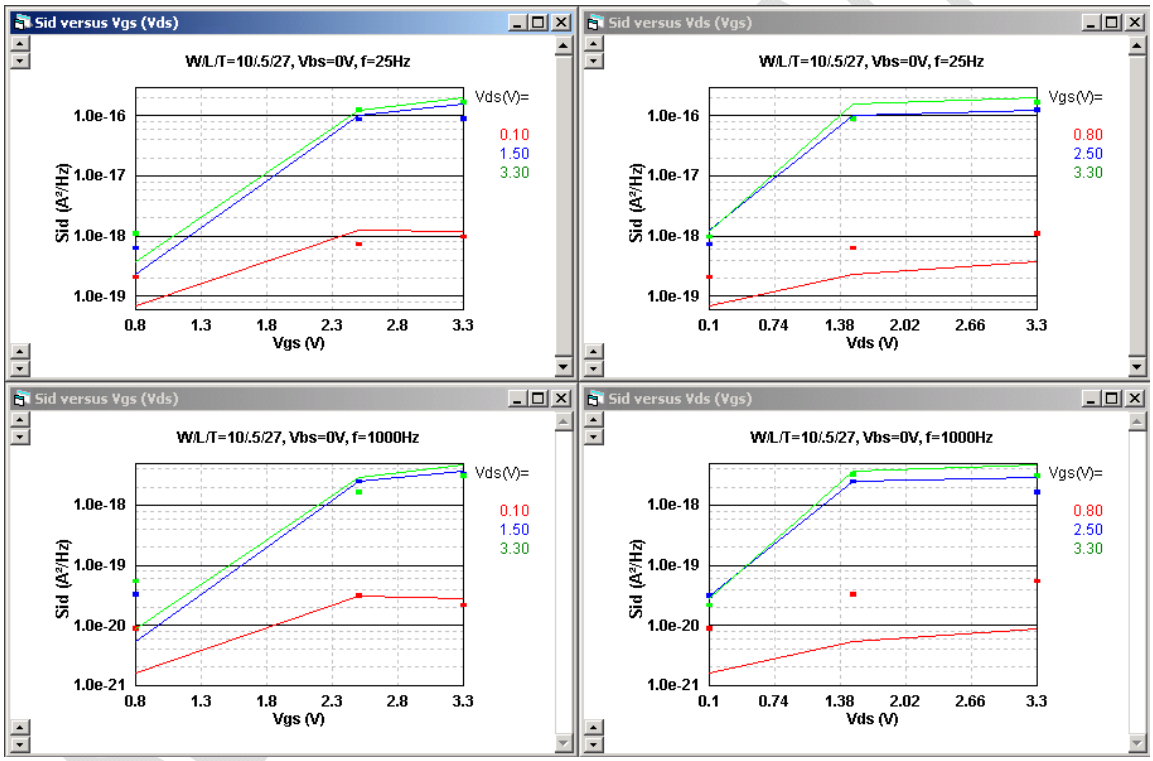

Figure 9-35 SID AT SAMPLING FREQUENCIES OF 25HZ AND 1000HZ

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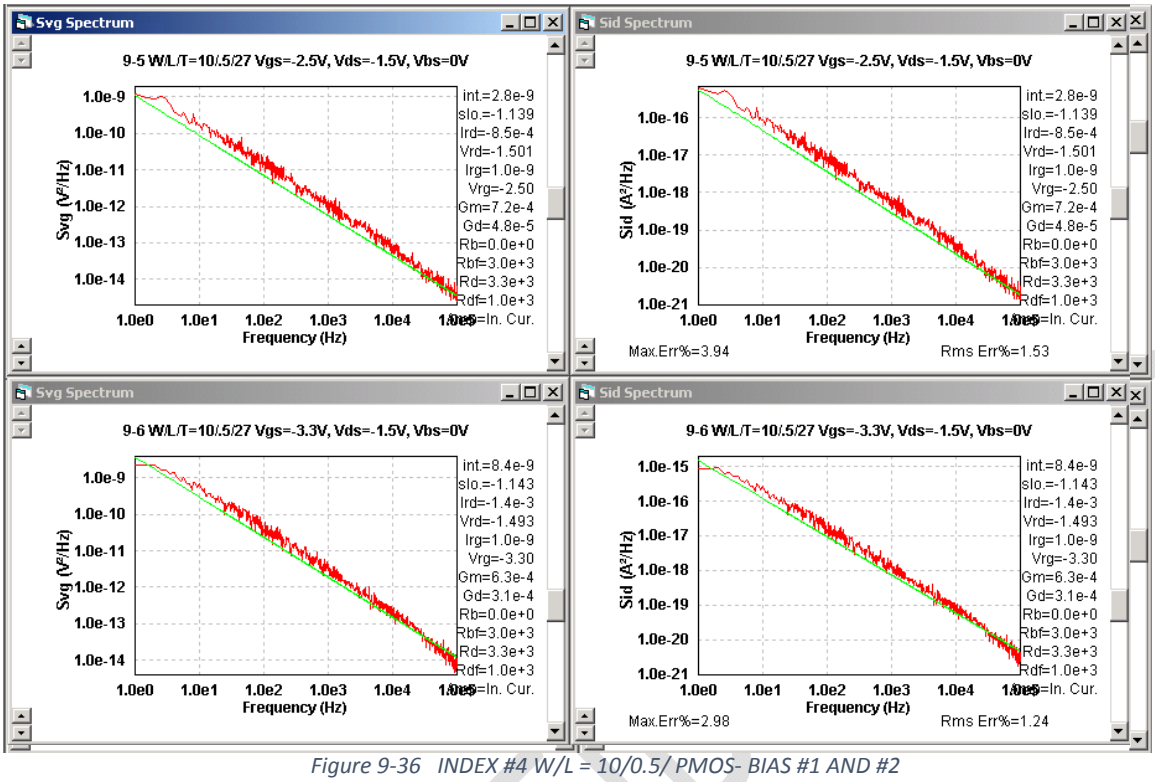

Figure 9-36 INDEX #4 W/L = 10/0.5/ PMOS- BIAS #1 AND #2

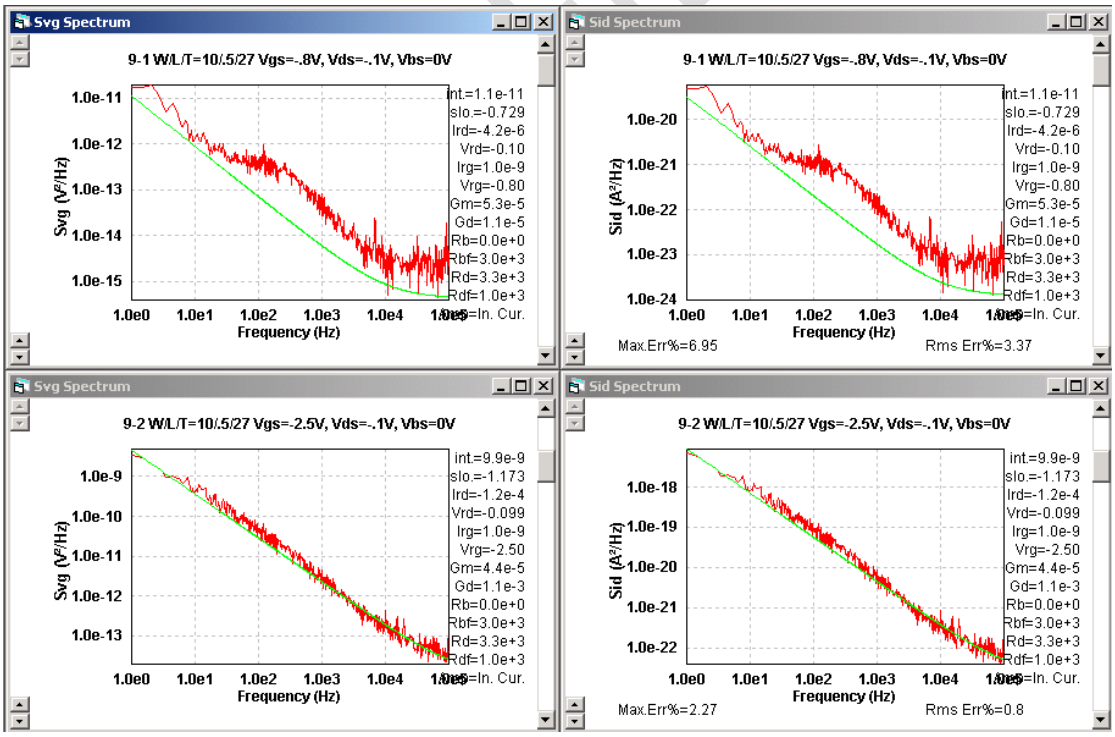

Figure 9-37 INDEX #4 W/L = 10/0.5/ PMOS - BIAS #3 AND #4

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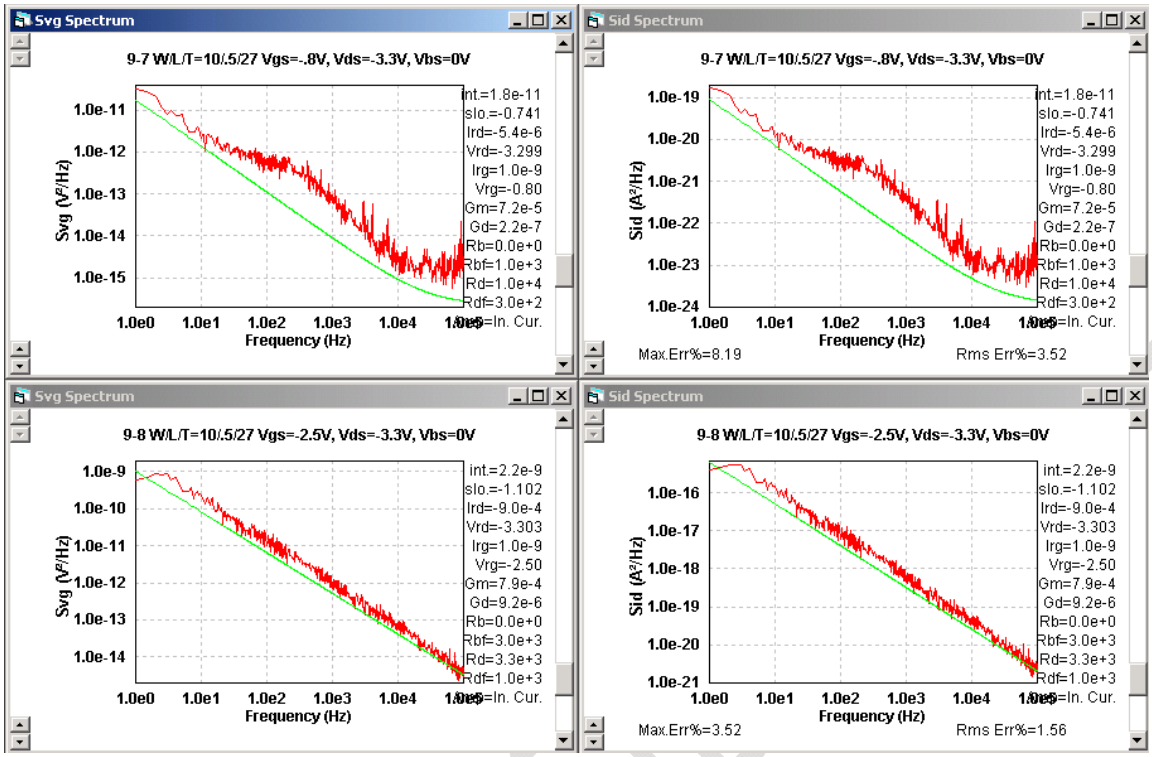

Figure 9-38 INDEX #4 W/L = 10/0.5/ PMOS- BIAS #7 AND #8

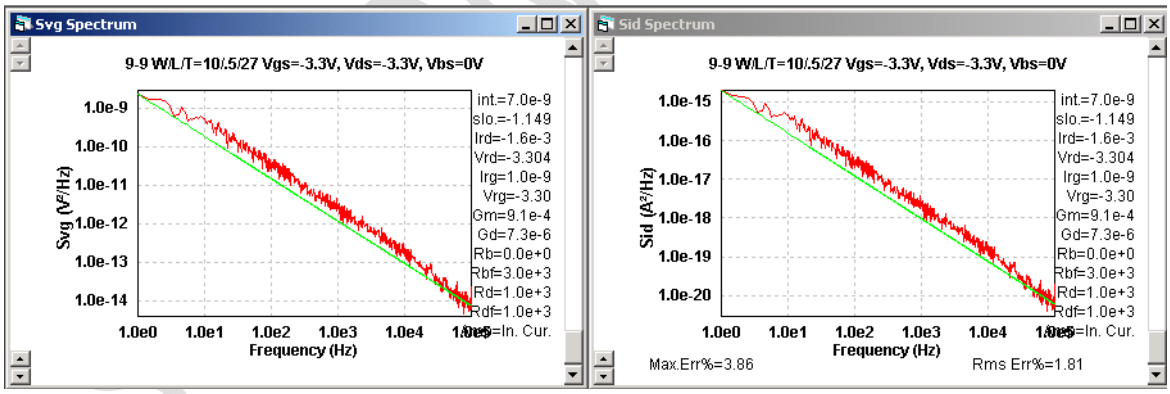

Figure 9-39 INDEX #4 W/L = 10/0.5/ PMOS- BIAS #9

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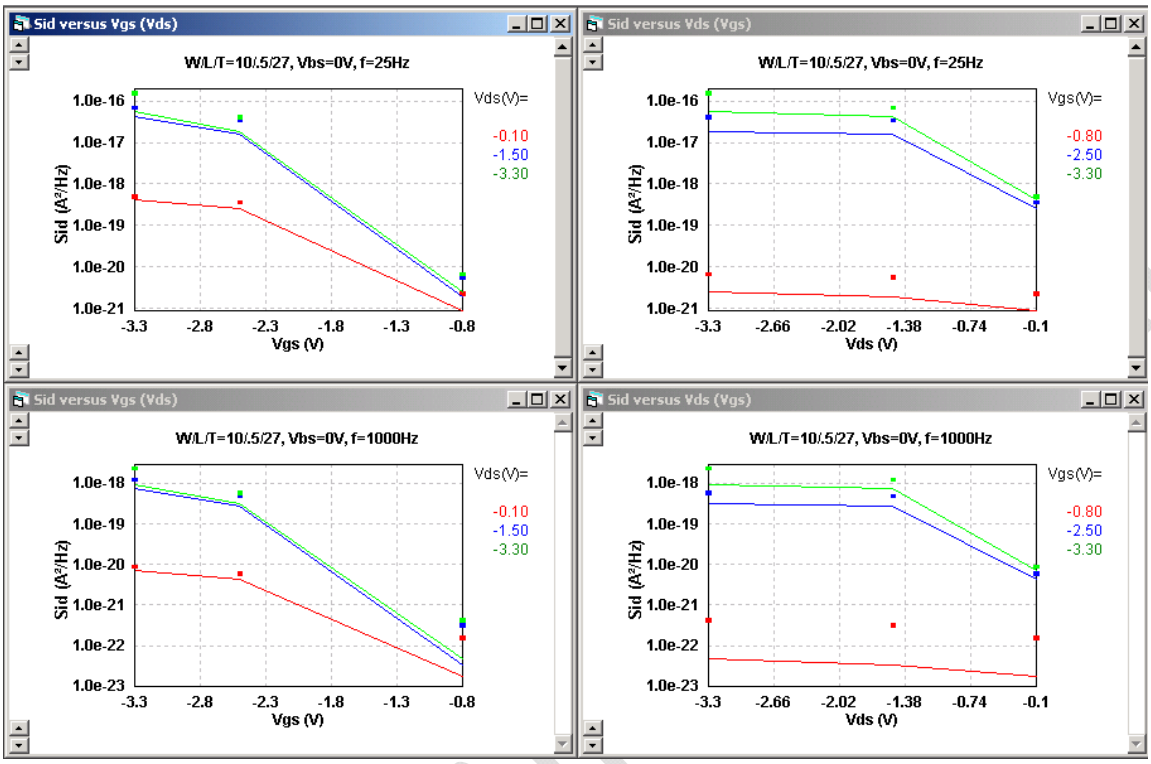

Figure 9-40 SID AT SAMPLING FREQUENCIES OF 25HZ AND 1000HZ

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9.6.3 5V MOS – PSP

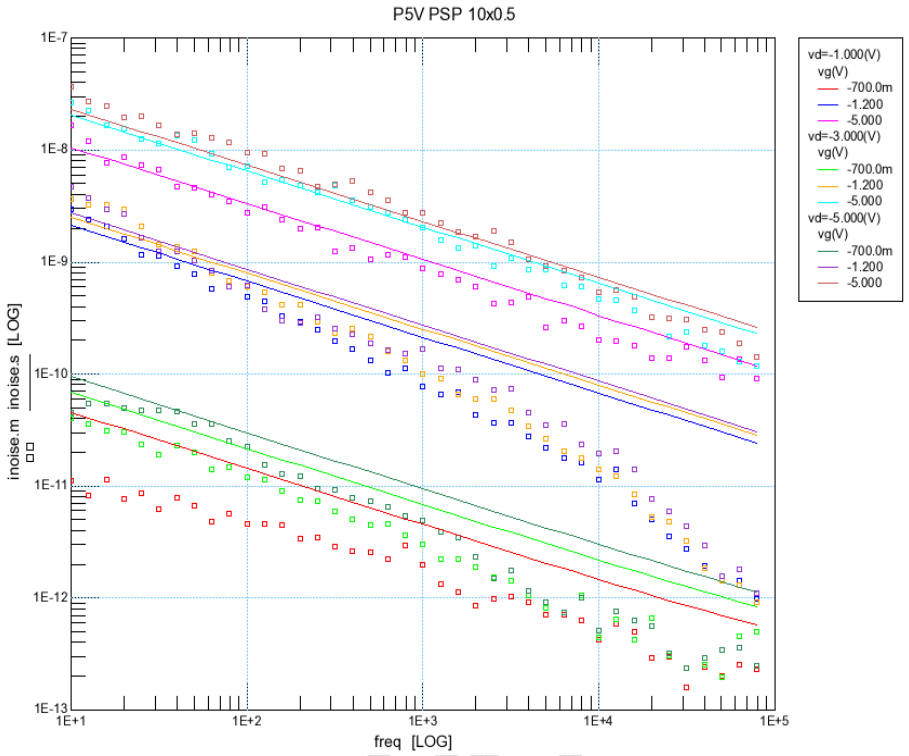

Figure 9-41 5V NMOS PSP 1/F MEASURED VS. SIMULATED DATA W/L = 10/0.6, DIFFERENT BIAS POINTS

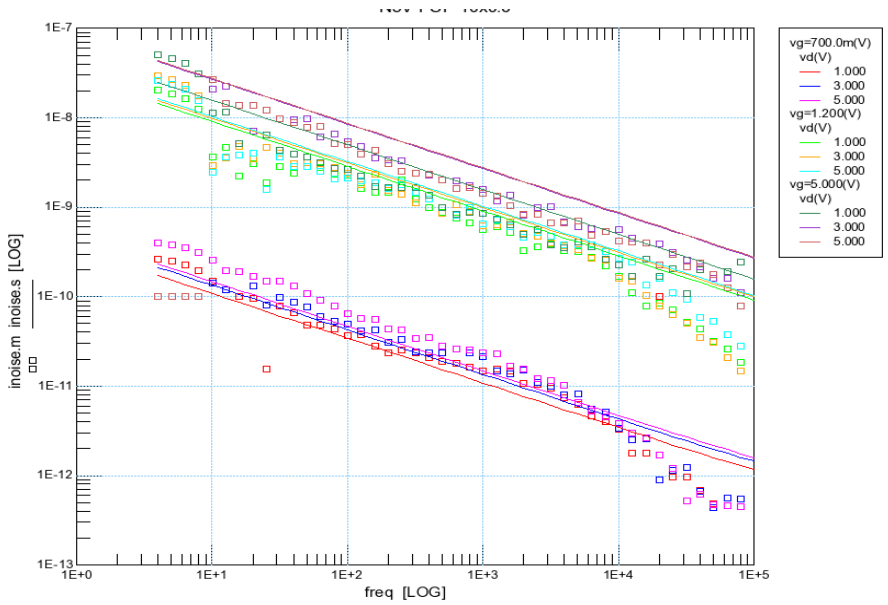

Figure 9-42 5V PMOS PSP 1/F MEASURED VS. SIMULATED DATA W/L = 10/0.5, DIFFERENT BIAS POINTS.

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Table 9-31 REPRESENTATION OF THE 1/F NOISE MODEL PARAMETER VALUES FOR 1.8V, 3.3V AND 5V MOSFET'S AT NOIMOD=1 AND NOIMOD=2

Noimod	Parameter	1.8V NMOS (Std Vth)	1.8V PMOS (Std Vth)	3.3 V NMOS	3.3V PMOS	5 V NMOS	5V PMOS
1	KF	1.58E-26	1.58E-23	4.54E-28	1.91E-25		
	AF	1.42E+00	1.84E+00	9.79E-01	1.52E+00		
	EF	9.52E-01	1.19E+00	9.35E-01	1.01E+00		
2	NOIA	7.94E+19	5.62E+19	2.56E+19	1.41E+19		
	NOIB	2.51E+05	1.53E+06	1.26E+05	2.40E+04		
	NOIC	-2.51E-12	3.63E-11	-3.69E-13	1.00E-11		
	EF	9.86E-01	1.24E+00	1.02E+00	1.10E+00		
	EM	5.92E+06	2.15E+07	6.64E+06	3.17E+07		

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9.7 Corner Policy

Corner models were generated by varying selected parameters (listed below) in the extracted (measured case) model to match the electrical process targets, Idsat and Vth specified in DR0018SL_SCL for nominal geometry devices. The Graphs below represent five corner models (typical, fast, slow, up and down) for MOSFETs of nominal geometry.

Model parameters used for corner generation:

VTH0, NFACTOR, NLX, LINT, WINT, TOX, U0, CGDO, CGSO, CJ, CJSW and other relevant parameter.

9.7.1 Simulated I-V Curves of Five Corner Models for MOSFETs of Nominal Geometry.

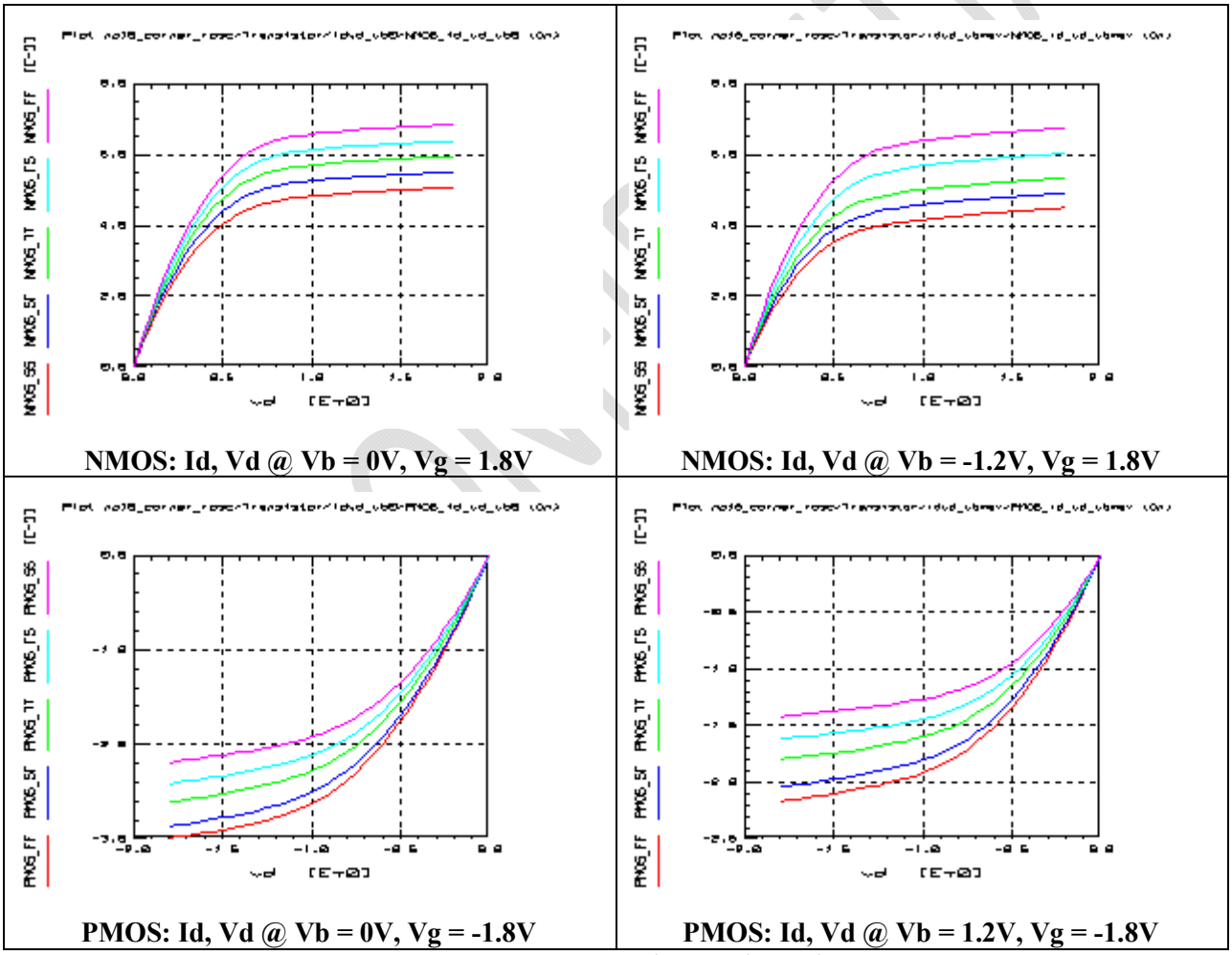

Figure 9-43 1.8 V STD. VTH N/PMOS W/L=10.0/0.18

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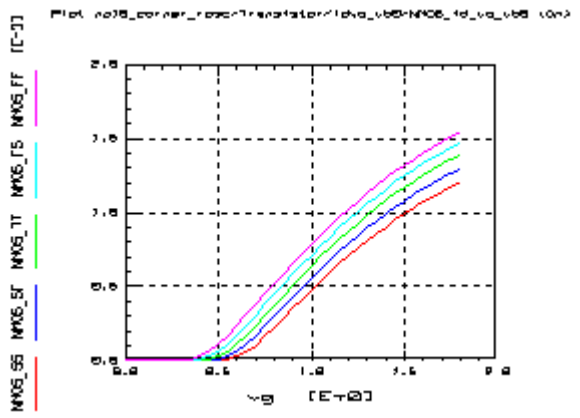
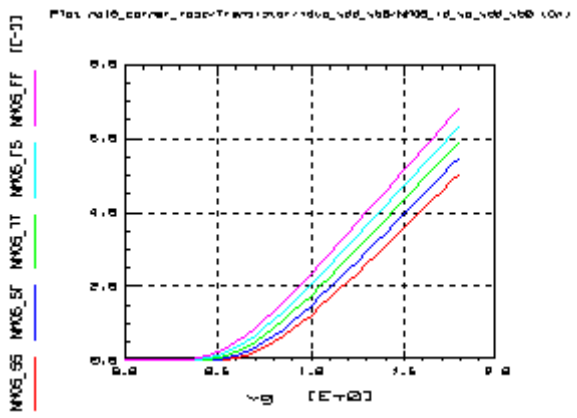
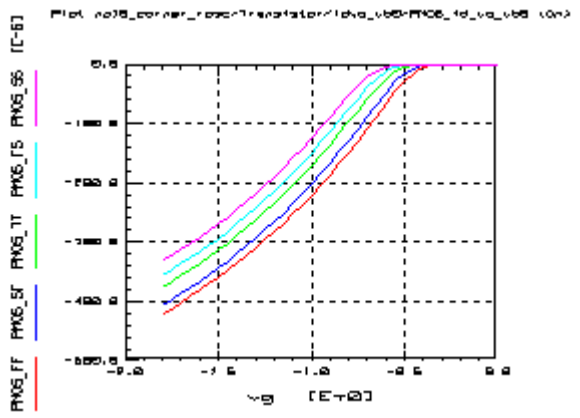


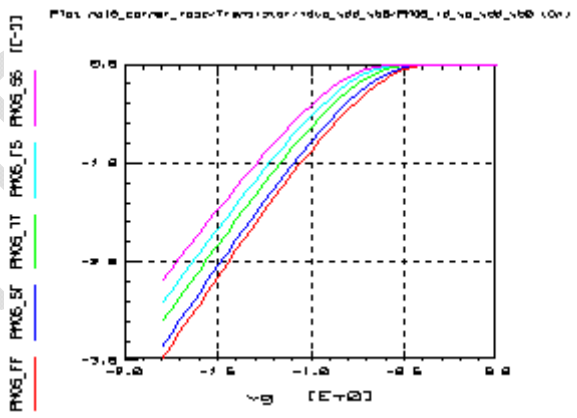
NMOS: Id, Vg @ Vb = 0V, Vd = 0.1V



NMOS: Id, Vg @ Vb = 0V, Vd = 1.8V



PMOS: Id, Vg @ Vb = 0V, Vd = -0.1V



PMOS: Id, Vg @ Vb = 0V, Vd = -1.8V

Figure 9-44 1.8 V STD. VTH N/PMOS W/L=10.0/0.18 [CONTINUED]

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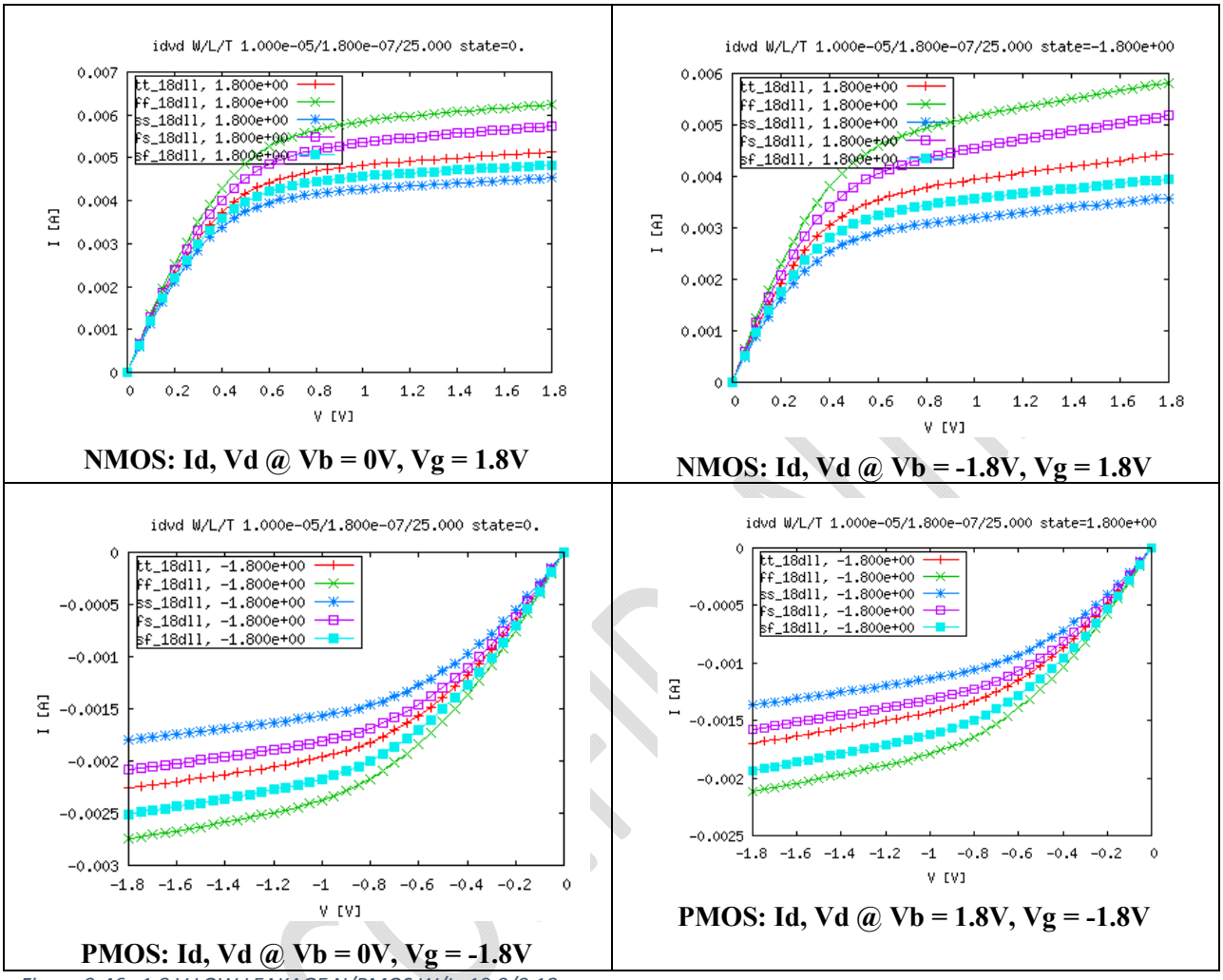

Figure 9-46 1.8 V LOW LEAKAGE N/PMOS W/L=10.0/0.18

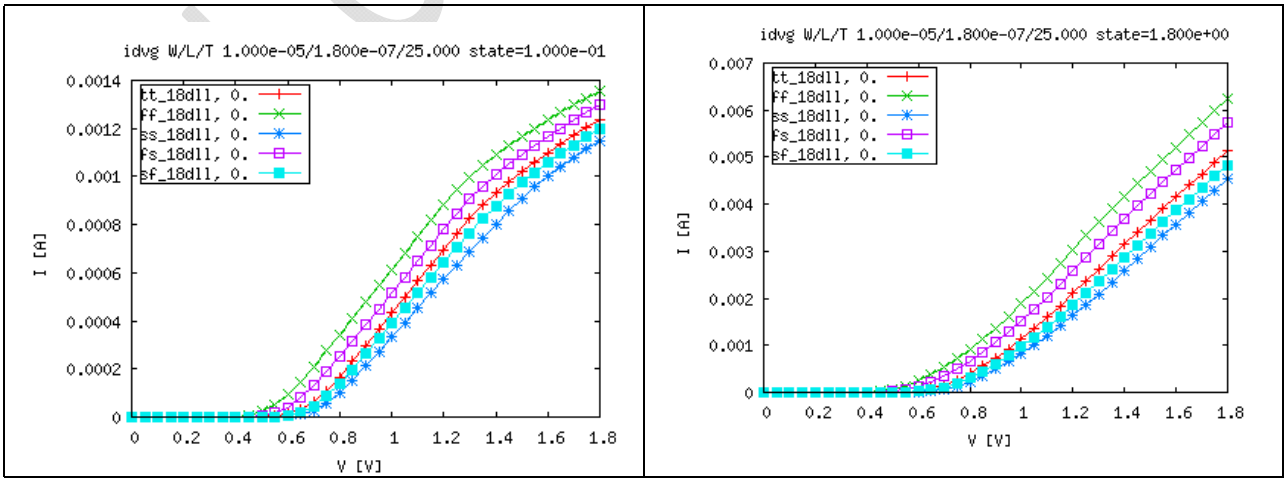

Figure 9-45 1.8 V LOW LEAKAGE N/PMOS W/L=10.0/0.18

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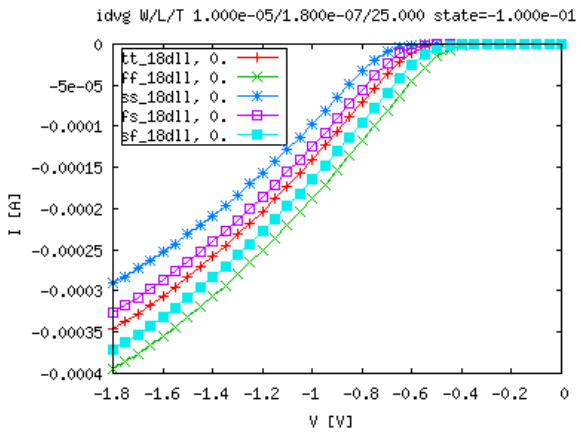
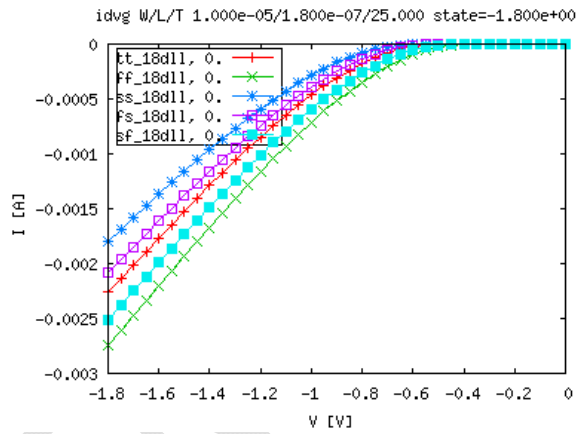
NMOS: Id, Vg @ Vb = 0V, Vd = 0.1V	NMOS: Id, Vg @ Vb = 0V, Vd = 1.8V
	
PMOS: Id, Vg @ Vb = 0V, Vd = -0.1V	**PMOS: Id, Vg @ Vb = 0V, Vd = -1.8V**

Figure 9-471.8 V LOW LEAKAGE N/PMOS W/L=10.0/0.18 [CONTINUED]

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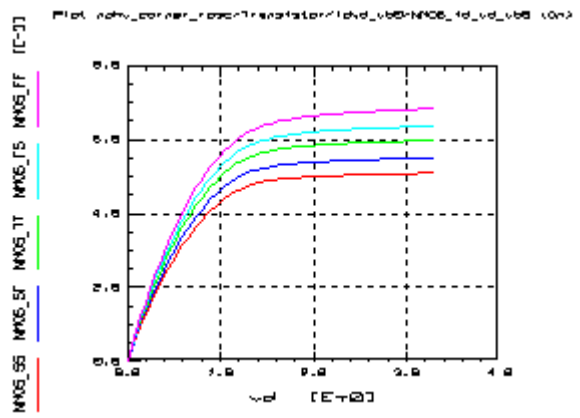
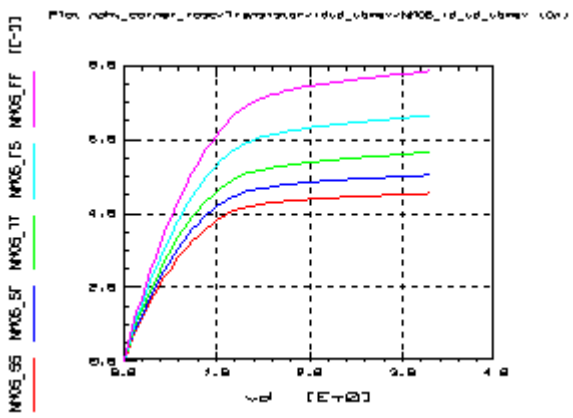
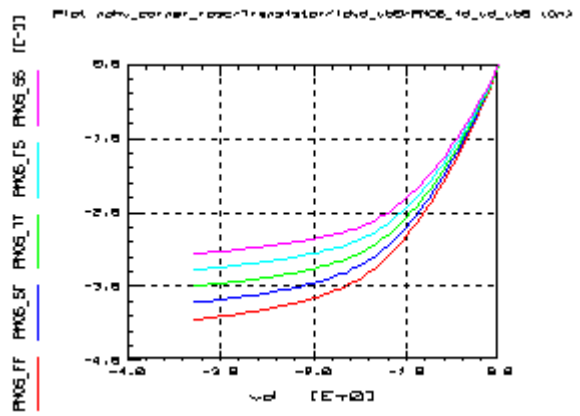


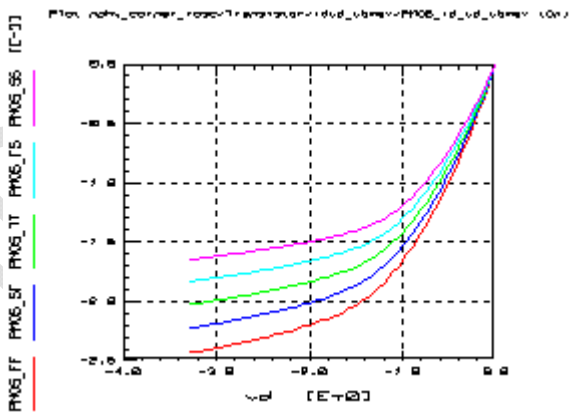
NMOS: Id, Vd @ Vb = 0V, Vg = 3.3V



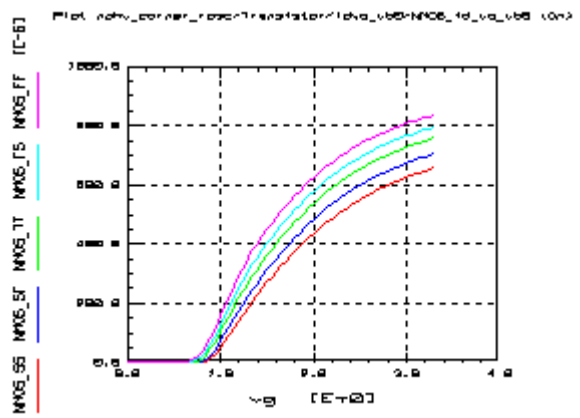
NMOS: Id, Vd @ Vb = -2.1V, Vg = 3.3V



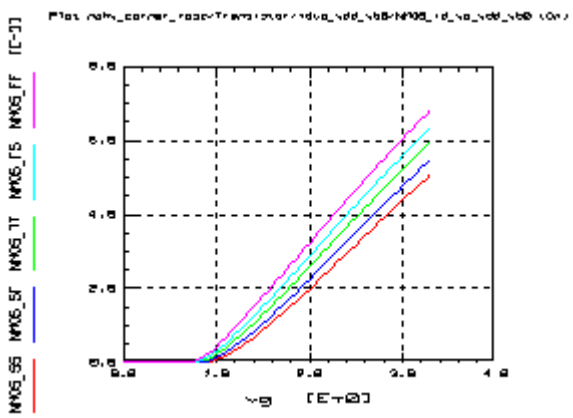
PMOS: Id, Vd @ Vb = 0V, Vg = -3.3V



PMOS: Id, Vd @ Vb = 2.1V, Vg = -3.3V



NMOS: Id, Vg @ Vb = 0V, Vd = 0.1V



NMOS: Id, Vg @ Vb = 0V, Vd = 3.3V

Figure 9-48 3.3 V N/PMOS W/L=10.0/0.35(0.3)

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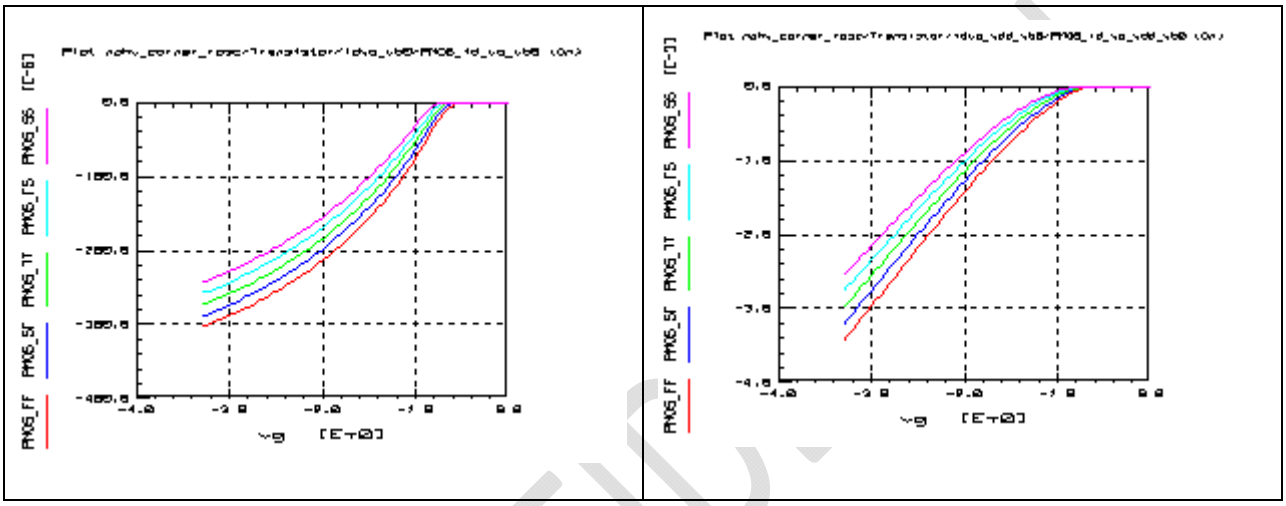
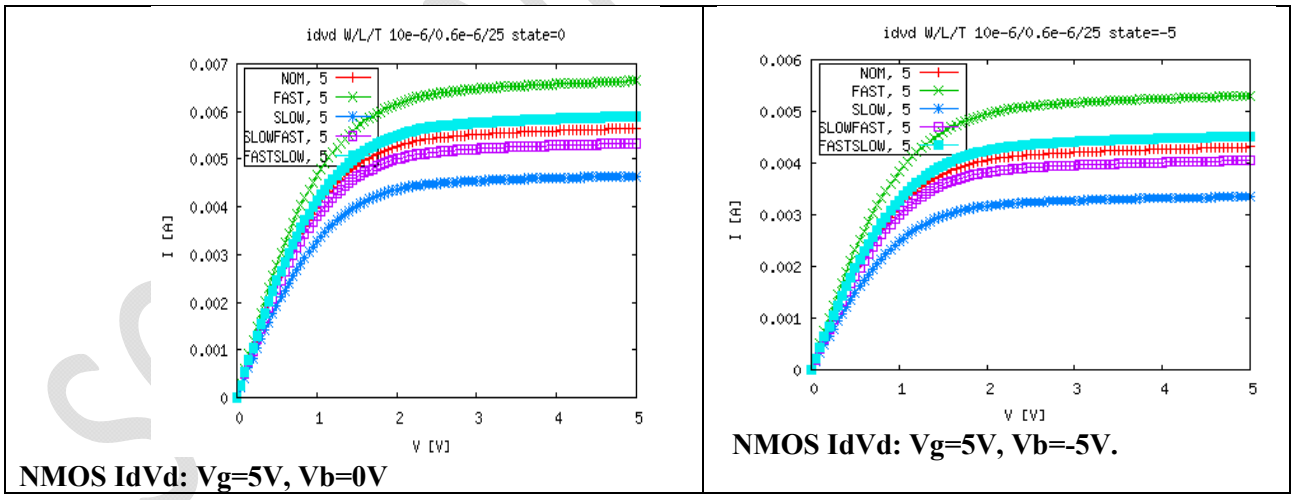


Figure 9-49 3.3 V N/PMOS W/L=10.0/0.35(0.3)



NMOS IdVd: Vg=5V, Vb=0V

NMOS IdVd: Vg=5V, Vb=-5V.

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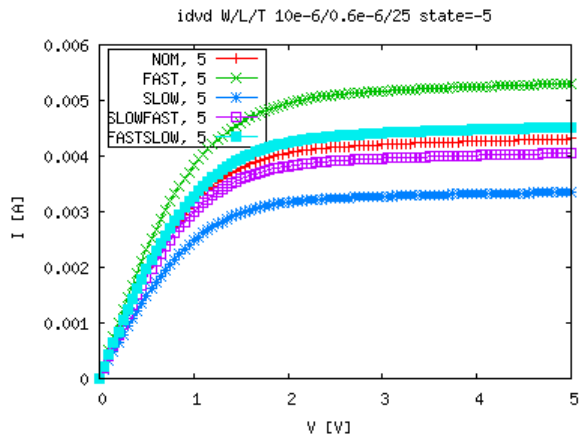
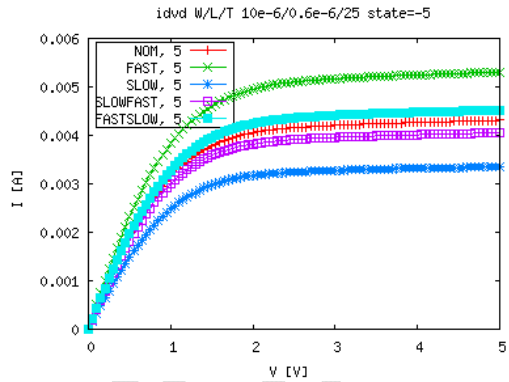


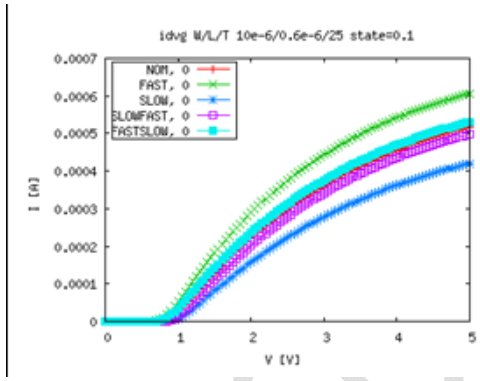
PMOS IdVd: Vg=-5V, Vb=0V



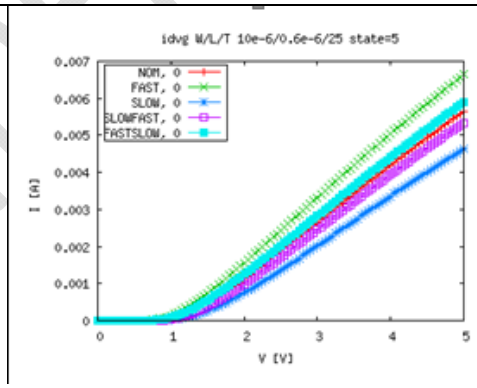
PMOS IdVd: Vg=-5V, Vb=5V.

Figure 9-50 5 V N/PMOS W/L=10.0/0.6

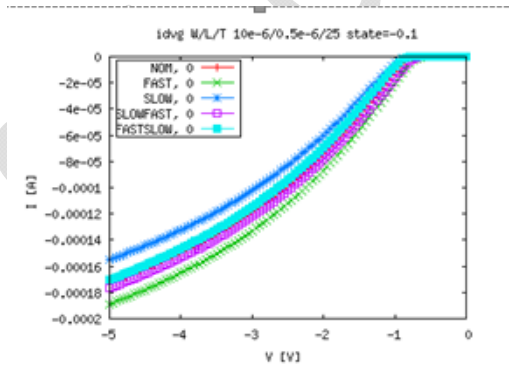
FIGURE 10-44: 5 V PSP NMOS,PMOS W/L=10.0/0.6, 10.0/0.5



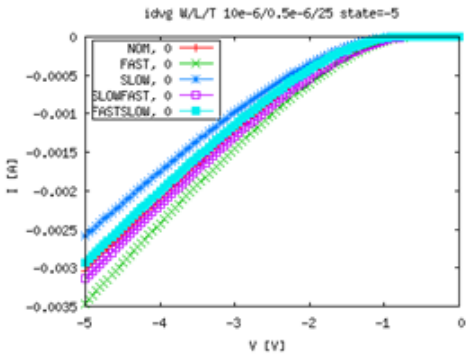
NMOS IdVg: Vd=0.1V, Vb=0V



NMOS IdVg: Vd=5V, Vb=0V.



PMOS IdVg: Vd=-0.1V, Vb=0V.



PMOS IdVg: Vd=-5V, Vb=0V.

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9.8 MOSFET matching

The following MOSFET types were characterized and modeled for mismatch: p18, n18, nhv & phv. Mismatch model is available using Monte-Carlo simulation for Hspice simulator. The results were compared to the measured data and to the theoretical calculation.

Note that there are 2 libraries for Hspice Simulation :

1. **tsl018_scl.lib**: This library includes all the devices of TS18SL and don't include mismatch modeling.
2. **tsl018_scl_mat.lib**: This library includes only Mosfets devices (Nmos and Pmos 1.8V/3.3V) as an example for mismatch modeling for HSPICE simulator. This modeling approach is not compatible with PCell library since uses subcircuit "wrapping" for MOSFET model. Netlist should be modified accordingly in order to run the simulation.

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10 Diode Scalable Junction Model

10.1 List of Legal Device

Table 10-1 LIST OF LEGAL DEVICE – DIODE SCALABLE JUNCTION MODEL

Device	SPICE Name	Voltage Range
Diode N^+/Pwell 1.8V	DN18	-1.5V to 10V
Diode P^+/Nwell 1.8V	DP18	-10 to 1.5V
Diode Nwell/Psub 1.8V	DNWELL	-1.5V to 13V
Diode N^+/Pwell 3.3V	DNH	-1.5V to 10V
Diode P^+/Nwell 3.3V	DPH	-10 to 1.5V
Diode Nwell/Psub 3.3V	DNWELL33	-1.5V to 13V
Diode IsoPwell/WB 1.8 V	DDWNPW18	-10 to 1.5V
Diode WB/Psub 1.8 V	DDWNPS18	-1.5V to 13V
Diode P+w/o salicidation/Nwell 1.8V	DPH	-10 to 1.5V
Diode N+w/o salicidation/Pwell 1.8V	DNH	-1.5V to 10V
Diode N+/Pwell 5V	DN50	-1.5V to 10V
Diode N+/Pwell 5V	DP50	-10 to 1.5V

10.2 Model description

Model parameter sets were extracted using N^+/Pwell, P^+/Nwell, Nwell/Psub WB/Psub and IsoPwell/WB test structures with different area/periphery ratio. Models are valid for temperature range –40°C to 125°C.The details regarding DUT's and extraction conditions are described in the tables below.

Table 10-2 LOW VOLTAGE 1.8V N+/WP DIODE

Device name	Area (um2)	Perimeter (um)	Temperature (°C)	Measurement
DUT1	47152	131149	25, 125, -40	IV, CV
DUT2	65490	1346	25, 125, -40	IV, CV

Table 10-3 LOW VOLTAGE 1.8V P+/WN DIODE

Device name	Area (um2)	Perimeter (um)	Temperature (°C)	Measurement
DUT1	47152	131149	25, 125, -40	IV, CV
DUT2	65490	1346	25, 125, -40	IV, CV

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Table 10-4 LOW VOLTAGE 1.8V WN/PSUB DIODE

Device name	Area (um2)	Perimeter (um)	Temperature (°C)	Measurement
DUT1	193140	4026	25, 125, -40	IV, CV
DUT2	146520	73788	25, 125, -40	IV, CV
DUT3	132800	132800	25, 125, -40	IV, CV

Table 10-5 HIGH VOLTAGE 3.3V N+/WP DIODE

Device name	Area (um2)	Perimeter (um)	Temperature (°C)	Measurement
DUT1	47152	131149	25, 125, -40	IV, CV
DUT2	130980	2692	25, 125, -40	IV, CV

Table 10-6 HIGH VOLTAGE 3.3V P+/WN DIODE

Device name	Area (um2)	Perimeter (um)	Temperature (°C)	Measurement
DUT1	47152	131149	25, 125, -40	IV, CV
DUT2	130980	2692	25, 125, -40	IV, CV

Table 10-7 WB/PSUB DIODE

Device name	Area (um2)	Perimeter (um)	Temperature (°C)	Measurement
DUT1	55272	16184	25, 125, -40	IV, CV
DUT2	80656	1136	25, 125, -40	IV, CV

Table 10-8 ISOPWELL/WB DIODE

Device name	Area (um2)	Perimeter (um)	Temperature (°C)	Measurement
DUT1	55272	16184	25, 125, -40	IV, CV
DUT2	80656	1136	25, 125, -40	IV, CV

Table 10-9 HIGH VOLTAGE 5V N+/WP DIODE

Device name	Area (um2)	Perimeter (um)	Temperature (°C)	Measurement
DUT1	47152	130980	25, 125, -40	IV, CV
DUT2	131149	2692	25, 125, -40	IV, CV

Table 10-10 HIGH VOLTAGE 5V P+/WN DIODE

Device name	Area (um2)	Perimeter (um)	Temperature (°C)	Measurement
DUT1	47152	130980	25, 125, -40	IV, CV
DUT2	131149	2692	25, 125, -40	IV, CV

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10.3 Extracted Model vs. Measured Data

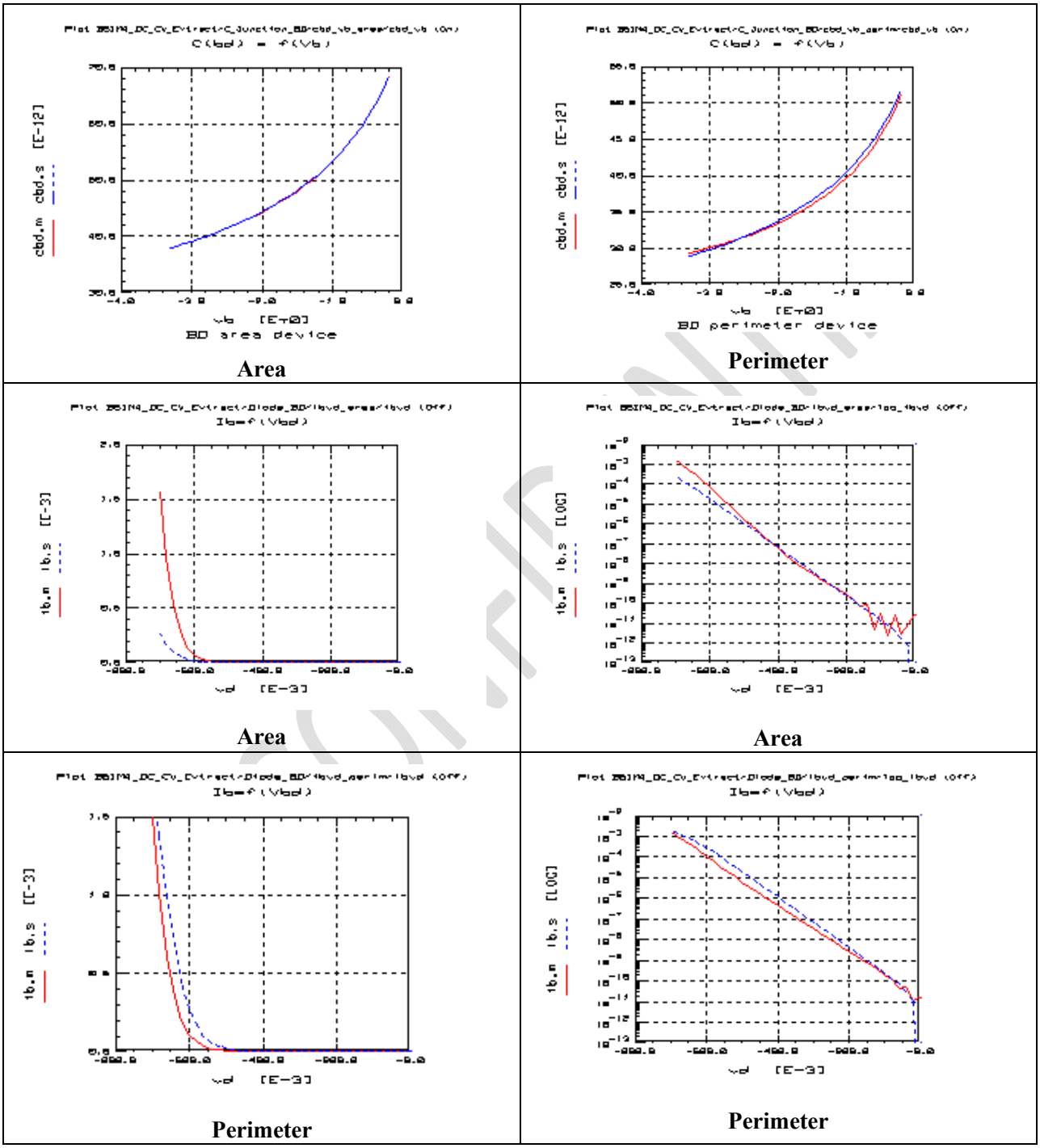

Figure 10-11.8V N+/PWELL DIODE C-V AND I-V CURVES @ T=25 C

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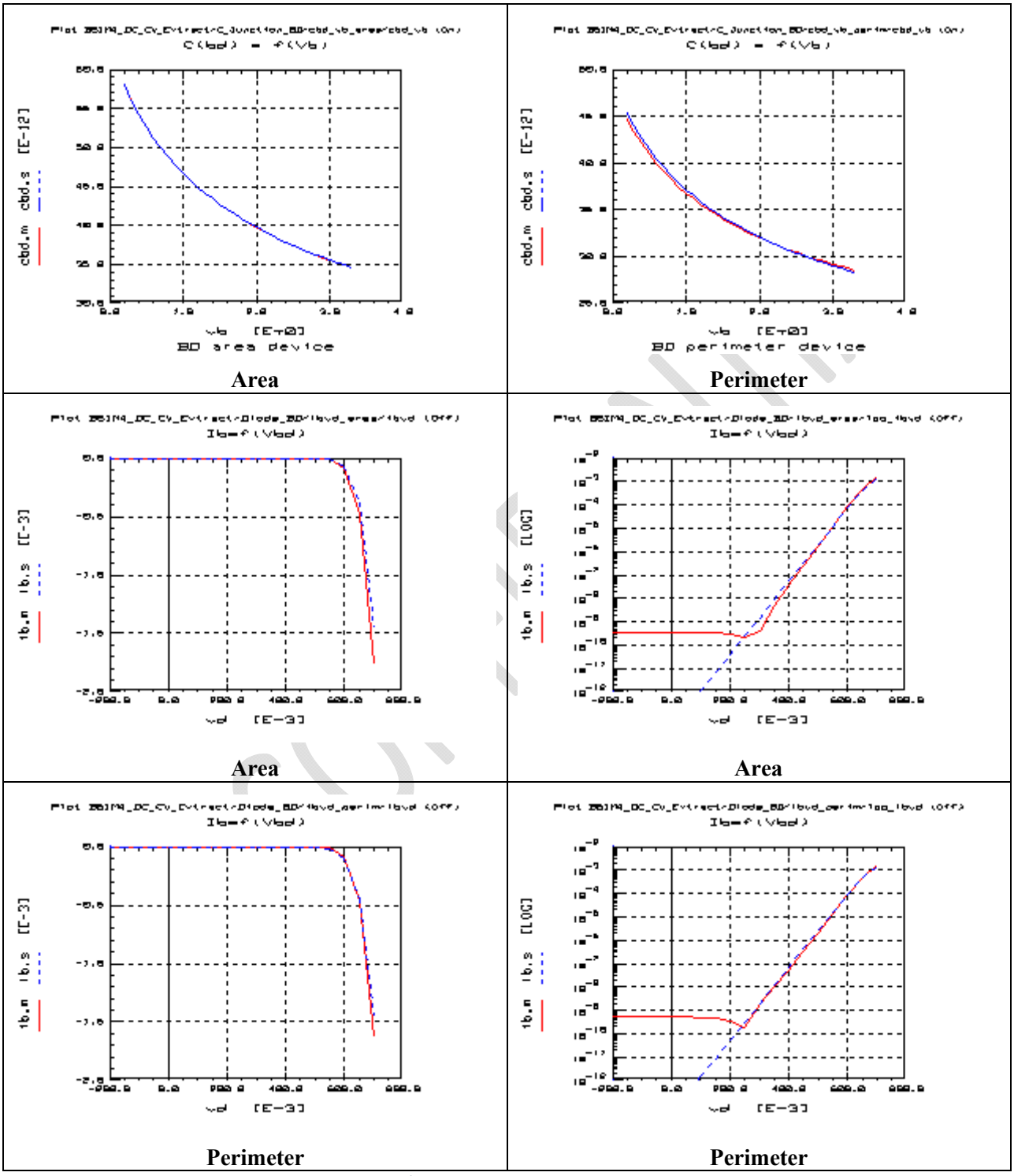

Figure 10-21.8V P+/NWELL DIODE C-V AND I-V CURVES @ T=25 C

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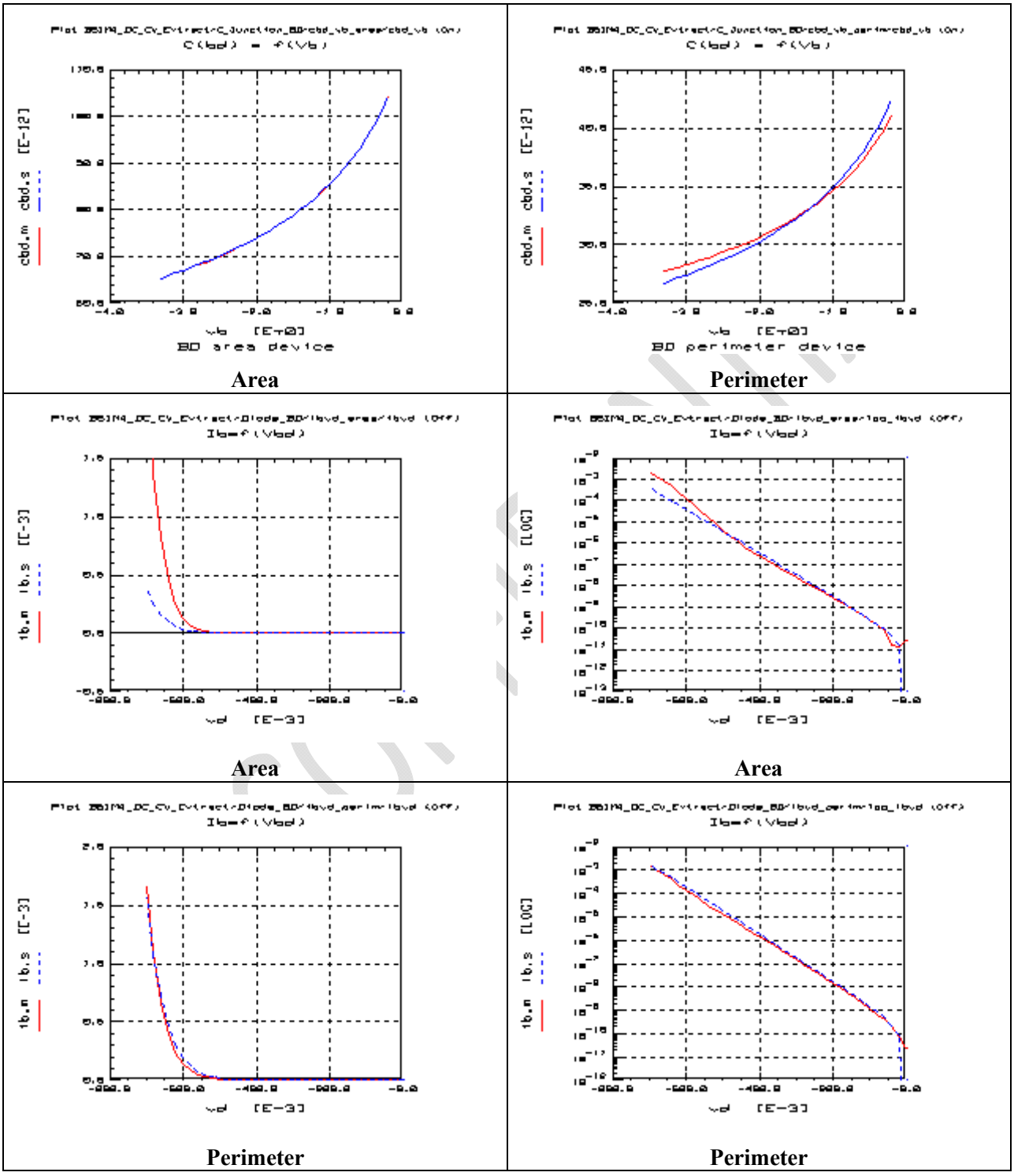

Figure 10-33.3V N+/PWELL & 1.8V N+W/O SALICIDATION DIODES, C-V AND I-V CURVES @ T=25 C

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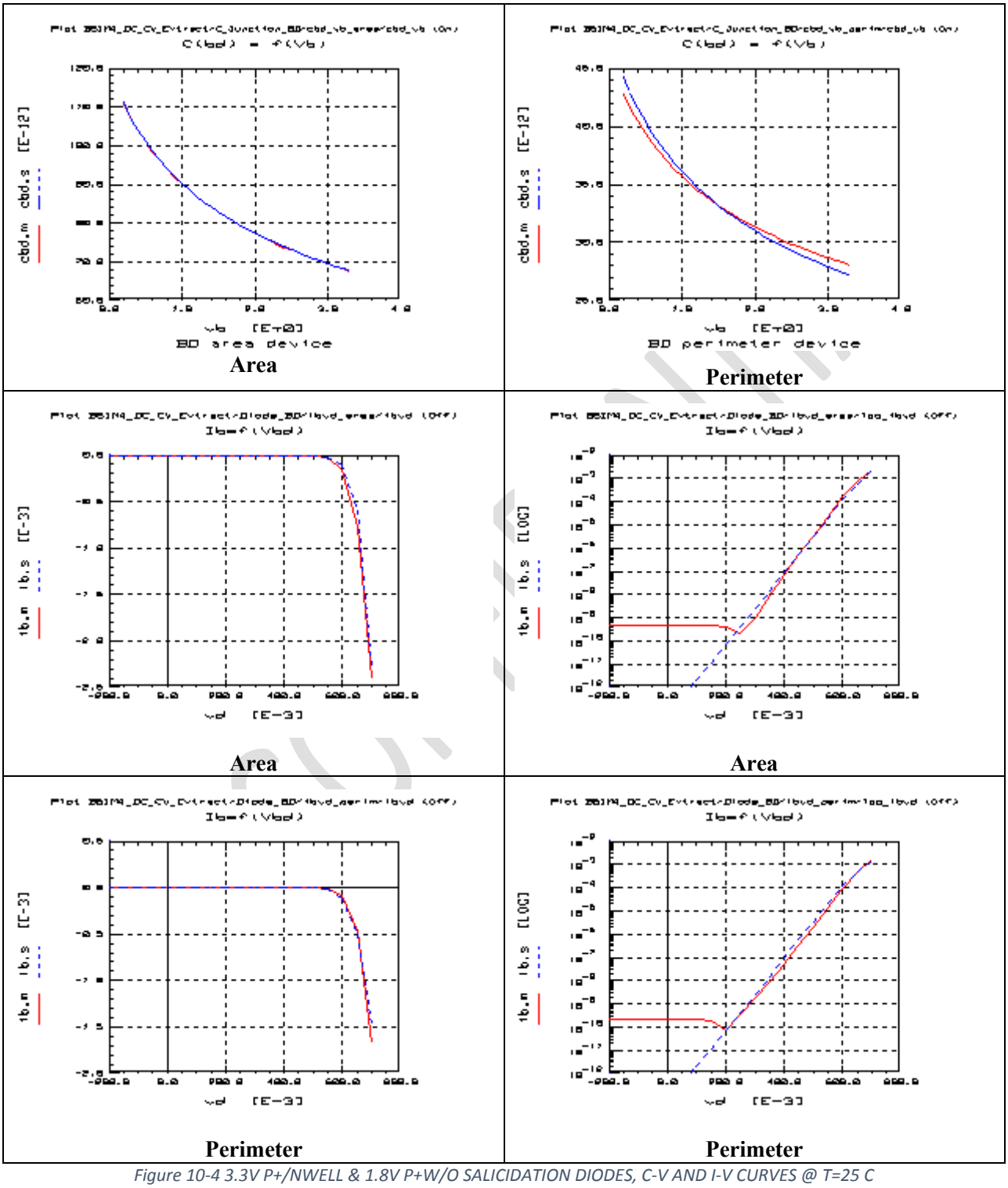

Figure 10-4 3.3V P+/NWELL & 1.8V P+W/O SALICIDATION DIODES, C-V AND I-V CURVES @ T=25 C

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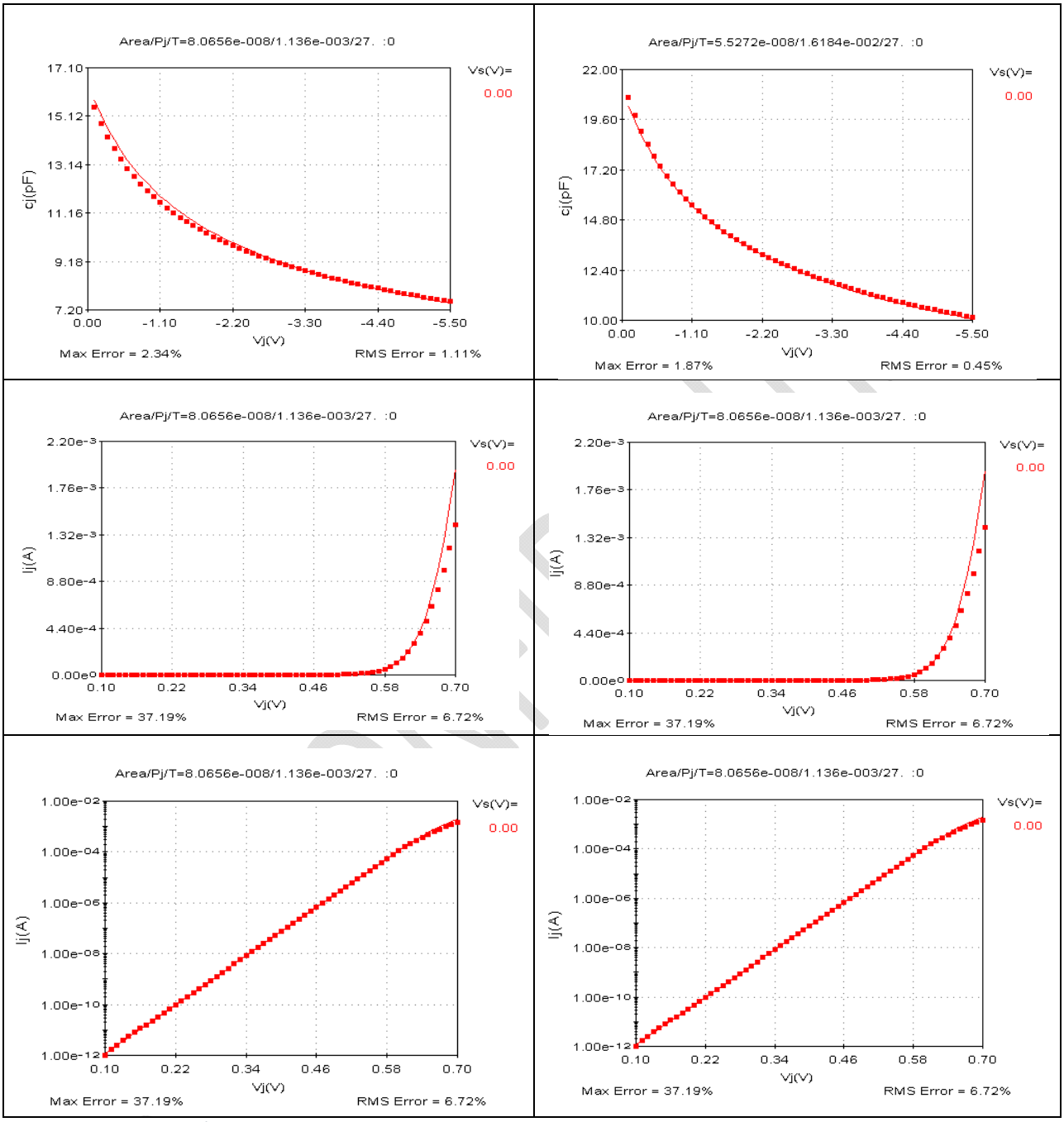

Figure 10-5WB/PSUB DIODE C-V AND I-V CURVES @ T=25 C

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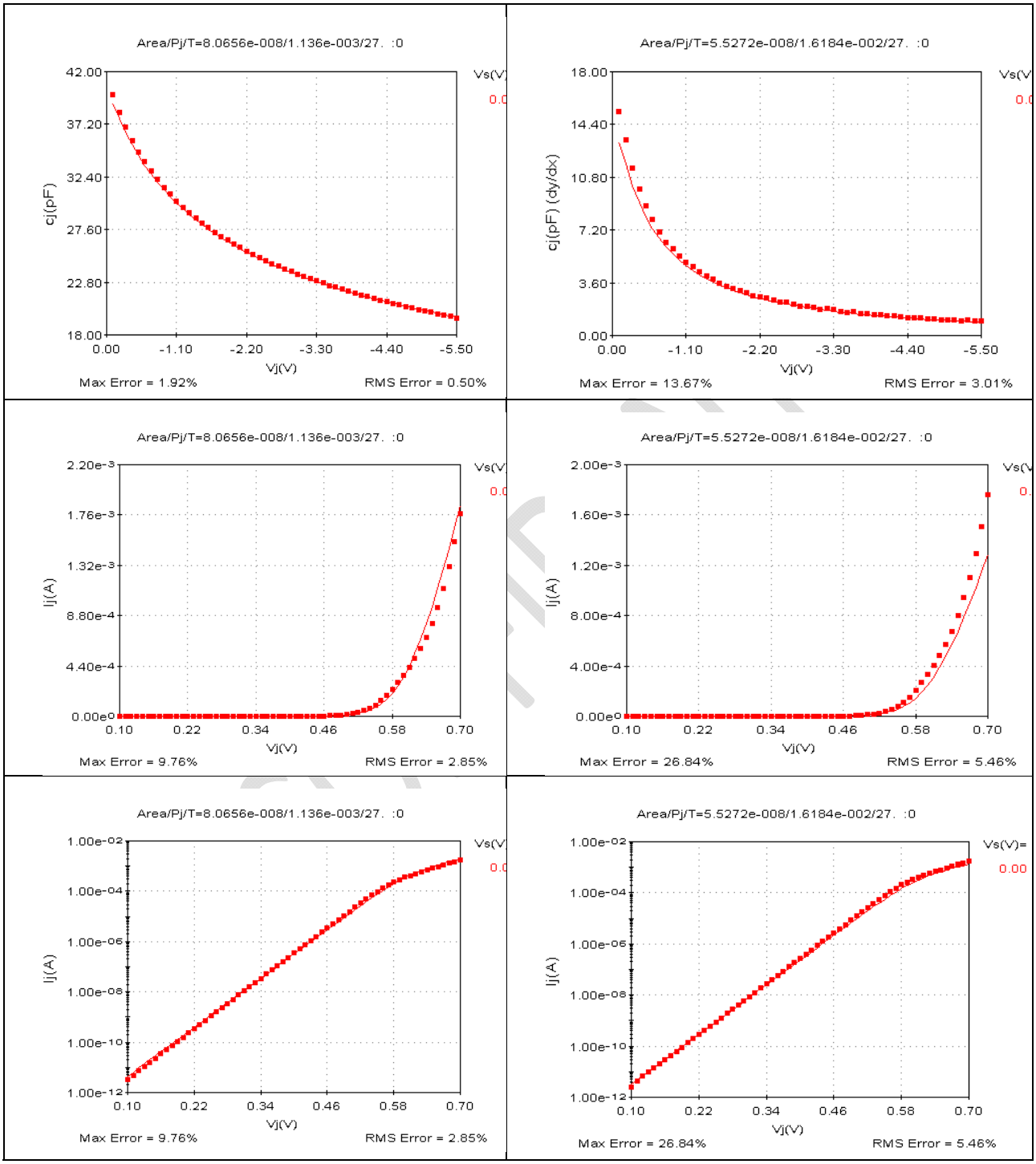

Figure 10-6 **I SOPWELL/WB DIODE C-V AND I-V CURVES @ T=25 C**

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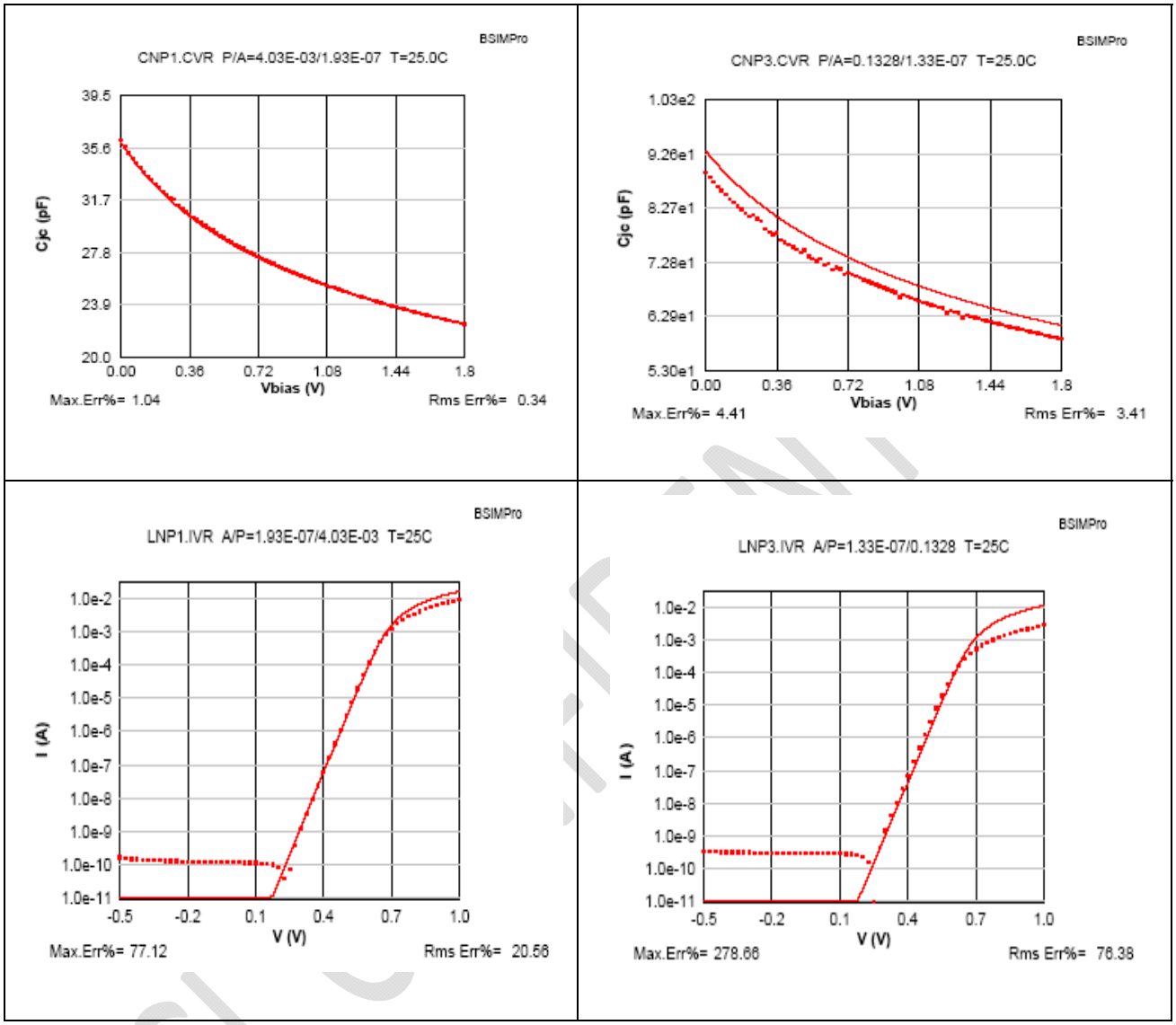

Figure 10-7WN/PSUB DIODE C-V AND I-V CURVES @ T=25 C

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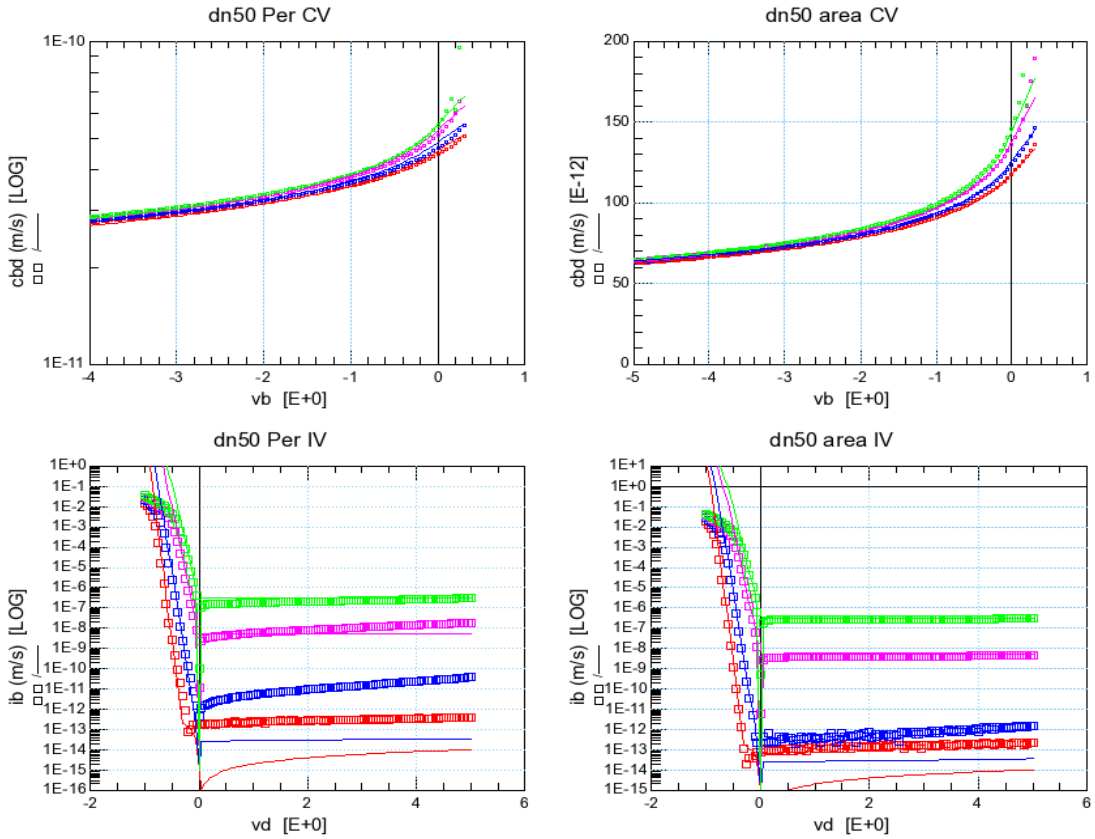

Figure 10-9 5V N+/PWELL DIODE C-V AND I-V CURVES @ T=-40,25,125 C

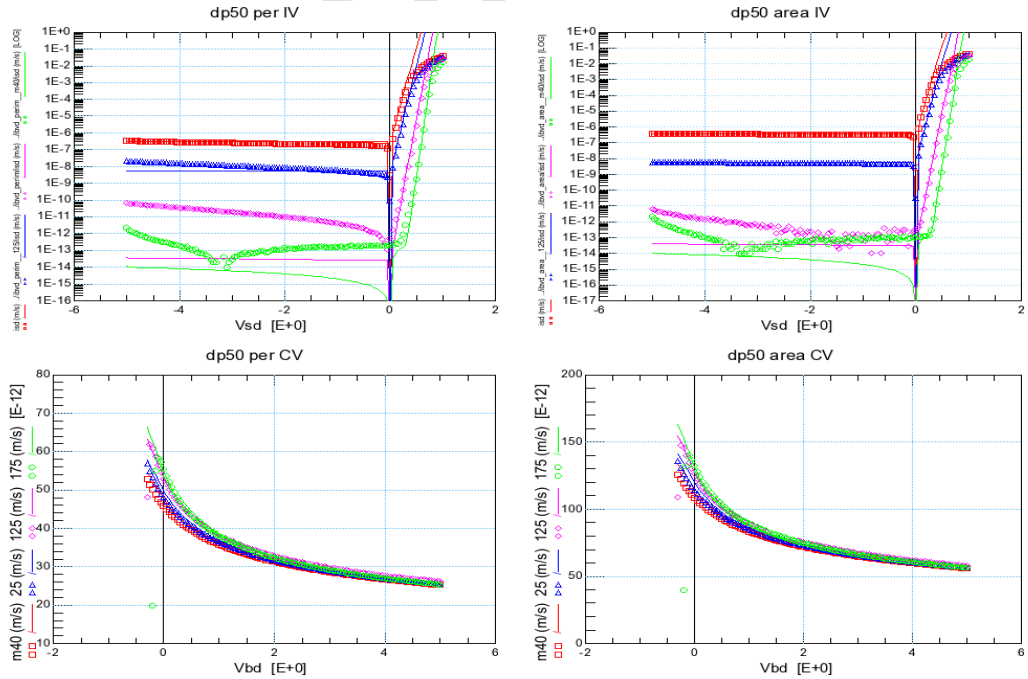

Figure 10-8 5V P+/NWELL DIODE C-V AND I-V CURVES @ T=-40,25,125 C

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11 Resistors

11.1 List of Legal Devices

Table 11-1 LIST OF LEGAL DEVICES - RESISTORS

Device	SPICE Name 2T	SPICE Name 3T
1k-HIPO–High Ohmic P-type Poly resistor	RPHPOLY2T	RPHPOLY3T
2k-HIPO–High Ohmic P-type Poly resistor	MR22T	MR23T
NMOPO–Medium Ohmic N Poly resistor	RNMPOLY2T	RNMPOLY3T
PMOPO–Medium Ohmic P Poly resistor	RPMPOLY2T	RPMPOLY3T
N+ AA Diffusion resistor	RNPLUS2T	RNPLUS3T
P+ AA Diffusion resistor	RPPLUS2T	RPPLUS3T
N-Well under AA resistor	RNWELLAA2T	RNWELLAA3T
N-Well under STI resistor	RNWELLSTI2T	RNWELLSTI3T
Metal 1 resistor	RM1	
Metal 2 resistor	RM2	
Metal 3 resistor	RM3	
Metal 4 resistor	RM4	
Metal 5 resistor	RM5	
Metal-Top resistor	RML	
N+LOPO (Salicided Poly) resistor	RNLPOLY2T	RNLPOLY3T
P+LOPO (Salicided Poly) resistor	RPLPOLY2T	RPLPOLY3T
N+ AA Diffusion (Salicided) resistor	NASR2T	NASR3T
P+ AA Diffusion (Salicided) resistor	PASR2T	PASR3T

11.2 Temperature and Voltage Coefficients

The model parameters described in this document are based on experimental data extracted from Tower Semiconductor facilities line.

Equivalent circuit of a **2-Terminal** resistor model:

For **2-Terminal** models except from metal & non-salicided resistors, the resistor values can be found according to the formula:

Rtot=2Rc+Rbody

Rc = Rc'/W

$Rbody=$

L- resistor Body length

W- resistor Body width

Rc' - contact resistance per width (= typical single contact resistance* contact pitch)

dT is T-Tnominal (25°C)

dV is the voltage across the resistor

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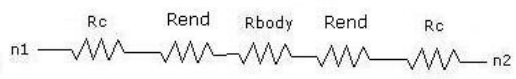
Equivalent circuit of a **2-Terminal** non-salicided poly resistor model (**HIPO, NMOPO, PMOPO**):



This model incorporates the interface resistance between the non-salicided area and the salicided area. The additional resistance, Rint [Ohm*um] is added.

For the non-salicided poly resistors (**HIPO, NMOPO, PMOPO**) the resistor values can be found according to the formula

$$Rtot = 2R_c + 2R_{end} + R_{body} \text{ ..... (1)}$$

$$R_c = \frac{R_c}{W} \ldots\ldots\ldots\ldots \quad (2)$$

$$R_{end} = \frac{R_{int}}{(W-\Delta W)} \ldots\ldots\ldots\ldots (3)$$

$$Rbody = R_{th} * \left(\frac{L}{(W-\Delta W)}\right) * (1 + VC1 * dV + VC2 * dV^2) * (1 + TC1 * dT + TC2 * dT^2) .. (4)$$

Where-

L- resistor Body length

W- resistor Body width

Rc' - contact resistance per width (= typical single contact resistance* contact pitch)

dT is T-Tnominal (25°C)

dV is the voltage across the resistor

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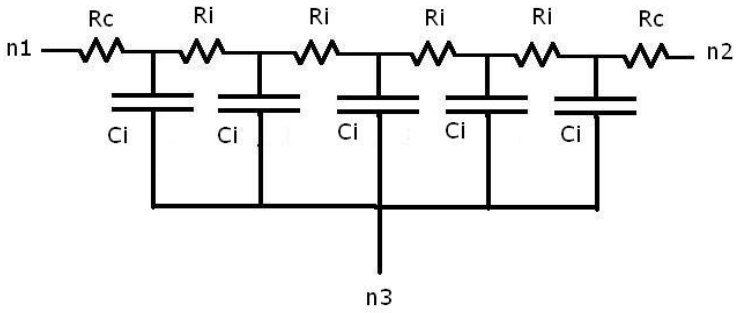
For **metal resistors** – the formula is:

$$Rtot = R_{body}$$

$$Rbody = R_{th} * \left(\frac{L}{(W - \Delta W)}\right) * (1 + VC1 * dV + VC2 * dV^2) * (1 + TC1 * dT + TC2 * dT^2)$$

3-Terminal Poly resistors:

A diagram of a 3-Terminal Poly resistor:



$$R_c = \frac{R_c}{W} \ldots\ldots\ldots\ldots \quad (1)$$

$$Rbody = R_{th} * \left(\frac{L}{4 * (W - \Delta W)}\right) * (1 + VC1 * dV + VC2 * dV^2) * (1 + TC1 * dT + TC2 * dT^2) \ldots (2)$$

$$Cj = (Ca * (W - dW) + 2 * Cp) * \left(\frac{L}{5}\right) \ldots (3)$$

L- resistor Body length
W- resistor Body width
Rc' - contact resistance per width (= typical single contact resistance* contact pitch)
dT is T-Tnominal (25°C)
dV is the voltage across the resistor

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Ca & Cp are extracted capacitance values from simulation. Ca [F/m*m] is the area capacitance; Cp [F/m] is the typical periphery capacitance for a 2um wide GC line, with 0.25um spacing between the lines.

For the **non-salicided poly resistors (HIPO, NMOPO, PMOPO),** the resistor values can be found according to the formula:

$$R_c = \frac{R_c}{W} + \frac{R_{int}}{(W - dW)} \ldots\ldots\ldots\ldots \quad (1)$$

(Rc is the total contact & interface resistance of one side; this model incorporates the interface resistance between the non-salicided area and the salicided area. The additional resistance, Rint [Ohm*um] is added)

$$Rbody = R_{th} * \left(\frac{L}{4 * (W - \Delta W)}\right) * (1 + VC1 * dV + VC2 * dV^2) * (1 + TC1 * dT + TC2 * dT^2) .. (2)$$

$$Cj = (Ca * (W - dW) + 2 * Cp) * \left(\frac{L}{5}\right) \ldots (3)$$

L- resistor Body length
W- resistor Body width
Rc' - contact resistance per width (= typical single contact resistance* contact pitch)
dT is T-Tnominal (25°C)
dV is the voltage across the resistor

Ca & Cp are extracted capacitance values from simulation. Ca [F/m*m] is the area capacitance; Cp [F/m] is the typical periphery capacitance for a 2um wide GC line, with 0.25um spacing between the lines.

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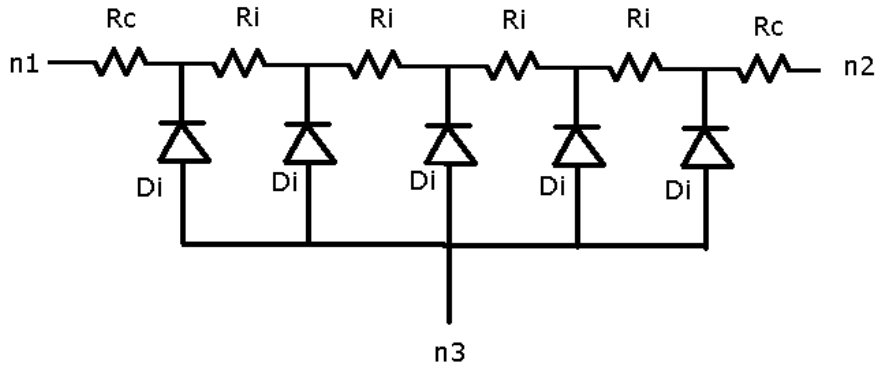
3-Terminal Diffusion/Implanted resistors:

A diagram of a 3-Terminal Diffusion/Implanted resistor:



In this example, the Diffusion is N-type (e.g. N+/AA resistor). For P-type (e.g. P+/AA resistor) the diodes will be drawn upside down.

$$R_c = \frac{R_c}{W} \ldots\ldots\ldots\ldots \quad (1)$$

$$Ri = R_{th} * \left(\frac{L}{4 * (W - \Delta W)}\right) * (1 + VC1 * dV + VC2 * dV^2) * (1 + TC1 * dT + TC2 * dT^2) .. (2)$$

Di area = (W-dW)*(L/5), periphery = (W-dW)+2*(L/5) for the two endpoint diodes and periphery = 2*(L/5) for the middle diodes

L- resistor Body length
W- resistor Body width
Rc' - contact resistance per width (= typical single contact resistance* contact pitch)
dT is T-Tnominal (25°C)
dV is the voltage across the resistor

The resistance is a function of temperature and the valid range is: -40°C-125°C

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11.3 Voltage Range

Table 11-2 VOLTAGE RANGE - RESISTORS

Resistor	Range
N^+ AA Diffusion resistor, N+ AA Diffusion (Salicided) resistor, N-Well resistor	From 0V to 3.3V
P^+ AA Diffusion resistor, P+ AA Diffusion (Salicided) resistor	From –3.3V to 0V
HIPO resistor, NMOPO resistor, PMOPO resistor.	From –8V to 8V
N+LOPO (Salicided Poly) resistor, P+LOPO (Salicided Poly) resistor, All Metal resistors	From –3.3V to 3.3V

11.4 Table of Model Parameters – Resistors

Table 11-3 MODEL PARAMETERS - RESISTORS

RESISTOR	RSH (OHM/SQ)			TC1 [1/C]	TC2 [1/C²]	VC1 [1/V]	VC2 [1/V²]	R CONTACT [OHM]	DW [UM]	RINT OHM*UM
	TYP	LOW	HIGH							
1k-HIPO–High Ohmic P-type Poly resistor	1000	880	1120	-987e-6	2.253e-6	193e-06	0	7	0.03	176
2k-HIPO–High Ohmic P-type Poly resistor	2000	1800	2200	1.69e-6	3.44e-6	-142e-6	-16.45e-6	7	0.02	-75.6
NMOPO–Medium Ohmic N Poly resistor	400	310	490	-1251e-6	2.04e-6	210e-6	-365e-6	7	0.055	93
PMOPO–Medium Ohmic P Poly resistor	310	310	310	310	310	310	310	7	310	158
AA Diffusion resistor	70	50	90	1.43e-3	1.03e-6	826e-6	7.6e-6	10	0.051	-
AA Diffusion resistor	120	80	160	1380e-6	0.27e-6	626e-6	-17.5e-6	10	-0.026	-
Nwell under AA	450	350	550	3173e-6	10.3e-6	4636e-6	-63.86e-6	10	0.52	-
Nwell under STI	1000	1000	1000	1000	1000	1000	1000	10	-	-
Metal 1 resistor	80e-3	40e-3	120e-3	3248e-6	-0.43e-6	-	-	-	-	-
Metal 2 resistor	80e-3	40e-3	120e-3	3248e-6	-0.43e-6	-	-	-	-	-
Metal 3 resistor	80e-3	40e-3	120e-3	3248e-6	-0.43e-6	-	-	-	-	-
Metal 4 resistor	80e-3	40e-3	120e-3	3248e-6	-0.43e-6	-	-	-	-	-
Metal 5 resistor	80e-3	40e-3	120e-3	3248e-6	-0.43e-6	-	-	-	-	-
Metal Last resistor	40e-3	30e-3	50e-3	3215e-6	-3.87e-6	-	-	-	-	-
LOPO (Salicided Poly) resistor	6	1	20	2744e-6	0.323e-6	14.26e-6	6673e-6	7	-	-
LOPO (Salicided Poly) resistor	5	1	20	3260e-6	0.915e-6	-1.41e-6	6917e-6	7	-	-
AA Diffusion (Salicided) resistor	7	1	15	2789e-6	0.477e-6	195.6e-6	1319e-6	10	-	-
AA Diffusion (Salicided) resistor	5	1	15	3306e-6	0.516e-6	-714.3e-6	1753e-6	10	-	-

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11.5 Comparison Between Modeling Results and Experimental Data (On Wafer) - Resistors

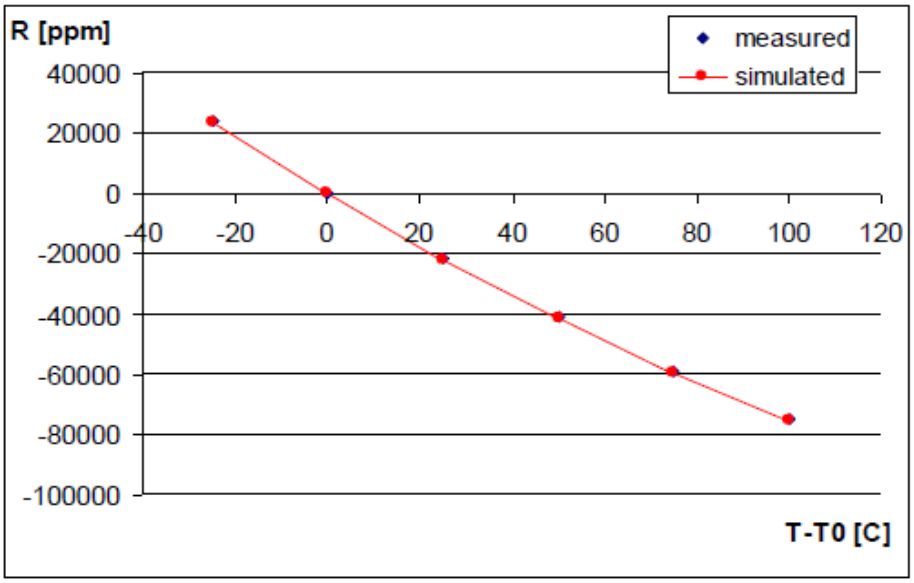

Figure 11-1 HIPO–HIGH OHMIC P-TYPE POLY RESISTOR

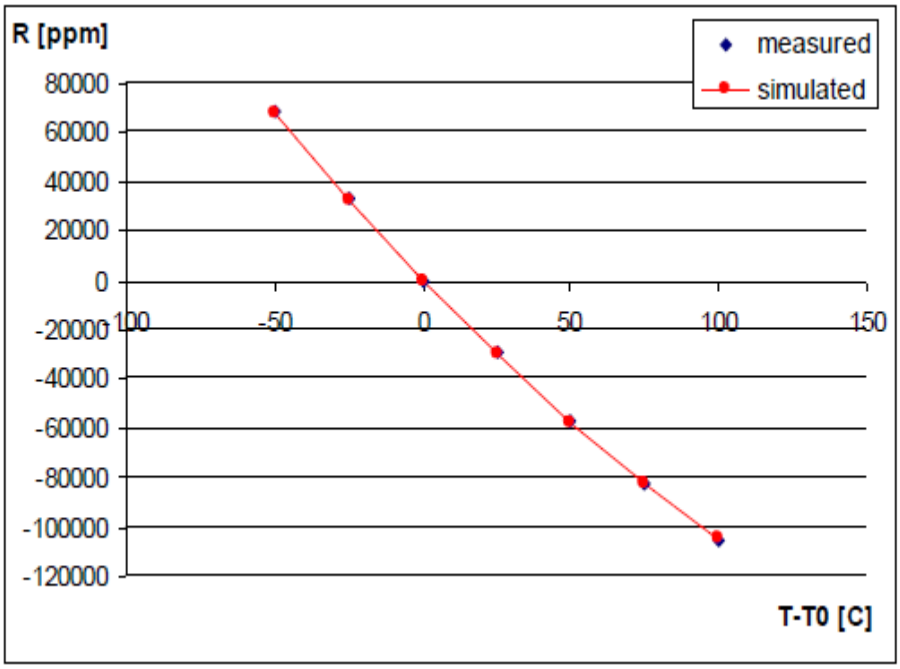

Figure 11-2 NMOPO-MEDIUM OHMIC N POLY RESISTOR

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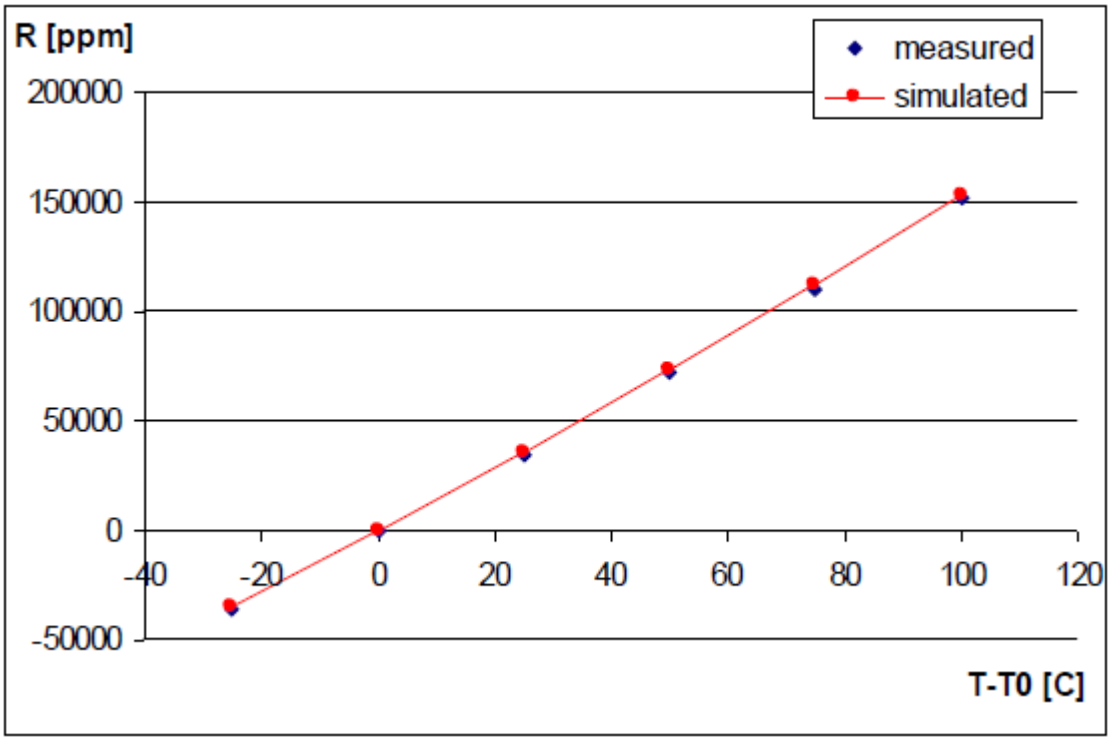

Figure 11-3 PMOPO-MEDIUM OHMIC P POLY RESISTOR

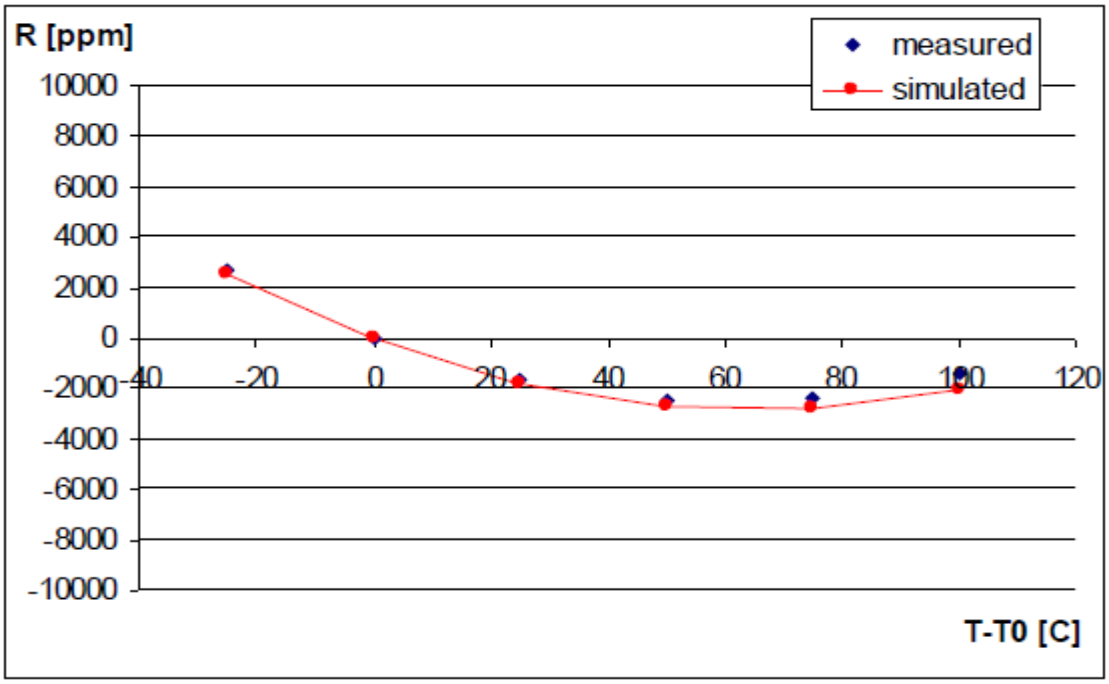

Figure 11-4 N+ AA DIFFUSION RESISTOR

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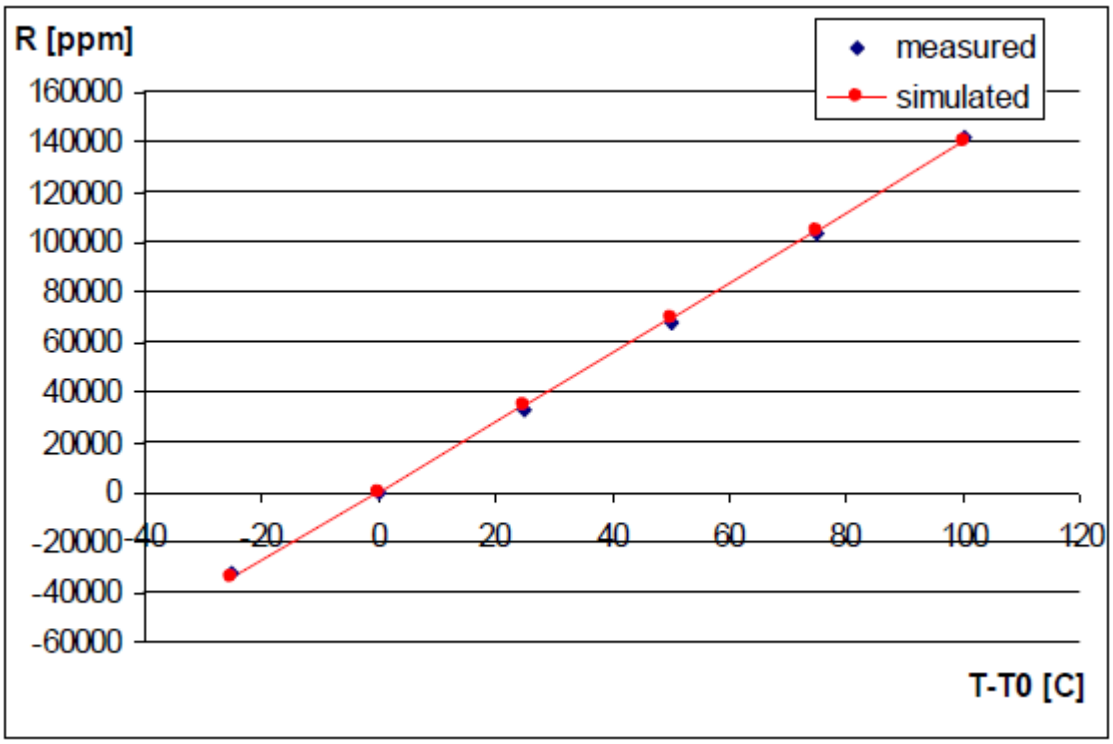

Figure 11-5 P+ AA DIFFUSION RESISTOR

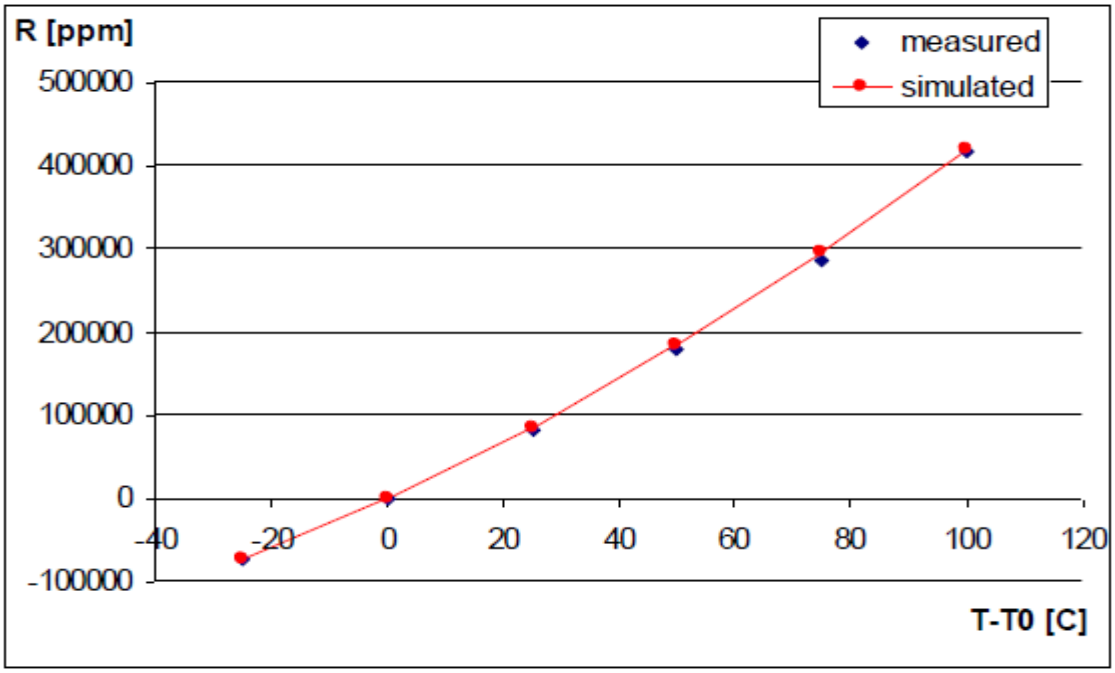

Figure 11-6 N-WELL RESISTOR UNDER AA

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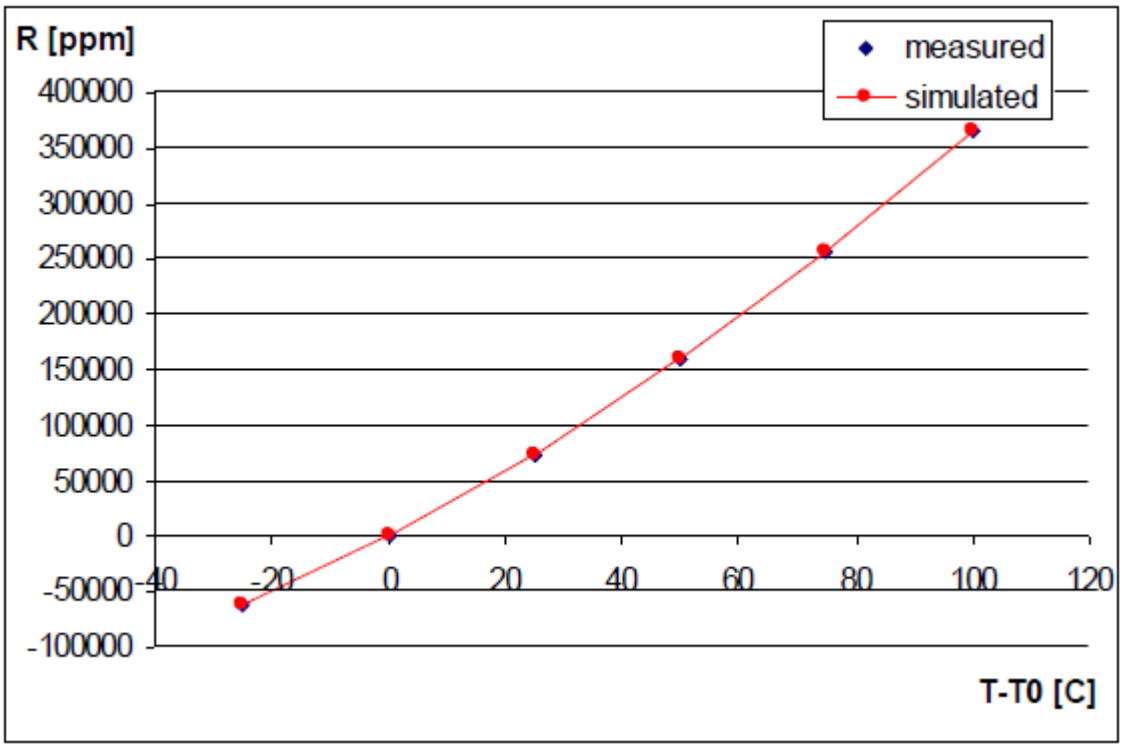

Figure 11-7 N-WELL RESISTOR UNDER STI

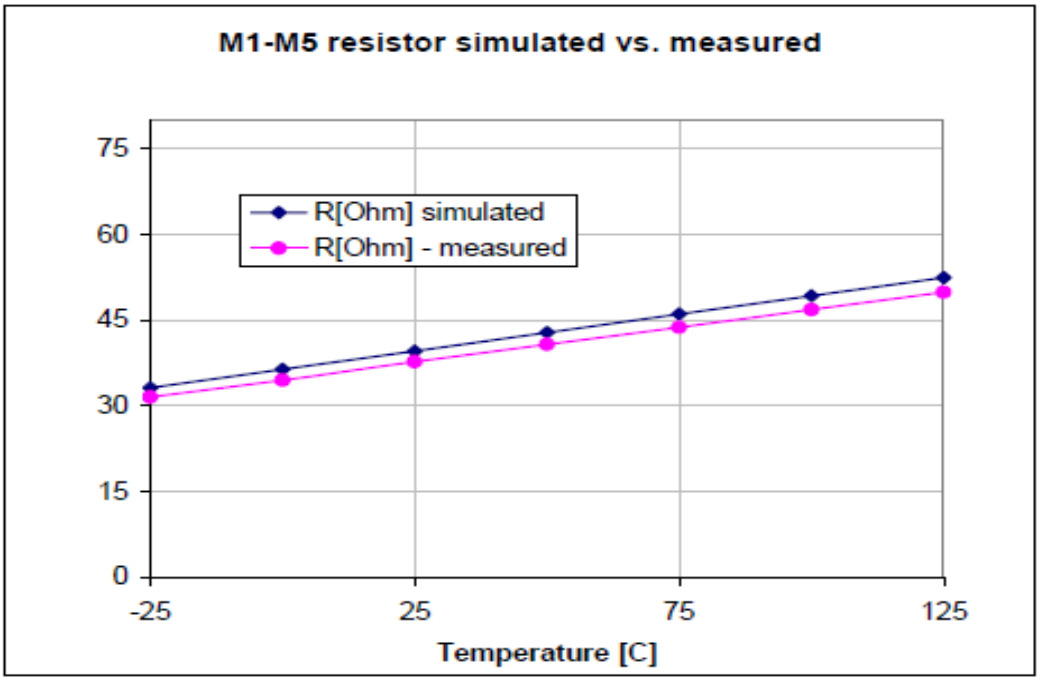

Figure 11-8 M1-M5 RESISTOR

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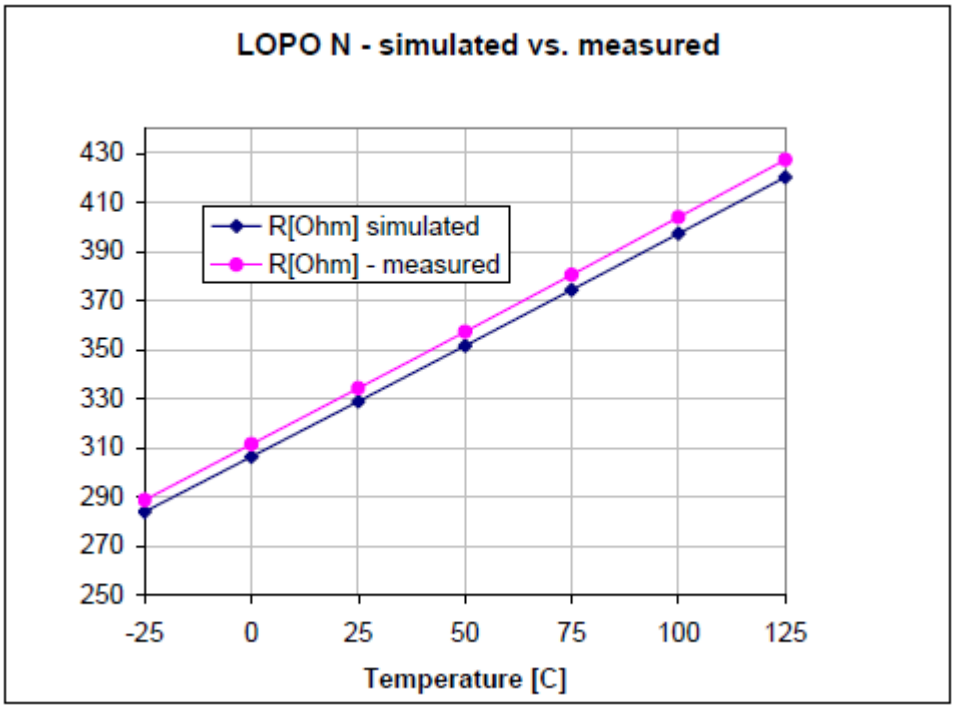

Figure 11-9 N+LOPO (SALICIDED POLY) RESISTOR

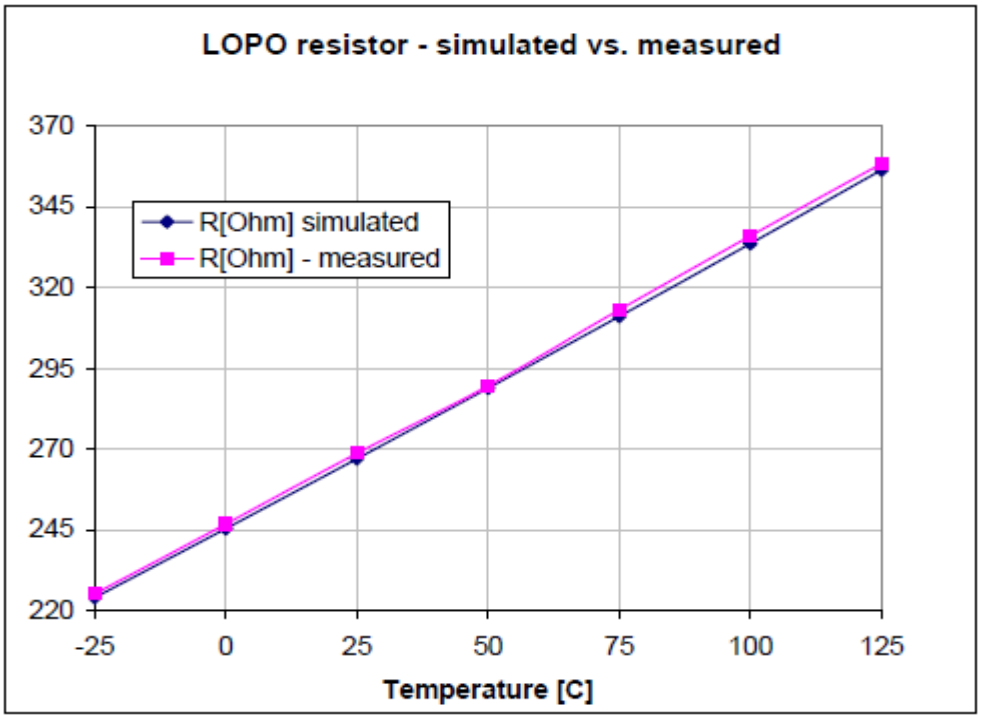

Figure 11-10 P+LOPO (SALICIDED POLY) RESISTOR

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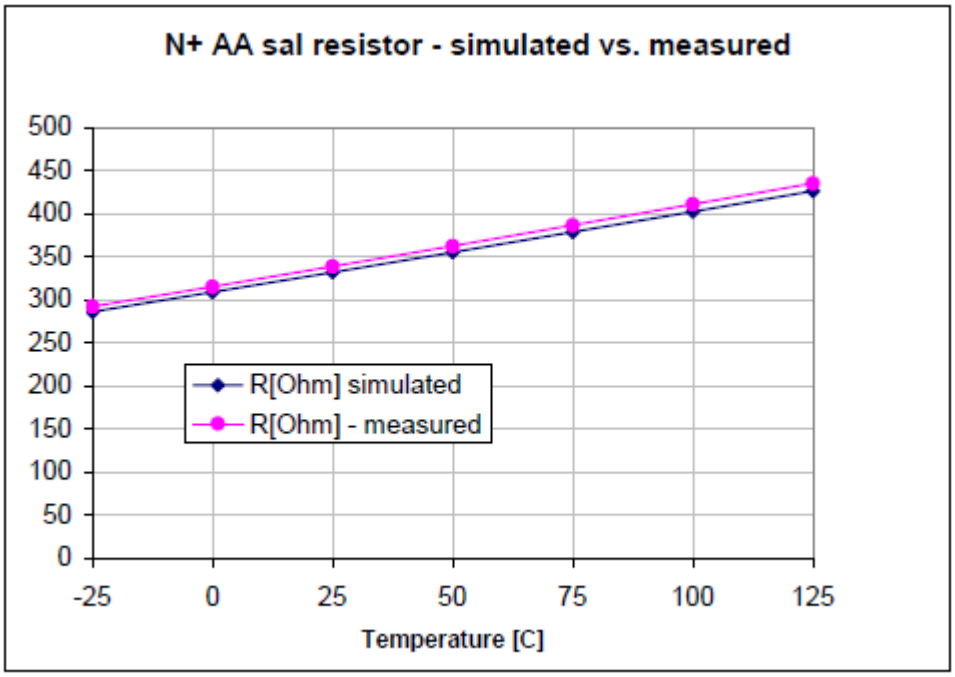

Figure 11-12 N+ AA DIFFUSION (SALICIDED) RESISTOR

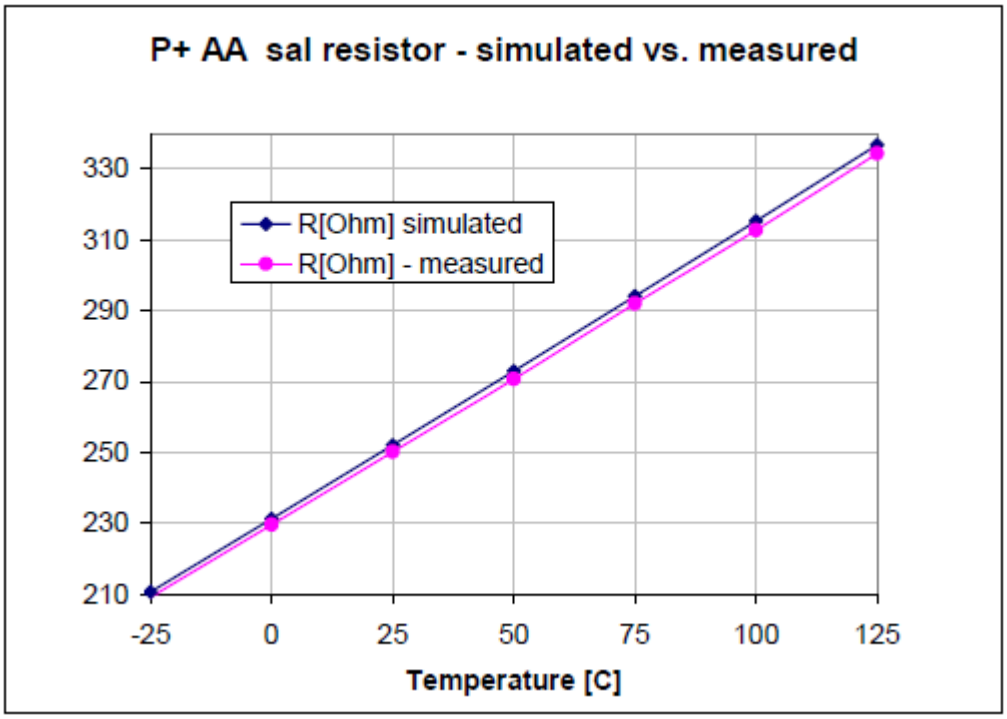

Figure 11-11 P+ AA DIFFUSION (SALICIDED) RESISTOR

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11.6 Resistors Matching

The following resistor types were characterized and modeled for mismatch: HIPO, NMOPO, PMOPO, N+ AA, P+ AA. Mismatch model is available using Monte-Carlo simulation. The results were compared to the measured data and to the theoretical calculation.

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12 Bipolar Junction Transistor Gummel-Poon Model

12.1 List of Legal Devices

Table 12-1 LIST OF LEGAL DEVICES - BIPOLAR JUNCTION TRANSISTOR GUMMEL-POON MODEL

Device	SPICE Name	Voltage Range
1.8V VPNP 10um x10 um	VD10	-3.3V to 3.3V
1.8V VPNP 5um x 5 um	VD5	-3.3V to 3.3V
1.8V VPNP 2um x 2um	VD2	-3.3V to 3.3V
1.8V VNPN 5um x 5 um	VDN5	-3.3V to 3.3V
3.3V VPNP 10um x10 um	VD10	-3.3V to 3.3V
3.3V VPNP 5um x 5 um	VD5	-3.3V to 3.3V
3.3V VPNP 2um x 2um	VD2	-3.3V to 3.3V

*VPNP 1.8V & 3.3V use same model.

12.2 Model description

The parameters were extracted from the vertical P+/Nwell/Psub & N+/Pwell/Deep-Nwell test structures at different temperatures in order to obtain temperature depended parameters. The extraction was performed using ICCAP software. The models are valid in the range of –40 to 125 C. The layout and cross-sectional view of the typical VPNP test structure is shown below. Ring base configuration has been used. These models are also applicable for LV VPNP & LV VNPN BJT transistors with voltage range –1.8V to 1.8V.

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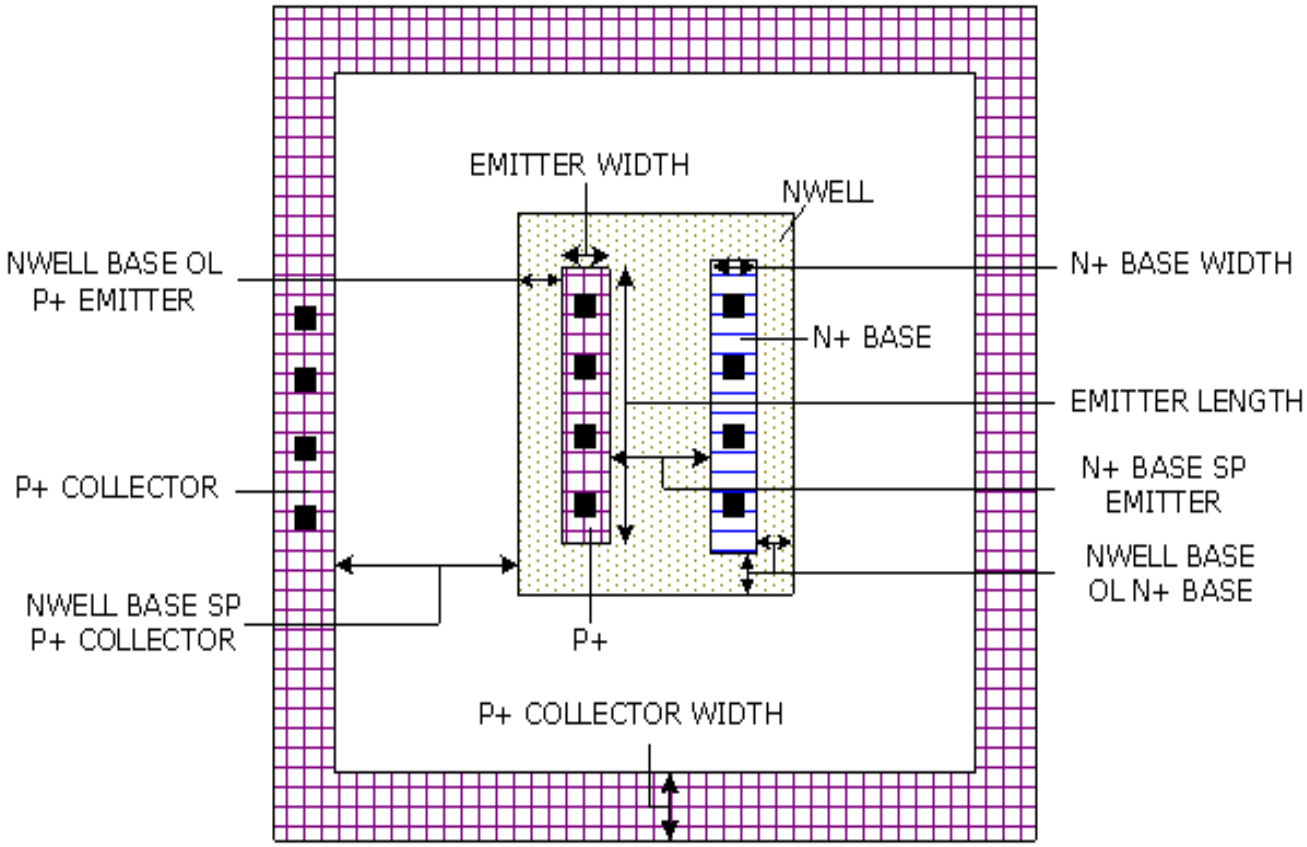
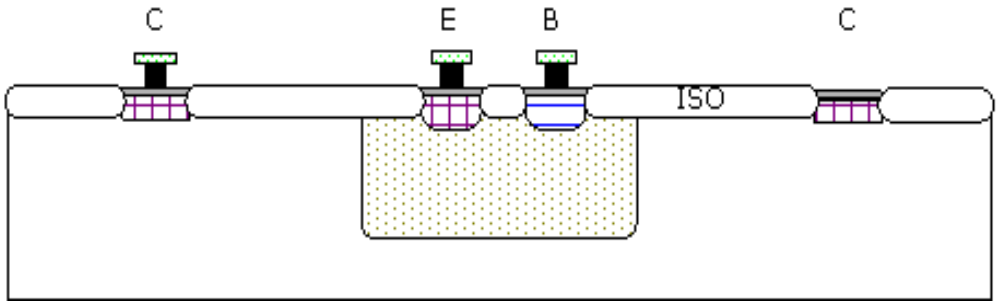
Top View



Cross Section



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12.2.1 Process Variation

Process variation is modeled using corner cases models. The BJT values for low/high corners are specified in the table below

Table 12-2 VALUES FOR MODEL PARAMETERS - BJT MODELS

BJT TYPE	BF	VAF	RB	RE	NF	CJE	CJC
VPNP 1.8V 10um x10	3.08	214	37.5	3.9	0.9802	9.8e-14	7.88e-14
VPNP 1.8V 5um x 5	2.99	204	53.02	5.062	1.003	2.99e-	4.45e-14
VPNP 1.8V 2um x	2.87	196	76.56	10.69	1.04	4.5e-15	6.37e-13
FAST	+25%	-32.5%-	-20%-	-20%-	-1%	-5%	-5%
SLOW	-25%	+32.5%	+20%	20%+	+1%	+5%	+5%

12.3 Simulated Results

Table 12-3

Device name	Parameter	Unit	Conditions/Criteria	Simulation		
				Typical	FAST	SLOW
VPNP 1.8V 10um x10 um	BETA		Vce=-0.9V;Vbe=-0.65V	2.97	3.72	2.24
	Icdrive	uA	I-collector @ Vce=-3.3V ;Vbe=-0.8	2.62E-4	3.53E-4	1.94E-4
VPNP 1.8V 5um x 5 um	BETA		Vce=-0.9V;Vbe=-0.65V	2.92	3.65	2.19
	Icdrive	uA	I-collector @ Vce=-3.3V ;Vbe=-0.8	7.9E-5	1.07E-4	5.84E-5
VPNP 1.8V 2um x 2um	BETA		Vce=-0.9V;Vbe=-0.65V	2.75	3.44	2.07
	Icdrive	uA	I-collector @ Vce=-3.3V ;Vbe=-0.8	1.67E-5	2.25E-5	1.24E-5

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12.4 Comparison Between Modeling Results and Experimental Data (On Wafer)

Table 12-4

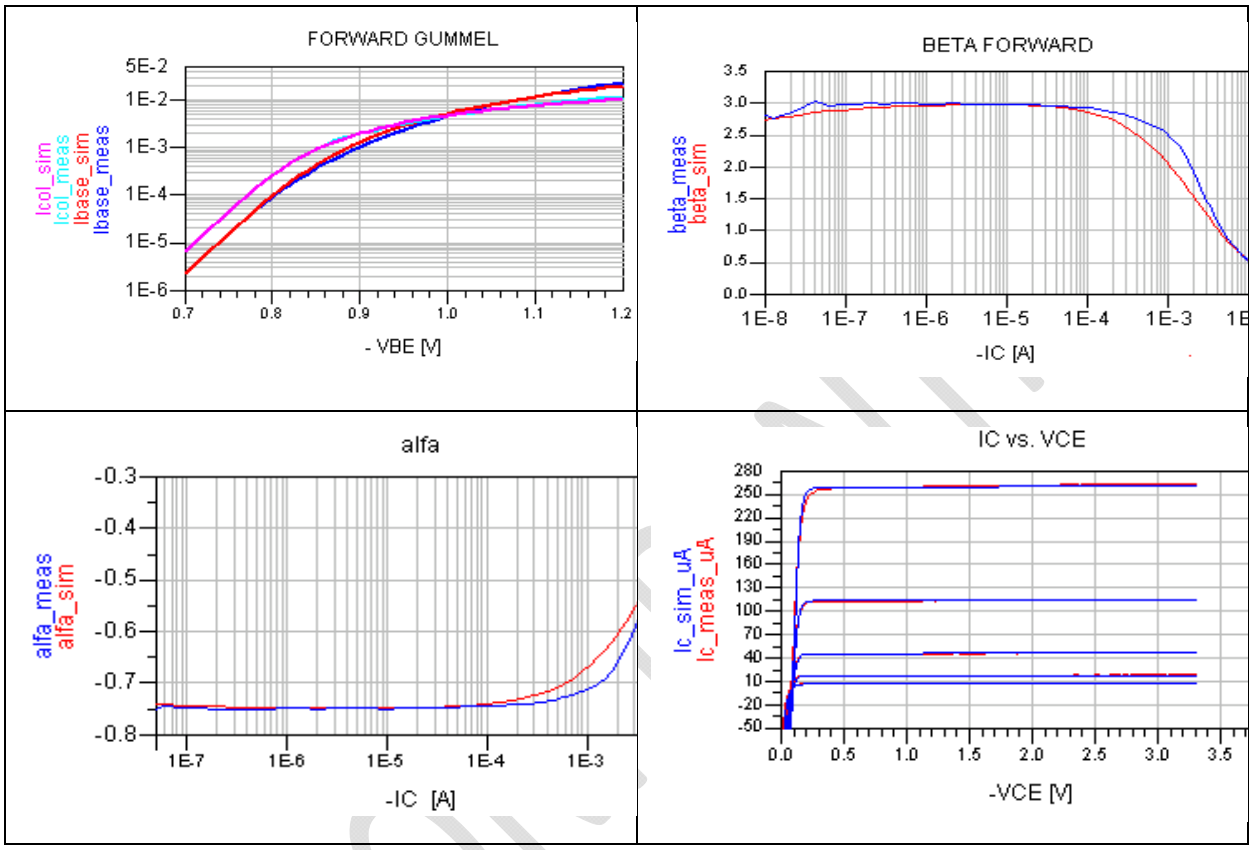

Figure 12-1 10X10 VPNP 25 C (DOTTED CURVE – MEASURED DATA, SOLID CURVE – SIMULATED DATA)

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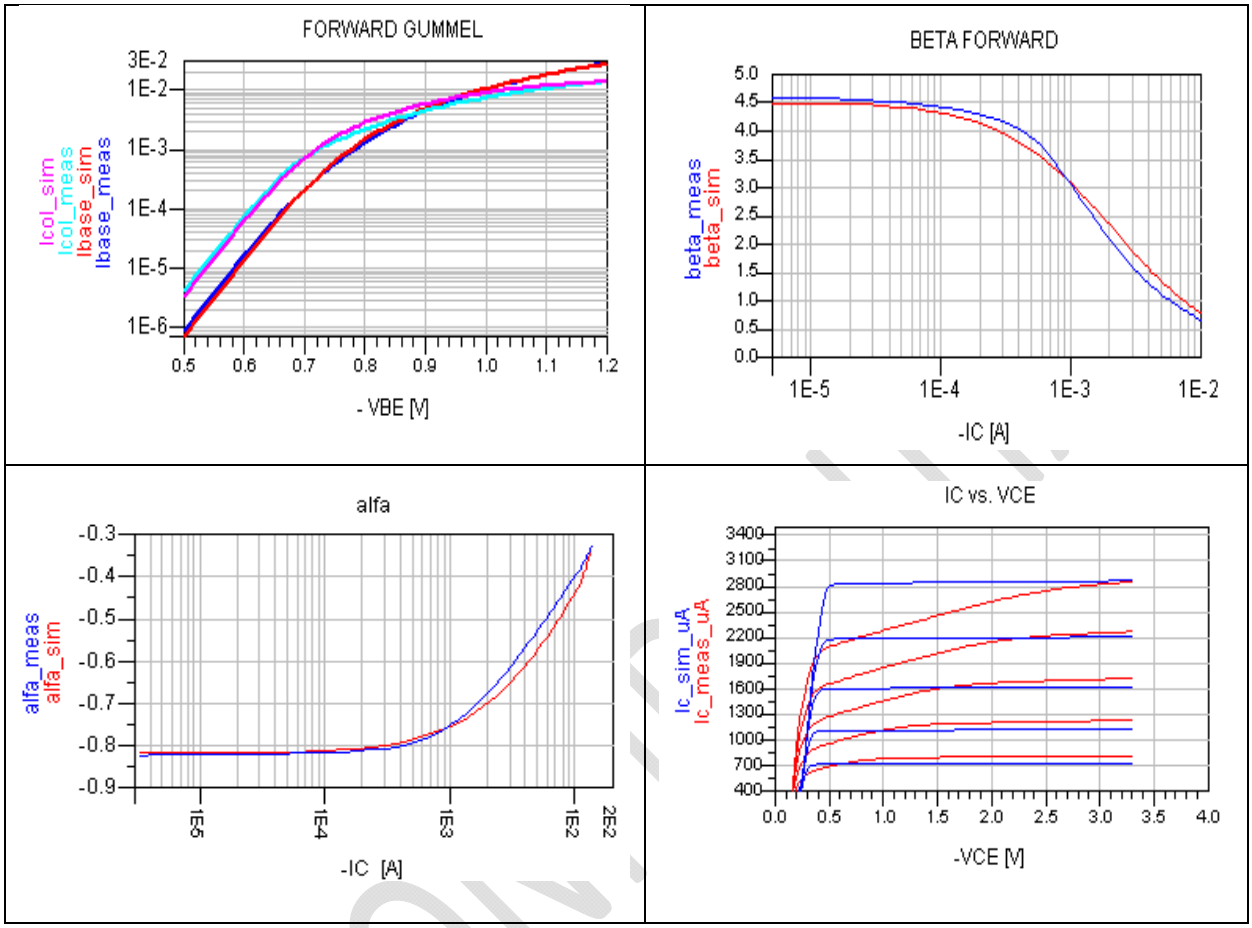

Figure 12-2 10X10 VPNP 125 C (DOTTED CURVE – MEASURED DATA, SOLID CURVE – SIMULATED DATA)

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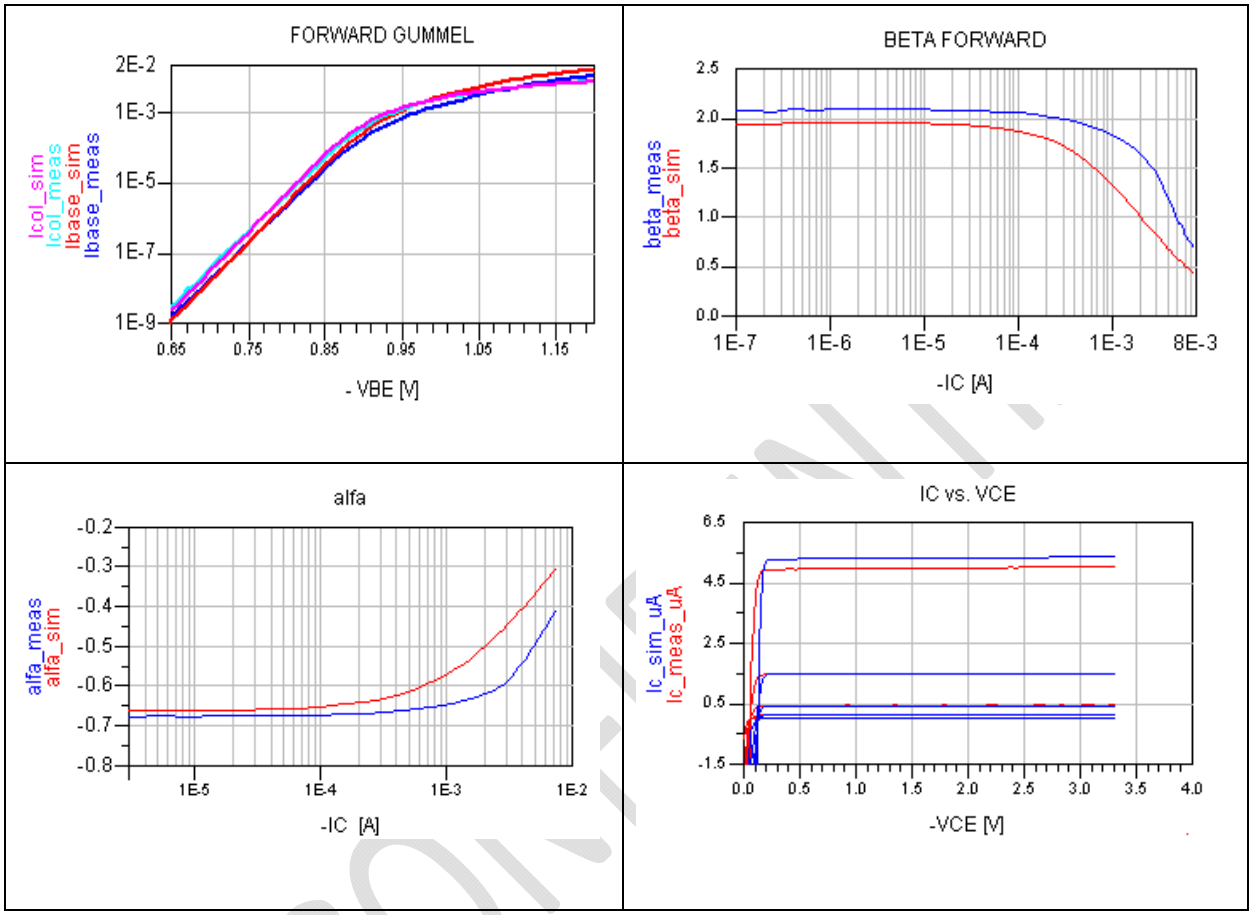

Figure 12-3 10X10 VPNP -40C (DOTTED CURVE – MEASURED DATA, SOLID CURVE – SIMULATED DATA)

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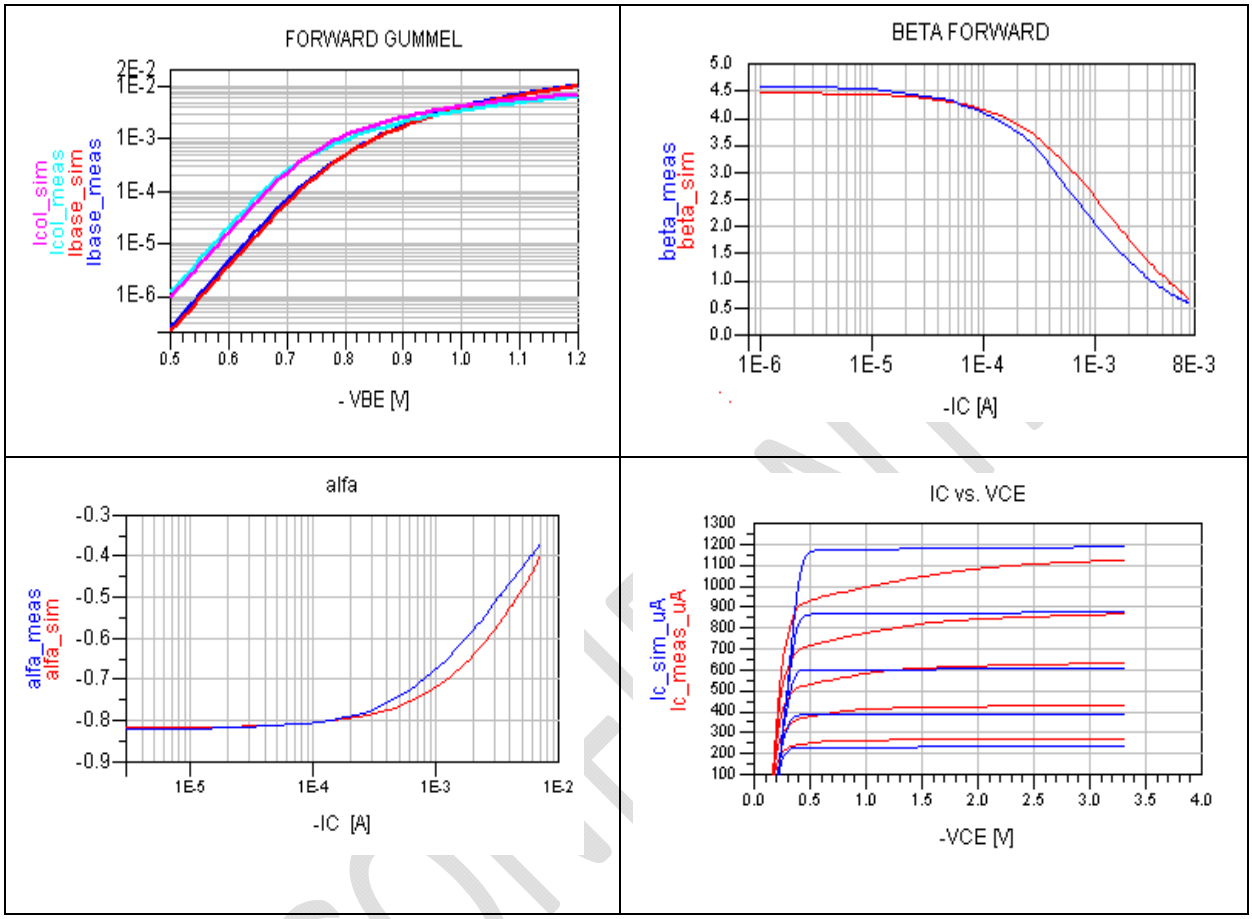

Figure 12-4 5X5 VPNP 125 C (DOTTED CURVE – MEASURED DATA, SOLID CURVE – SIMULATED DATA)

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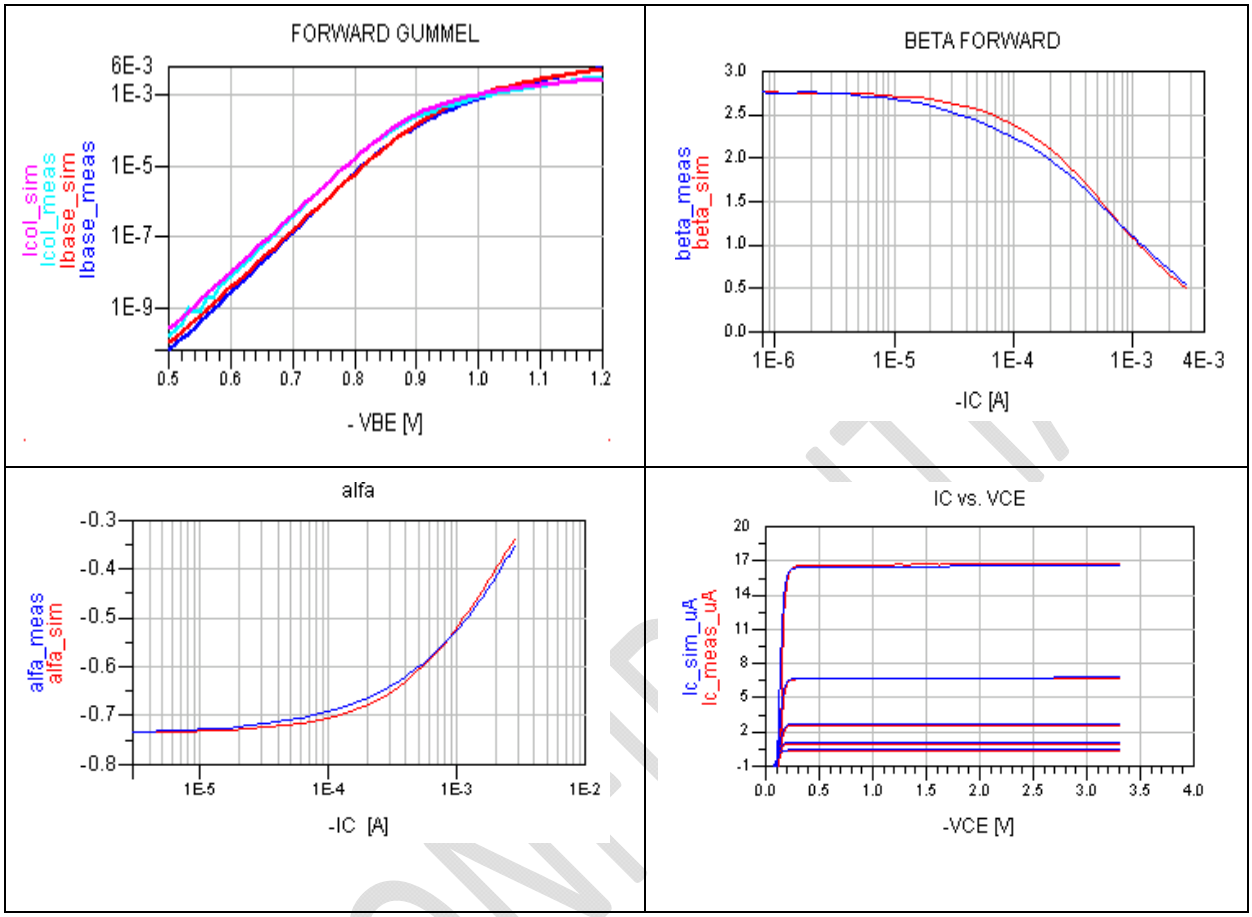

Figure 12-5 2X2 VPNP 25 C (DOTTED CURVE – MEASURED DATA, SOLID CURVE – SIMULATED DATA)

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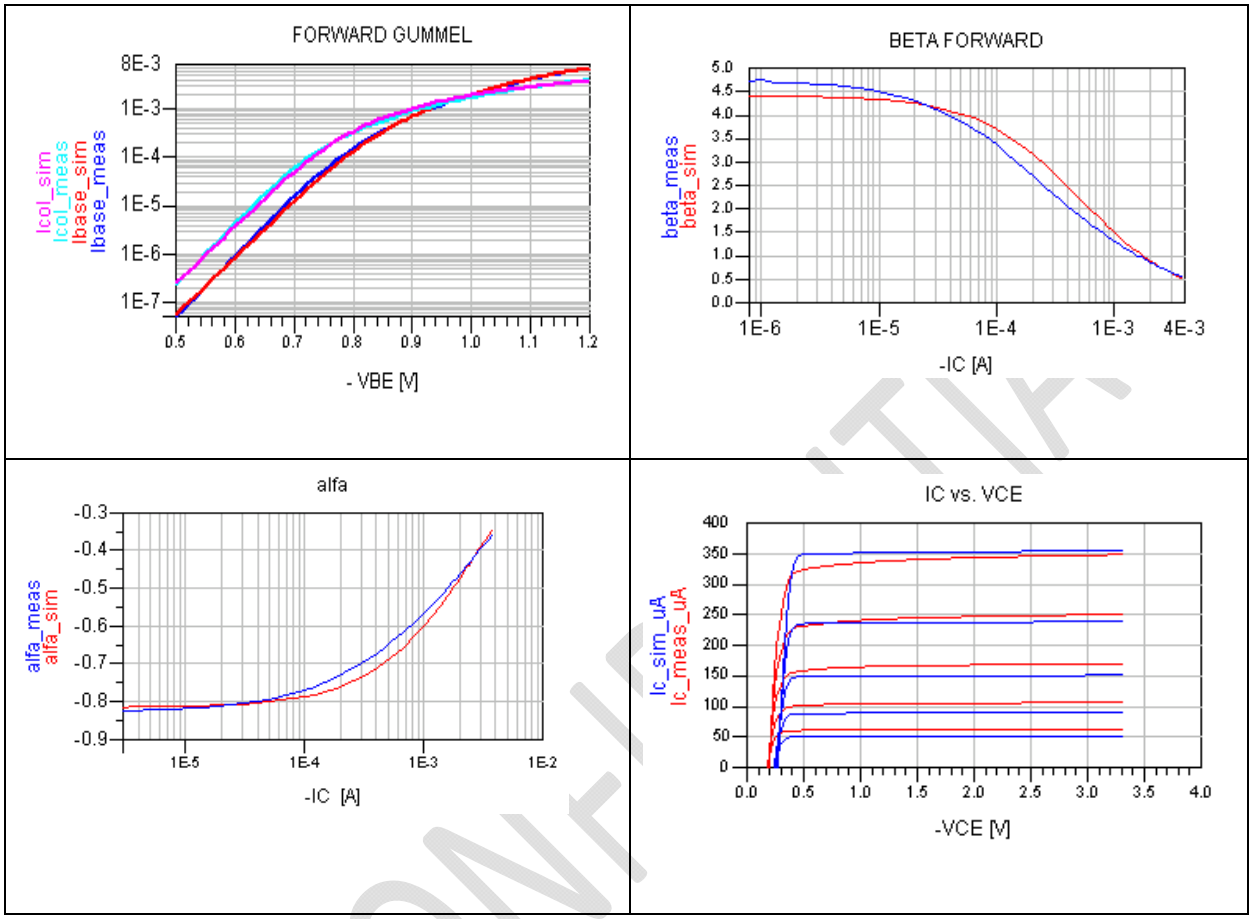

Figure 12-6 2X2 VPNP 125 C (DOTTED CURVE – MEASURED DATA, SOLID CURVE – SIMULATED DATA)

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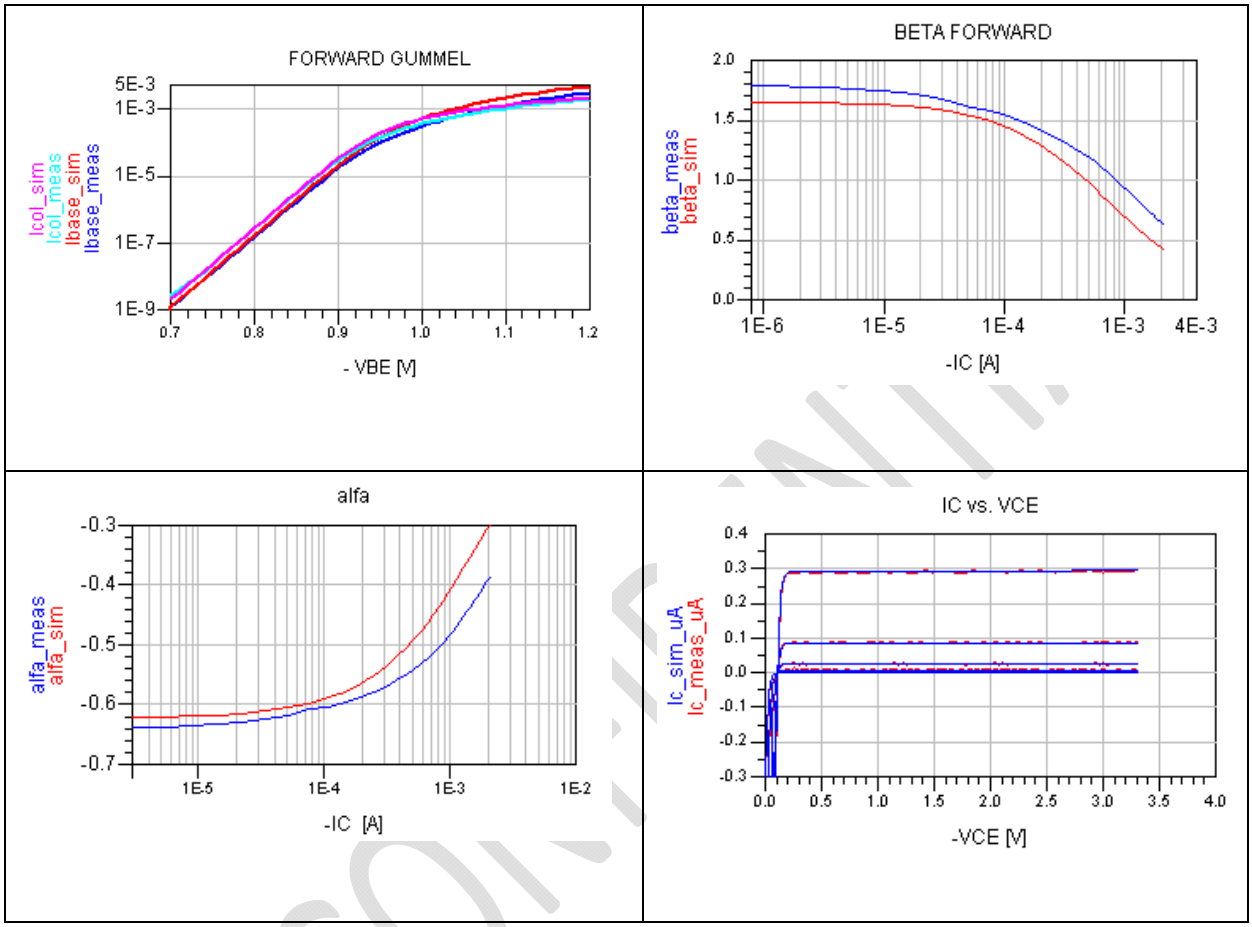

Figure 12-7 2X2 VPNP -40 C (DOTTED CURVE – MEASURED DATA, SOLID CURVE – SIMULATED DATA)

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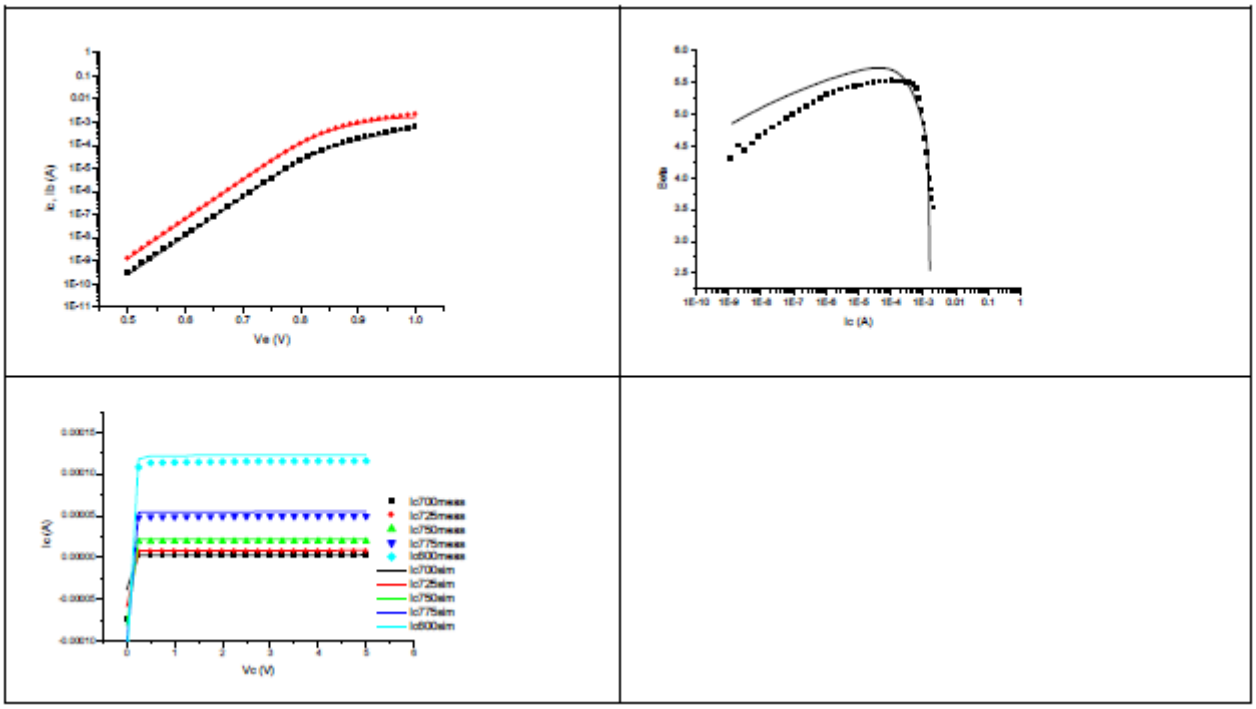

Figure 12-8 2X2 VPNP -40 C (DOTTED CURVE – MEASURED DATA, SOLID CURVE – SIMULATED DATA)

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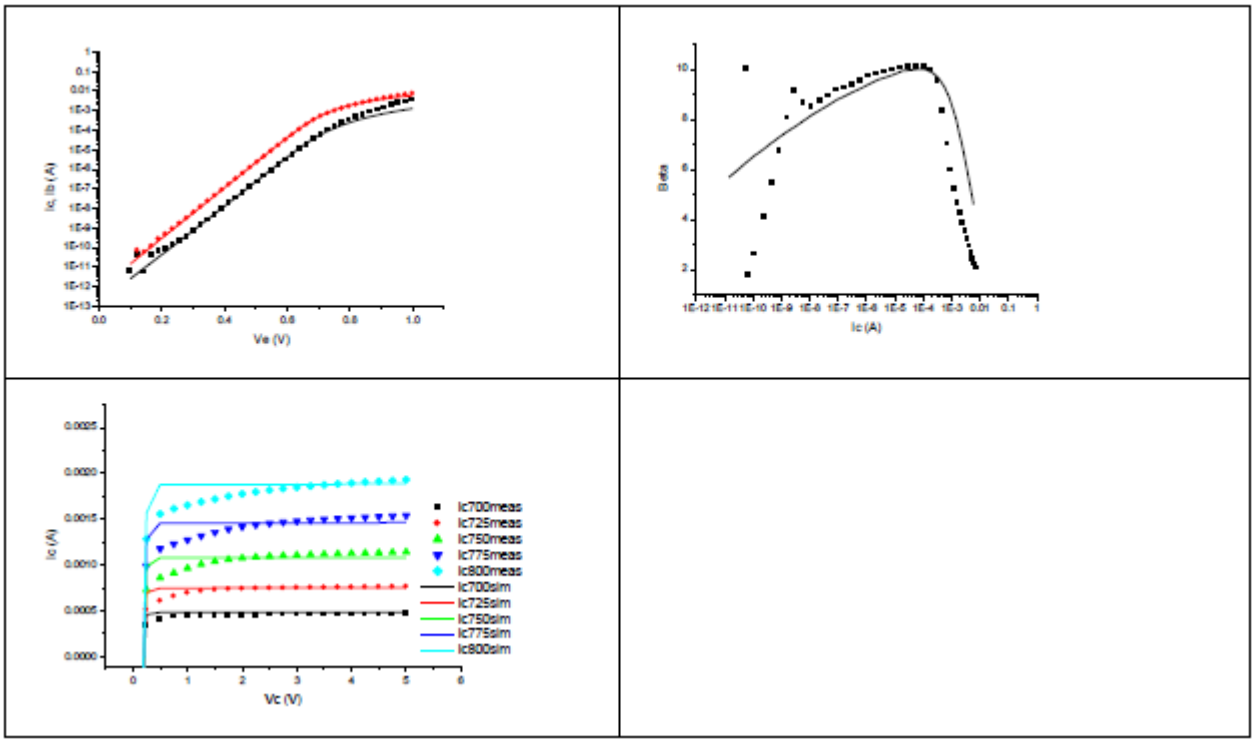

Figure 12-9 5X5 VNPN 25 C (DOTTED CURVE – MEASURED DATA, SOLID CURVE – SIMULATED DATA)

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13 Accumulation Capacitor Model

13.1 List of Legal Devices

Table 13-1 LIST OF LEGAL DEVICES - ACCUMULATION CAPACITOR MODEL

Device	SPICE Name 2T	SPICE Name 3T	Voltage Range
Accumulation capacitor 1.8V (NW)	NWCAP2T	NWCAP3T	0 to 1.8V
Accumulation capacitor 3.3V (NW)	NWCAPH2T	NWCAPH3T	0 to 3.3V
Accumulation capacitor 5V (NW)		CH53T	0 to 5V

13.2 Model description

Model parameter sets were extracted using 1.8V and 3.3V Nwell test structures with different area/periphery ratio. Eight different 1.8V and 3.3V n-well devices were characterized and model parameters were extracted for temperatures at –40, 25, and 125C. The accumulation capacitor sizes used for extraction (W/L in um) are listed in tables below:

Table 13-2 LIST OF DEVICES USED FOR EXTRACTION

Type	Geometries (W/L um)	No. of Geometries
1.8V Nwell	10/0.18, 10/0.25, 10/0.36, 10/0.72, 10/1.8, 10/10, 5/5, 2/2	8
3.3V Nwell	10/0.3, 10/0. 5, 10/0.72, 10/1.8, 10/3.6, 10/10, 5/5, 2/2	8
5V Nwell	10/0.3, 10/0. 5, 10/0.72, 10/1.8, 10/3.6, 10/10, 5/5, 2/2	8

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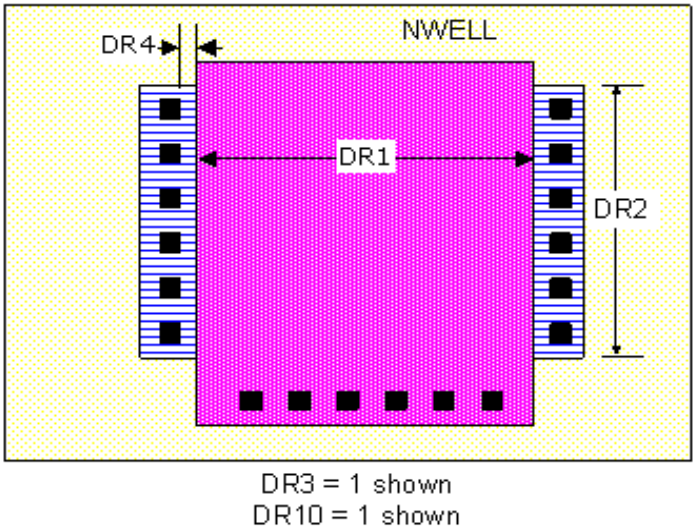
The layout below is an example for Accumulation capacitor 1.8V (NW).

Block



ACCUMULATION CAPACITOR LAYOUT

DR1=DR2=5um

DR4=0.16um

The model is scalable with L & W (DR1, DR2) and temperature dependent. Models are valid for temperature range –40°C to 125°C. The equivalent circuits are shown below.

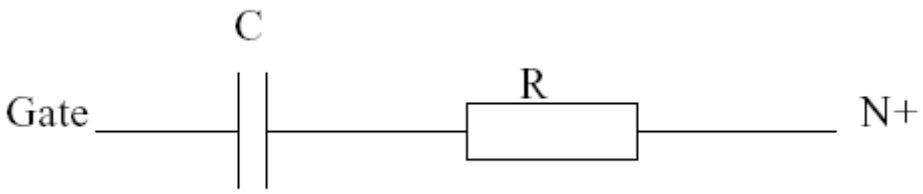


Figure 13-1EQUIVALENT CIRCUIT FOR 2 TERMINAL NWELL ACCUMULATION CAPACITOR

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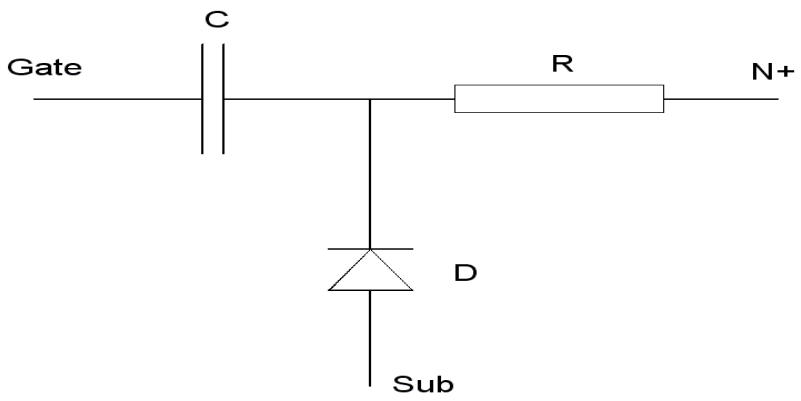

Figure 13-2 EQUIVALENT CIRCUIT FOR 3 TERMINAL NWELL ACCUMULATION CAPACITOR

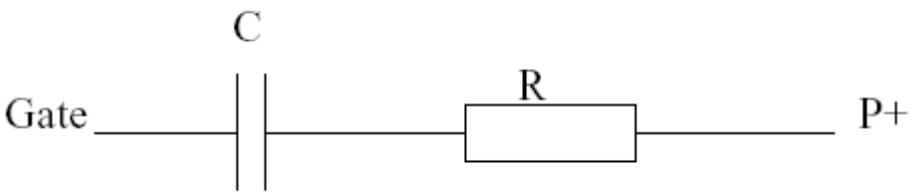

Figure 13-3 EQUIVALENT CIRCUIT FOR 2 TERMINAL PWELL ACCUMULATION CAPACITOR

13.2.1 Process Variation

Model parameters were extracted from measured case and three corners (typ/high/low) were generated to match DR2 0018 SL_SCL values by multiplying capacitance values by a factor.

Process variation is modeled as followed: The total capacitance value for low voltage capacitors (pwcap, nwcap) is ±6% for high/low corners respectfully. The total capacitance value for high voltage capacitors (pwcaph, nwcaph) is ±10% for high/low corners respectfully. Process variability is based mainly on *Tox* variability limits. Parasitic resistance variability based on data of well resistors variation

Table 13-3 DEVICE PARAMETERS FOR ACCUMULATION CAPACITOR

Device	Parameter	Unit	Typical	Low Limit	High Limit
1.8V Nwell, L=5um, W=5um	Capacitance in accumulation	fF / um^2	9.15	8.6	9.70
3.3V Nwell, L=5um, W=5um	Capacitance in accumulation	fF / um^2	5.10	4.55	5.65
5V Nwell, L=5um, W=5um	Capacitance in accumulation	fF / um^2	3.11	2.77	3.45

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13.2.2 Nwell Capacitor

The capacitance is simulated using the equation:

C= C0+C1*tanh((Vc1+Vgs)/Nc1)

Where parameters C0 and C1 are scalable with L and W (in micron):

$C0=C0_W*W+C0_L*L+C0_{Area}*W*L$

$C1=C1_W*W+C1_L*L+C1_{Area}*W*L$

And Vc1 and Nc1 are temperature dependent (*temper* is temperature in ^{0}C):

$Vc1=Vc1_0*(1+Vc1_T*(temper-25))$

$Nc1=Nc1_0*(1+Nc1_T*(temper-25))$

The resistor is modeled as following:

R=R0*(L/W)

For the 3 terminal models, the parasitic diode is modeled as following:

Diode area=(W+2*0.43u) * (L+2*0.43u+2*0.48u)

Diode periphery=2*((W+2*0.43u) + (L+2*0.43u+2*0.48u))

13.3 Table of Model Parameters

Table 13-4 MODEL PARAMETERS - ACCUMULATION CAPACITOR MODEL

Device	$C0_W$ [F/m]	$C0_L$ [F/m]	$C0_{AREA}$ [F/m2]	$C1_W$ [F/m]	$C1_L$ [F/m]	$C1_{AREA}$ [F/m2]
nwcap	1.8808E-10	6.8344E-10	0.0054382	-1.1966E-10	-3.6962E-11	0.0035703
nwcaph	3.2476E-10	6.6222e-010	0.0031521	-1.9614e-011	-2.4526E-11	0.0017597
ch53t	3.0687E-10	0	0.0018520	-6.1707E-11	0	0.0012194

Table 13-5 MODEL PARAMETERS - ACCUMULATION CAPACITOR MODEL (CONT.)

Device	$Vc1_0$ [V]	$Vc1_T$ [1/^{0}C]	$Nc1_0$ [1/^{0}C]	$NC1_T$ [1/^{0}C]	R0 [Ω/sq]
nwcap	0.10889	0.0019048	0.43674	0.0014088	113
nwcaph	0.21576	0.0016562	0.76716	0.0015273	113
ch53t	0.50816	0.0016562	0.83909	0.0016037	113

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13.4 Comparison Between Modeling Results and Experimental Data

13.4.1 Voltage Dependence

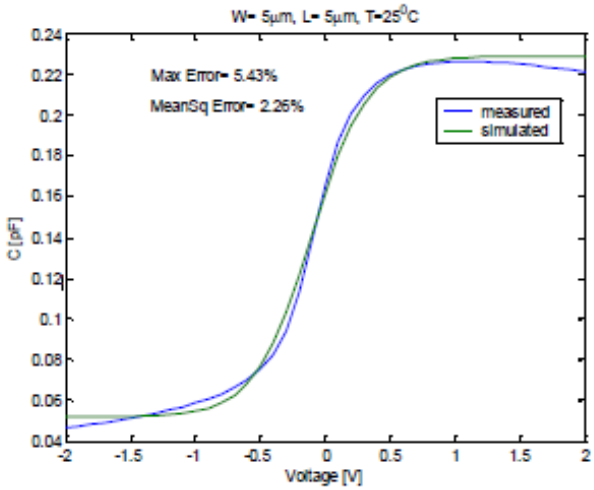

Figure 13-5 3.3V NWELL (5UMX5UM) CAPACITOR C-V

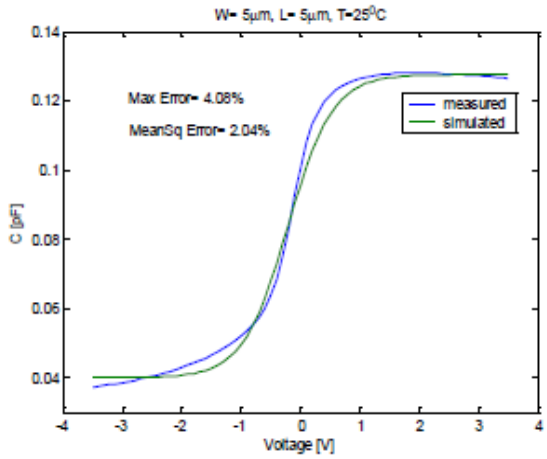

Figure 13-4 1.8V NWELL (5UMX5UM) CAPACITOR C-V

13.4.2 Temperature Dependence

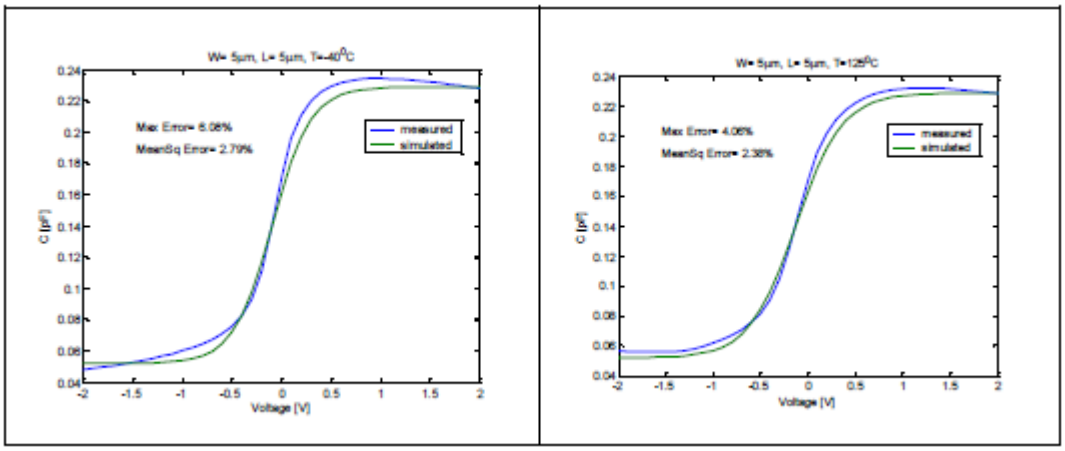

Figure 13-6 NWCAP (5UMX5UM) TEMPERATURE DEPENDENCE

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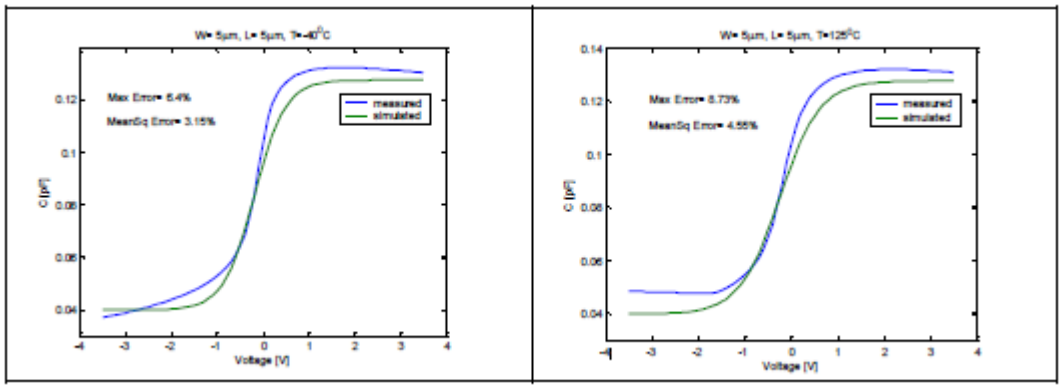

Figure 13-7 NWCAPH (5UMX5UM) TEMPERATURE DEPENDENCE

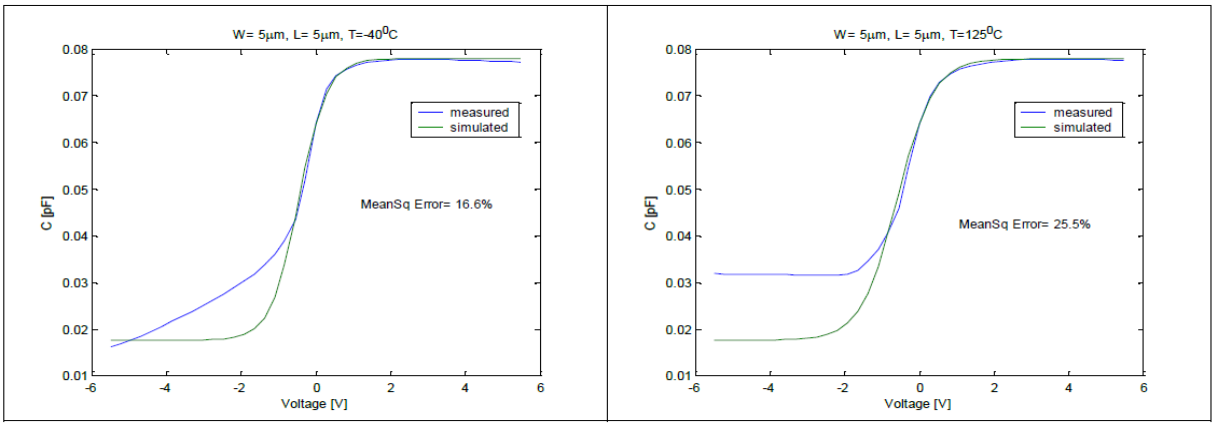

Figure 13-8 CH53T (5UMX5UM) TEMPERATURE DEPENDENCE

13.5 Accumulation Capacitors Matching

All Accumulation capacitor types were characterized and modeled for mismatch. Mismatch model is available using Monte-Carlo simulation for Hspice simulator. The results were compared to the measured data and to the theoretical calculation.

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14 MIM Capacitor Models

The MIM capacitor is a commonly used capacitor in Mixed Signal and RF designs. It is composed of an ONO (Silicone Oxide – Nitride – Oxide) insulator layer between two metal plates ML-1 and CE (the CE plate is connected to ML with vias). It has a nominal value of 1fF/um^2.

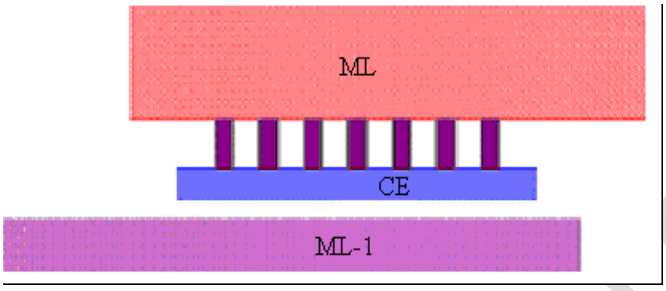

Figure 14-1 CROSS SECTION OF MIM CAPACITOR

14.1 List of Legal Devices

Table 14-1 LIST OF LEGAL DEVICES - MIM CAPACITOR MODELS

Device	SPICE Name	Voltage Range
MIM Capacitor-Units	CMIM_SQ	-5V to 5V

14.2 Typical layout of Legal Capacitors

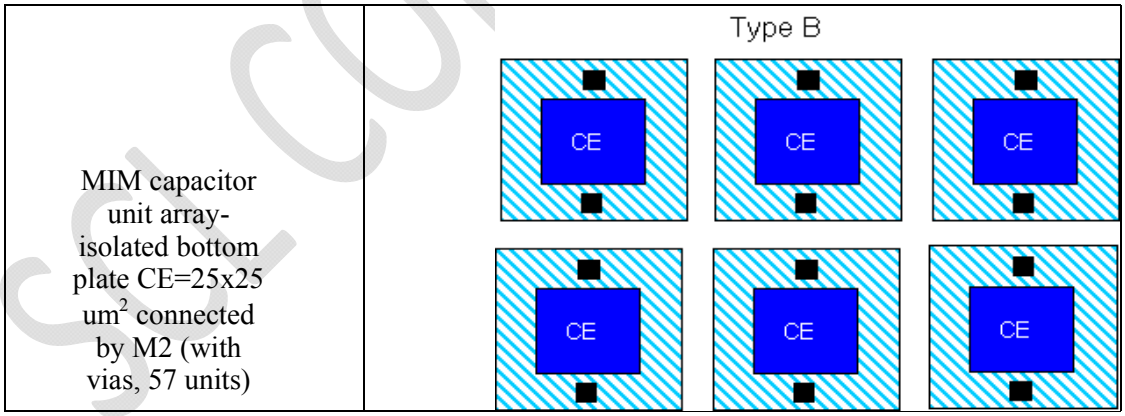

Figure 14-2 MIM CAPACITOR-ISOLATED BOTTOM PLATE

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14.3 Model description

The model parameters were extracted from the test structures with CE dimensions specified above at different temperatures in order to obtain temperature depended parameters. The models are valid in the range of –40 to 125 °C.

The capacitance value can be found using following formula:

$$C = (c_{area} * W * L + C_{periphery} * 2 * (W + L)) * (1 + VC1 * dV + VC2 * dV^2) * (1 + TC1 * dT + Tc2 * dT^2)$$

Where:

Carea – Area Capacitance
Cperiphery – Periphery Capacitance
VC1-linear voltage coefficient
VC2-first non-linear voltage coefficient
TC1-linear temperature coefficient
TC2-first non-linear temperature coefficient

Table 14-2 VALUES FOR MODEL PARAMETERS - MIM CAPACITOR MODELS

Device Type	CArea [F/m²]			Cper [F/m]	TC1 1/C (x1e-6)	TC2 1/C² (x1e-6)	VC1 1/V (x1e-6)	VC2 1/V² (x1e-6)
	Typical	Low	High					
MIM Capacitor-Isolated bottom plate	1.0e-3	0.83e-3	1.17e-3	0.11e-9	-1.475	0.016	19.98	-1.65

14.3.1 Process Variation

Process variation is modeled. The variation is separated into area variation (according to table above) and periphery variation.

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14.4 Comparison Between Modeling Results and Experimental Data (On Wafer)

14.4.1 MIM Capacitor –Units

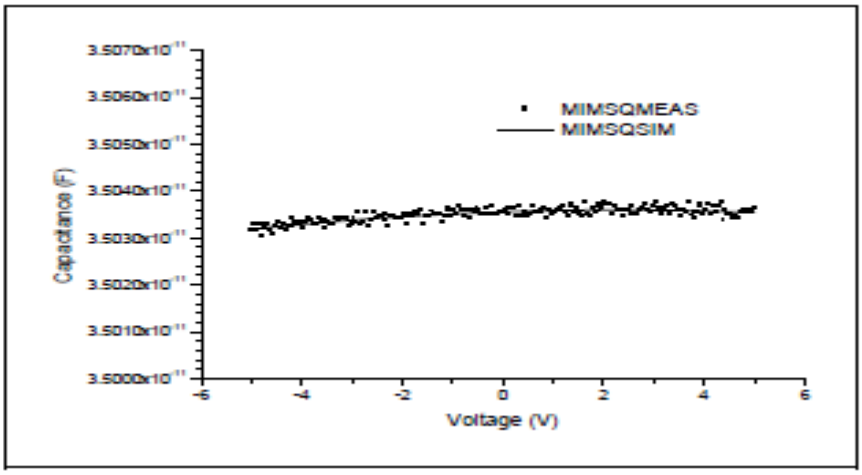

Figure 14-3 MIM CAPACITOR-UNITS - VOLTAGE (V)

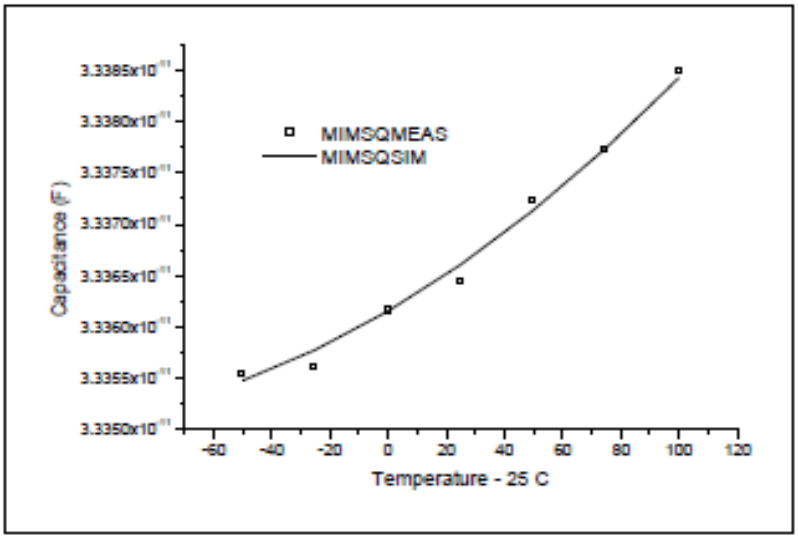

Figure 14-4 MIM CAPACITOR-UNITS – TEMPERATURE – 25C

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